A Comprehensive Interpretation of the Experimental Results

An illustrated, model-by-model research narrative covering every graph

Illicit transaction detection with classical machine learning, boosted trees, and graph neural networks.

Source: EXPOSED_DIAGNOSTIC_20261029T100035Z.zip | Eleven model configurations | Two frozen decision policies | 82 original figures

Prepared October 29, 2026.
All numerical comparisons in this document refer to the official Elliptic++ experiment rather than synthetic Smoke-test outputs.

Scientific qualification: periods 35-49 were exposed during research development.
They are a retrospective diagnostic window, not a fresh, blinded or prospective test.

Executive summary

The Elliptic++ experiment asks a deceptively difficult question: how well can models identify transactions labeled as illicit when most transactions have no verified label, the illicit class is relatively rare among known labels, and the statistical character of the network changes over time?

Eleven configurations were evaluated, including five conventional feature-based approaches and six graph-neural-network or architecture-matched graph-control variants.

The results are rich, but they do not support a single model ranking that is valid for every objective.

A model that ranks risky transactions well may produce too many false alarms at an operational threshold, while a very selective model may attain remarkable precision and still miss many illicit transactions.

The exposed diagnostic window contains **16,670** known-label transactions: **1,083** illicit and **15,587** licit.

CatBoost obtains the highest pooled Average Precision (AP = **0.757**), narrowly ahead of XGBoost (**0.752**) and LightGBM (**0.745**).

Under the historically chosen F1 policy, however, the MLP has the highest F1 score (**0.733**) and a precision of **0.987**, largely because it produces only eight false positives.

That selectivity comes with a cost: its recall is **0.583**, meaning it misses **452** of the **1,083** known illicit transactions.

At the opposite end, logistic regression detects **813** known illicit transactions but also flags **2,419** known licit ones.

Neither pattern is intrinsically preferable without specifying the review capacity and the relative consequences of missed detections and false alerts.

An even more important result is the sharp deterioration in per-period ranking after time step **42**.

For LightGBM, the unweighted mean of period-wise AP is approximately **0.880** over periods **35**-**42** and **0.067** over periods **43**-**49**.

This contrast coincides with changes in the feature distribution and in how many illicit labels are observed.

The joint occurrence is consistent with several explanations - including distribution shift, class-prevalence effects, and selective labeling - but it does not isolate a causal mechanism.

In late periods containing only a handful of known illicit cases, period-level estimates are especially unstable.

The graph comparisons also resist simplistic interpretation.

Bidirectional graph aggregation looks beneficial relative to its corresponding no-message control in several historical development comparisons, the incoming-only architecture does not consistently show the same improvement.

Rewiring experiments provide additional evidence that original network connectivity matters in at least some settings.

Still, graph contribution depends on the development fold, architecture, and definition of the comparison.

The report therefore interprets the graph experiments as controlled evidence about association with structure, not as proof of a universal or causal graph advantage.

The following sections explain the evaluation protocol and the metric definitions, present structured numerical comparisons, then walk through all 82 source figures.

Every figure has a reading guide, an interpretation grounded in the saved CSVs, and a statement about what the chart cannot establish.

The report closes by synthesizing the results into scientifically supportable findings and explicit limitations.

The primary selected research model is LightGBM, based on its historical development mean AP of **0.8997**. CatBoost is the observed leader in pooled diagnostic ranking, MLP leads in diagnostic F1, and the preferred model for ranked alert review changes with K. Section 10 explains these conditional choices, while Section 11 states the final model selection rule.

1. Experimental context and the meaning of the labels

The full Elliptic++ transaction dataset contains **203,769** transactions linked by **234,355** directed edges.

Of these transactions, **4,545** have an illicit label, **42,019** have a licit label, and **157,205** have no known class label.

The experiment uses **93** transaction-level numerical features from the selected source feature group.

Importantly, an unknown label is not a licit label.

This distinction affects almost every downstream interpretation, from supervised model training and confusion matrices to the meaning of a top-ranked alert.

The research protocol separates training (steps **1**-**26**), early stopping (**27**-**29**), probability calibration (**30**-**31**), decision-policy selection (**32**-**34**), and exposed retrospective diagnostics (**35**-**49**).

The two threshold policies are not re-optimized against the late diagnostic labels.

The first maximizes historical F1, the second targets five false positives per **10,000** known licit examples on the historical policy window.

A policy name records the historical selection objective, not an enforceable guarantee on later time steps.

Chronological window	Role in the experiment	Permitted use of labels
1-26	Model fitting	Training model parameters
27-29	Early stopping	Selecting stopping behavior
30-31	Probability calibration	Fitting the calibrator
32-34	Decision-policy window	Fixing thresholds before evaluation
35-49	Exposed retrospective diagnostics	Describing already-fixed predictions, not model selection

All **234,355** recorded graph edges in the data-quality report connect transactions belonging to the same time step.

That makes a batch, end-of-period interpretation more appropriate than an instantaneous real-time detection claim: an edge inside a given period need not have been known at the moment the earlier transaction appeared.

Similarly, the publication of a derived feature does not by itself establish the precise timestamp when it became available in a live setting.

The study can evaluate the archived retrospective problem without implying that it has independently validated point-in-time feature provenance.

2. Reading the metrics correctly

Confusion matrices report counts conditional on known labels.

True positives (TP) are known illicit transactions correctly flagged, false negatives (FN) are known illicit transactions missed.

False positives (FP) are known licit transactions incorrectly flagged, and true negatives (TN) are known licit transactions correctly rejected.

Precision = TP / (TP + FP) asks whether flagged, labeled transactions are actually illicit.

Recall = TP / (TP + FN) asks what fraction of known illicit transactions are retrieved.

F1 = 2TP / (2TP + FP + FN) balances those two objectives, while F2 = 5TP / (5TP + 4FN + FP) puts greater weight on recall.

ROC-AUC and AP do something different: they describe how scores order known-label examples over many possible thresholds.

ROC-AUC corresponds to the probability that a randomly selected known illicit example scores above a randomly selected known licit example, with an appropriate convention for ties.

Average Precision summarizes the precision-recall curve, which is often particularly informative in the presence of class imbalance.

Neither number tells us exactly how many alerts are produced at a deployed threshold.

Brier score and reliability diagrams examine yet another property: whether the probability assigned to an example is numerically consistent with the observed known-label frequency.

Top-K results describe a limited review budget.

A model first ranks all candidate transactions in a time step by predicted illicit probability, then its K highest-scoring alerts are inspected.

The experiment records which selected transactions are known illicit, known licit, or unknown.

The displayed “precision among verified alerts” divides known illicit alerts by known illicit plus known licit alerts among those selected.

It is therefore conditional on verification and must not be presented as the true precision among all K alerts, where unknown labels are abundant.

3. Cross-model results and the trade-offs they reveal

The pooled diagnostic performance table below compares eleven models under the same fixed, historically selected F1 policy.

AP and ROC-AUC are score-ranking metrics, precision, recall, F1, and the false-positive count depend on the chosen operating threshold.

As a result, a difference between columns is not a contradiction: the columns answer different questions.

Model	AP	ROC-AUC	Precision	Recall	F1	FP
LOGIT	**0.171**	**0.821**	**0.252**	**0.751**	**0.377**	**2,419**
XGB	**0.752**	**0.921**	**0.542**	**0.731**	**0.623**	**668**
LGB	**0.745**	**0.911**	**0.650**	**0.676**	**0.662**	**395**
CAT	**0.757**	**0.922**	**0.650**	**0.695**	**0.672**	**406**
MLP	**0.732**	**0.872**	**0.987**	**0.583**	**0.733**	**8**
SAGE_BATCH	**0.693**	**0.889**	**0.423**	**0.715**	**0.531**	**1,056**
SAGE_NO_GRAPH	**0.737**	**0.871**	**0.306**	**0.758**	**0.436**	**1,863**
SAGE_REWIRED	**0.361**	**0.822**	**0.393**	**0.419**	**0.406**	**702**
BIDIR_SAGE	**0.620**	**0.881**	**0.389**	**0.640**	**0.484**	**1,090**
BIDIR_NO_GRAPH	**0.731**	**0.875**	**0.315**	**0.758**	**0.445**	**1,783**
BIDIR_REWIRED	**0.479**	**0.852**	**0.537**	**0.575**	**0.556**	**537**

The three boosted-tree models form a tight cluster in score-ranking quality.

CatBoost leads narrowly in pooled AP, but the margins among CatBoost, XGBoost, and LightGBM are too small to establish a robust statistical superiority without a suitably designed uncertainty analysis.

Their F1-policy operating points differ more visibly: XGBoost has higher recall but substantially more false positives than LightGBM, whereas CatBoost sits between them in recall and very close to LightGBM in false-positive burden.

These differences are operationally meaningful even when the underlying ranking scores are close.

The MLP illustrates why operating-point analysis is indispensable.

Its near-perfect positive predictive value among labeled cases is striking, but it identifies fewer known illicit transactions than either the strongest boosted-tree models or several graph controls.

Logistic regression demonstrates the reverse problem: a comparatively high fraction of illicit transactions is retrieved, but positive alerts are so numerous that precision falls to about **25%**.

As the evaluation set is dominated by known licit transactions, even moderate false-positive rates generate large numbers of false alarms.

3.1 The historically defined false-positive-budget policy

The second policy uses a stringent historical false-positive objective.

The next table reports what happens when its frozen threshold is applied to the exposed later diagnostic data, values here are outcomes, not threshold-fitting measurements.

A false-positive rate below five per 10,000 was not maintained by every model.

In particular, logistic regression, XGBoost, LightGBM, CatBoost, and bidirectional rewired GraphSAGE exceed the target in the diagnostic window.

Model	Frozen threshold	TP	FP	Recall	FP per 10k licit
LOGIT	**0.990**	**0**	**181**	**0.000**	**116.12**
XGB	**0.763**	**694**	**101**	**0.641**	**64.80**
LGB	**0.919**	**664**	**11**	**0.613**	**7.06**
CAT	**0.728**	**692**	**75**	**0.639**	**48.12**
MLP	**0.803**	**424**	**4**	**0.392**	**2.57**
SAGE_BATCH	**0.914**	**74**	**0**	**0.068**	**0.00**
SAGE_NO_GRAPH	**0.844**	**418**	**4**	**0.386**	**2.57**
SAGE_REWIRED	**0.389**	**17**	**3**	**0.016**	**1.92**
BIDIR_SAGE	**0.927**	**142**	**1**	**0.131**	**0.64**
BIDIR_NO_GRAPH	**0.815**	**393**	**4**	**0.363**	**2.57**
BIDIR_REWIRED	**0.841**	**27**	**23**	**0.025**	**14.76**

These deviations should be interpreted as evidence about policy transfer rather than as automatic software failures.

The thresholds were chosen before the later labels were examined.

Raising a threshold until the later observations meet a desired false-positive budget would answer a different, post-hoc question and would contaminate any claim of independent evaluation.

There is also substantial sampling granularity in the relatively small historical threshold window: a target of only five false positives per **10,000** may effectively require zero false positives among the available known licit historical cases.

3.2 Top-K results

Model	Known illicit alerts	Known licit alerts	Unknown alerts	Recall@100	Verified precision
LOGIT	**19**	**320**	**1161**	**0.018**	**0.056**
XGB	**347**	**21**	**1132**	**0.320**	**0.943**
LGB	**386**	**11**	**1103**	**0.356**	**0.972**
CAT	**372**	**6**	**1122**	**0.343**	**0.984**
MLP	**191**	**5**	**1304**	**0.176**	**0.974**
SAGE_BATCH	**266**	**7**	**1227**	**0.246**	**0.974**
SAGE_NO_GRAPH	**160**	**6**	**1334**	**0.148**	**0.964**
SAGE_REWIRED	**132**	**67**	**1301**	**0.122**	**0.663**
BIDIR_SAGE	**273**	**32**	**1195**	**0.252**	**0.895**

Model	Known illicit alerts	Known licit alerts	Unknown alerts	Recall@100	Verified precision
BIDIR_NO_GRAPH	**197**	**6**	**1297**	**0.182**	**0.970**
BIDIR_REWIRED	**133**	**59**	**1308**	**0.123**	**0.693**

The K = **100** rows aggregate up to **100** highest-scoring transactions from each of **15** diagnostic time steps, for up to **1,500** reviewed alerts per model.

For example, LightGBM retrieves **386** verified illicit transactions, **11** verified licit transactions, and **1,103** unverified transactions.

The verified precision among the **397** labeled alerts is high, but the **1,103** unknown transactions prevent estimating precision across all **1,500** from those counts alone.

A credible alert-review discussion should report all three label-status counts together.

3.3 Model choice in historical development

Model	Mean development AP	Lowest fold AP	Fold SD
LGB	**0.900**	**0.732**	**0.145**
XGB	**0.896**	**0.730**	**0.145**
CAT	**0.893**	**0.724**	**0.148**
BIDIR_SAGE	**0.787**	**0.633**	**0.161**
SAGE_NO_GRAPH	**0.694**	**0.531**	**0.231**
BIDIR_NO_GRAPH	**0.659**	**0.463**	**0.251**
MLP	**0.655**	**0.467**	**0.262**
SAGE_REWIRED	**0.651**	**0.527**	**0.213**
BIDIR_REWIRED	**0.615**	**0.478**	**0.215**
SAGE_BATCH	**0.610**	**0.422**	**0.191**
LOGIT	**0.532**	**0.432**	**0.164**

LightGBM was selected according to the predetermined development process, where its average AP of approximately **0.900** slightly exceeds XGBoost and CatBoost.

The development folds have overlapping historical training windows, so the three measured APs are not three independent samples.

It would be incorrect to choose CatBoost retroactively as the “winning” model solely because it has the highest AP in the already-exposed late diagnostic interval.

Such a statement can be made as a descriptive observation, but not as a new independent model-selection conclusion.

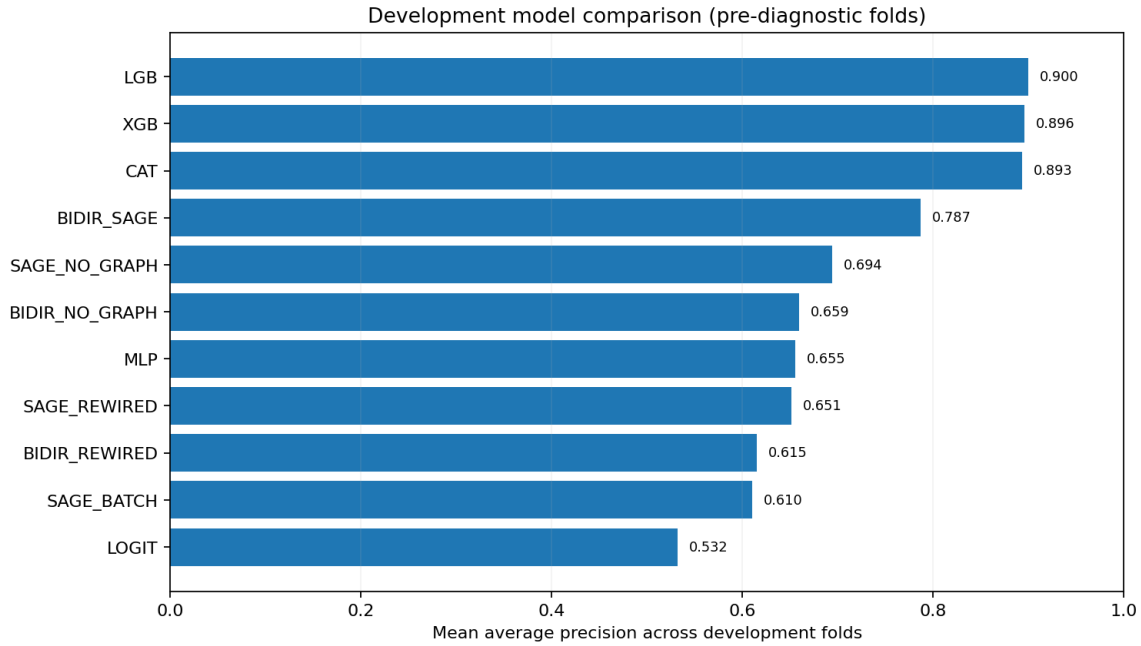
4. Reading the complete set of research figures

The 16 figures in this section provide a broad view of model selection, time-dependent degradation, graph controls, label coverage, feature drift, ranking, calibration and alert composition.

Some figures are different visualizations of the same underlying measurements, they should not be mistaken for independent replications.

The captions retain the original image files, and each interpretation explicitly identifies the evaluated population.

Figure 1. Historical development ranking



Source figure: figures/01_development_model_ranking.png

The horizontal bars compare mean Average Precision over three historical development folds.

LightGBM occupies the leading position at approximately 0.900, followed closely by XGBoost (0.896) and CatBoost (0.893).

It is worth emphasizing that the bars describe score-ranking discrimination rather than the number of alerts a threshold would generate.

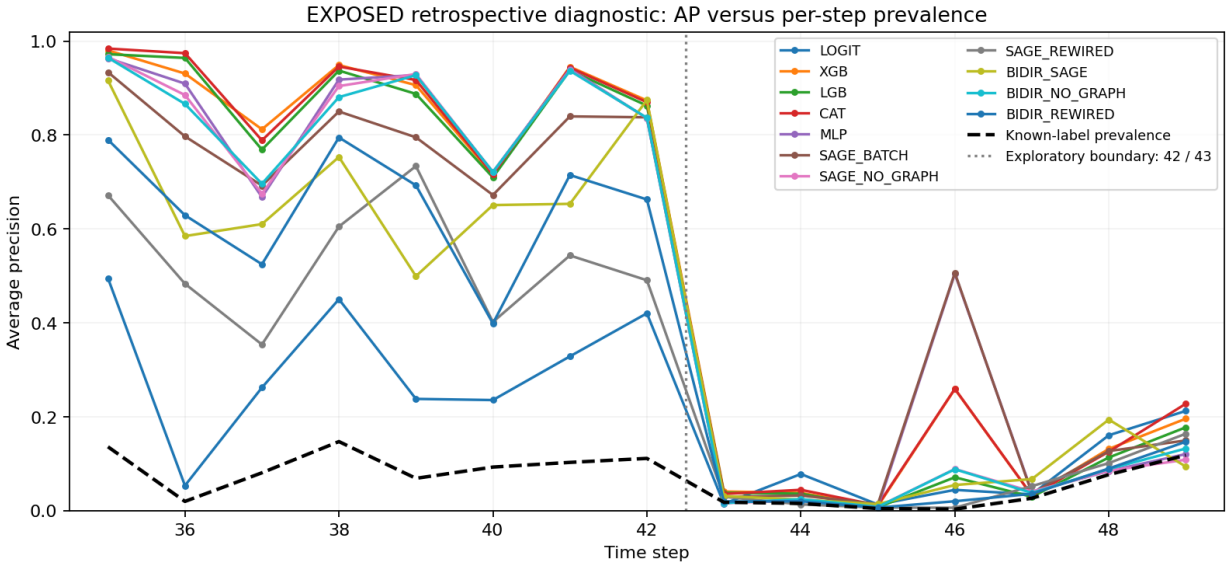
The visual ranking is best read as support for the historical model-selection procedure, not as a declaration that LightGBM dominates in every other metric.

The top three scores are close enough that the differences should be described as small and descriptive.

The development folds share some training history, and the variability displayed elsewhere in the development reports cautions against treating the average rankings as highly precise estimates of future behavior.

The proper next question is whether these models preserve their quality when the underlying transaction population changes.

Figure 2. Average Precision across diagnostic periods



Source figure: figures/02_step_average_precision.png

This temporal plot is one of the central figures in the study.

The horizontal axis lists the exposed diagnostic periods, from **35** to **49**, and the colored traces represent models' Average Precision within each individual period.

Performance is much stronger in the first group of periods and drops sharply after period **42**.

For LightGBM, the arithmetic mean of the eight early period-specific AP values is approximately **0.880**, for periods **43**-**49** it is approximately **0.067**.

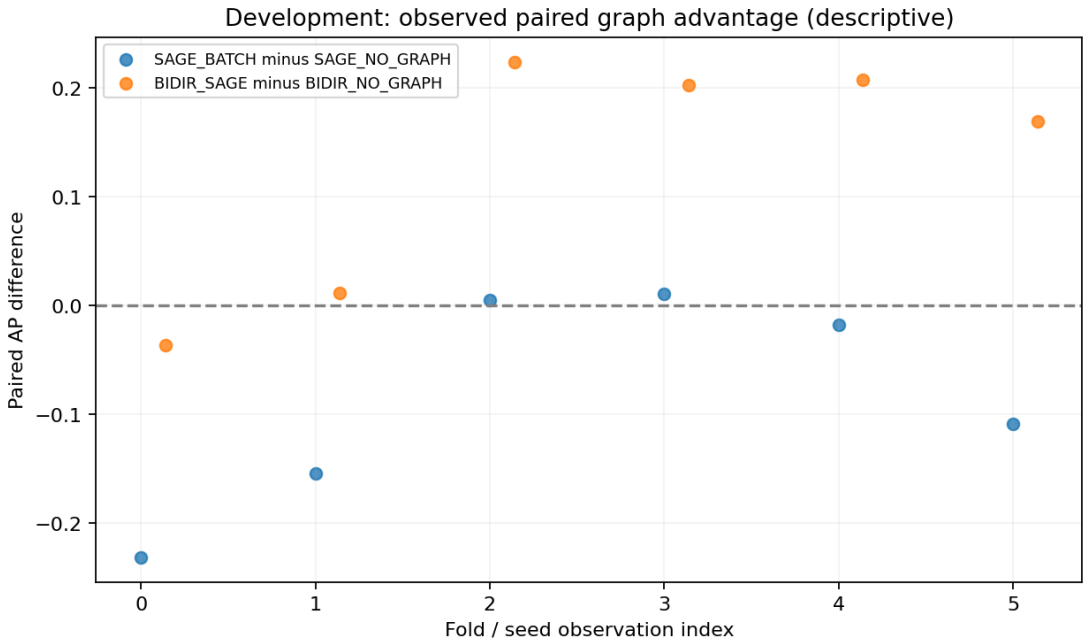
The dramatic visual break needs careful interpretation.

A period-level AP is calculated separately within that period and then displayed as a point, averaging those values is not the same as computing AP from pooled predictions over all periods.

Several later periods contain very few verified illicit cases, which makes the ranking score noisy and sensitive to individual examples.

The dashed prevalence line and the observed label counts help frame this difficulty, but the figure alone cannot determine whether feature drift, label observation, actual concept drift, or another factor caused the collapse.

Figure 3. Paired comparisons of original-graph and no-graph neural networks



Source figure: figures/03_paired_graph_seeds.png

The differences shown here compare closely matched neural-network designs rather than unrelated models.

A positive difference means that incorporating graph messages improved the selected development ranking metric relative to the matching no-graph control.

In historical development, bidirectional aggregation provides a positive descriptive advantage in several comparisons, the pooled AP difference reported for the paired bidirectional comparison is about **0.138**, with a period-bootstrap descriptive interval of approximately **0.043** to **0.236**.

Incoming-only GraphSAGE does not show the same consistent advantage, and its average paired difference is around -**0.021**.

The main interpretation is architectural rather than simply competitive.

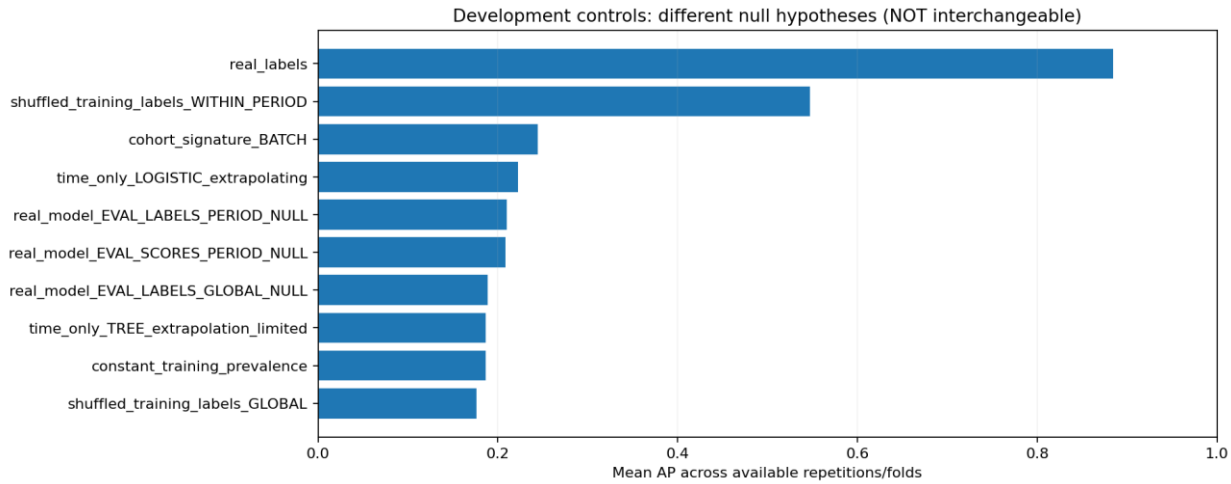
Incoming and outgoing transaction relationships may carry distinct information, and using both directions can change what the model learns.

Yet the comparisons are contingent on the selected folds, random seeds and bootstrap design.

A bootstrap interval constructed from the available historical periods should not be upgraded into a universal causal assertion.

The graph experiments are best seen as targeted controls that make a plausible structural contribution easier to investigate.

Figure 4. Negative controls and the role of temporal structure



Source figure: figures/04_negative_controls.png

This chart places the real-label development results alongside several versions of label randomization and other controls.

The real-label configurations achieve a much higher mean development AP, around **0.885** for the highlighted baseline, while global label permutation reduces AP to roughly **0.176**.

In contrast, a permutation confined within time periods can retain a noticeably higher AP, around **0.547**.

The contrast is not paradoxical: different null experiments preserve different parts of the original data structure.

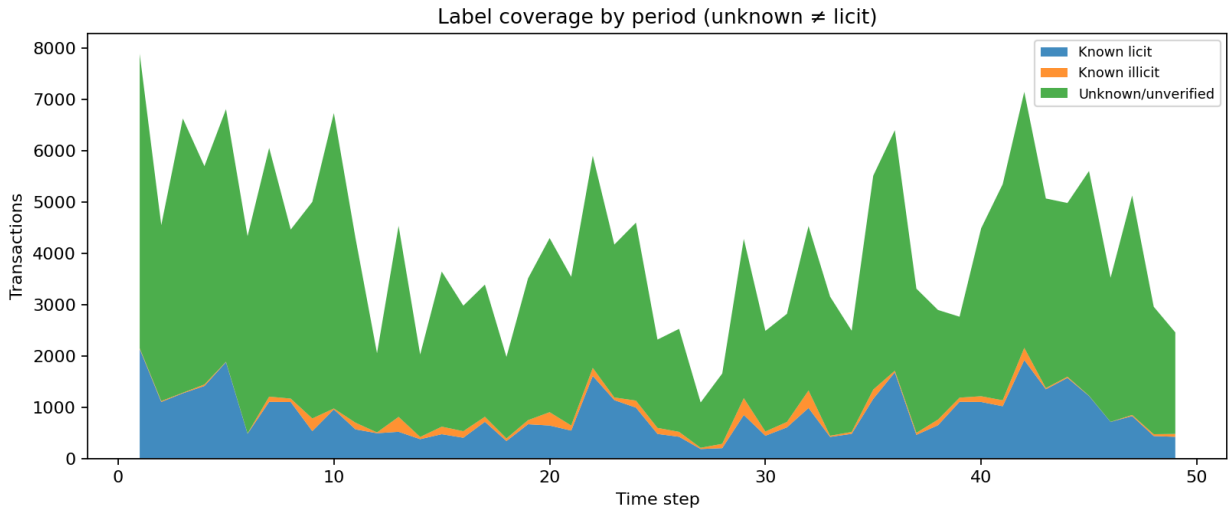
Global permutation disrupts both the direct relationship between features and labels and their relationship with historical period membership.

Within-period permutation destroys the individual transaction-label correspondence but retains period-specific label prevalence.

A model may still exploit the temporal cohort context.

Thus the plot warns against calling one particular randomized baseline “chance performance” without first defining what information it is designed to preserve.

Figure 5. Known illicit, known licit and unknown labels over time



Source figure: figures/05_label_coverage.png

The stacked areas display how many transactions fall into each observed label category across the **49** time steps.

The unknown-label region dominates much of the graph, while the two verified classes vary substantially between periods.

This is not merely a data-cleaning detail: the population used to calculate conventional supervised metrics is a subset of the full network.

A model can have strong performance among verified transactions without the evidence needed to estimate its error rate among all unverified transactions.

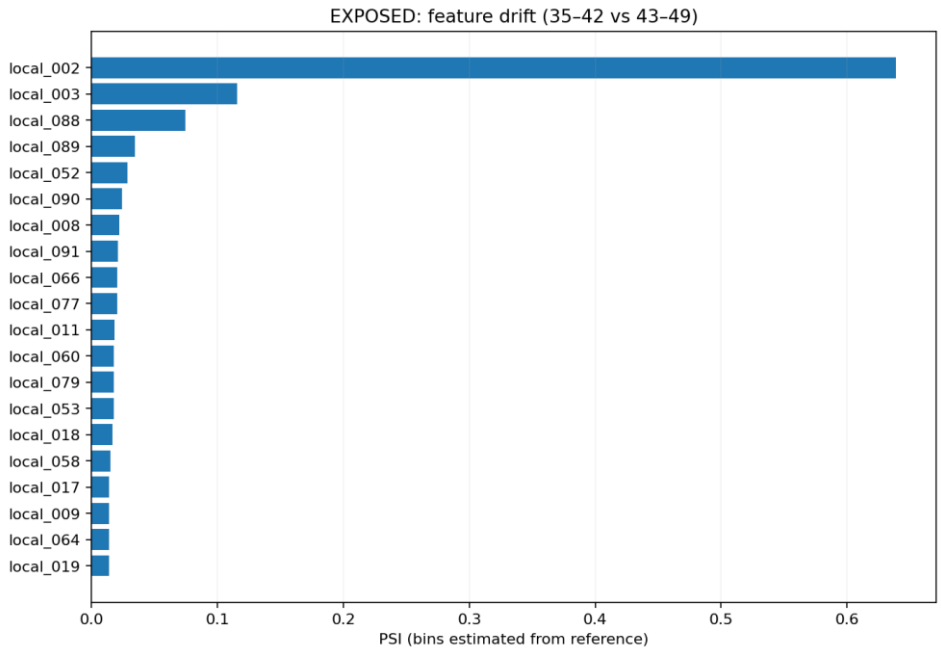
The late-period variation in the number of known illicit cases also changes the reliability of performance estimates.

When a time step has only a handful of positives, a single missed transaction has a large effect on recall and AP.

The chart should therefore accompany any figure that describes temporal performance.

Unknown labels must never be silently recoded as licit transactions.

Figure 6. Feature distribution changes measured with PSI



Source figure: figures/06_feature_drift.png

The horizontal bars quantify Population Stability Index values for selected features between the early (**35-42**) and late (**43-49**) diagnostic periods.

A larger PSI indicates a larger difference between the binned distributions of a feature in the two windows.

The feature denoted local_002 stands out with a PSI around 0.639, while some other features exhibit smaller, but still visible, shifts.

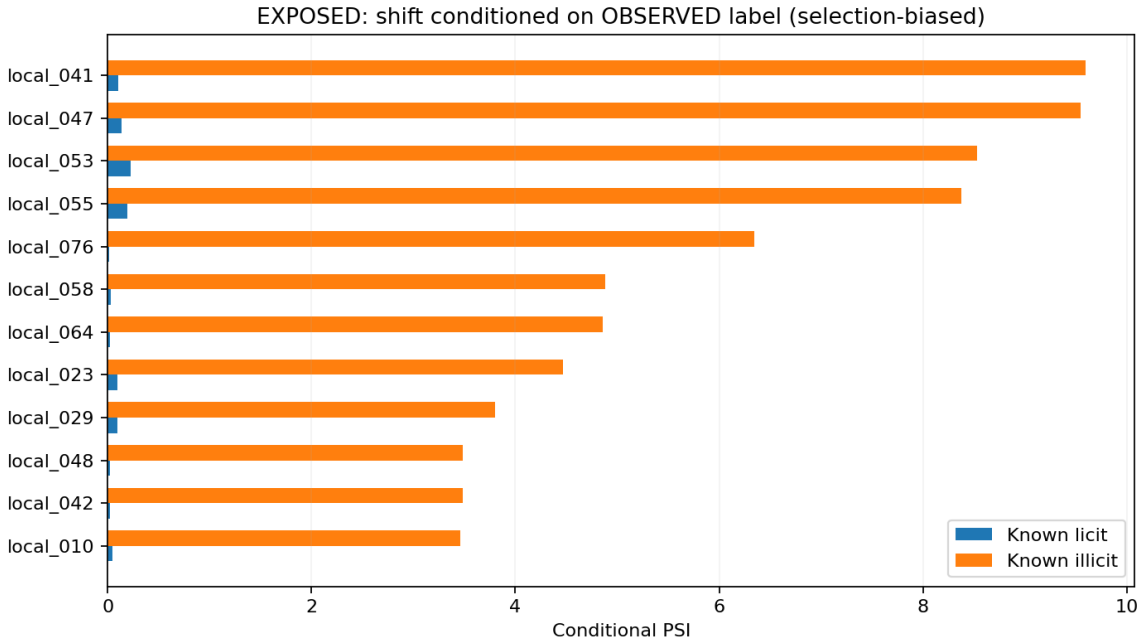
The pattern supports the descriptive claim that these windows do not contain identically distributed inputs.

It does not identify which feature, if any, explains the changes in illicit-detection performance.

A feature can drift without being particularly influential for a model, conversely, a model may be sensitive to a comparatively modest shift in a highly predictive feature.

The causal mechanism would require additional interventions or a clearly separated follow-up evaluation.

Figure 7. Feature drift within observed-label groups



Source figure: figures/07_label_conditional_drift.png

This chart stratifies feature-drift diagnostics by the observed licit and illicit categories.

Its purpose is to distinguish an overall change in the mixture of classes from changes that remain visible when one looks inside verified-label subsets.

Several features show meaningful drift even under this conditioning.

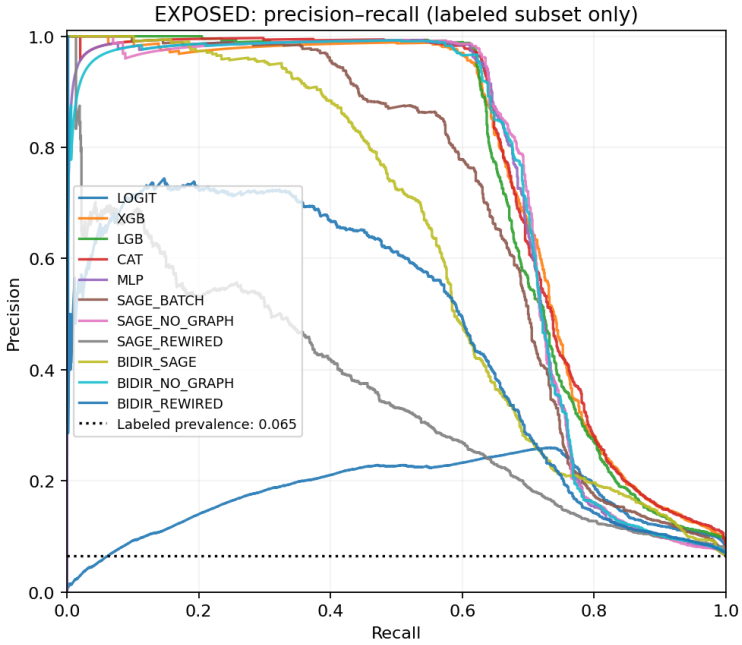
That makes it less satisfactory to explain the full pattern solely as a change in the proportion of known positives.

However, conditioning on the presence and value of an observed label introduces an important selection issue.

The verified transactions need not be representative of the network, and the process by which transactions become labeled may itself vary over time.

The figure is evidence about the recorded, labeled subsets, it does not identify population-wide concept drift or establish a cause for weaker model predictions.

Figure 8. Pooled precision-recall curves



Source figure: figures/08_precision_recall.png

All eleven models are drawn in a common Precision-Recall space for the known-label diagnostic set.

The x-axis shows recall and the y-axis shows precision as score thresholds vary.

A curve closer to the upper-right region indicates that high recall can be maintained with relatively little contamination among flagged known-label examples.

CatBoost has the highest pooled AP at approximately **0.757**, with XGBoost (**0.752**) and LightGBM (**0.745**) close behind.

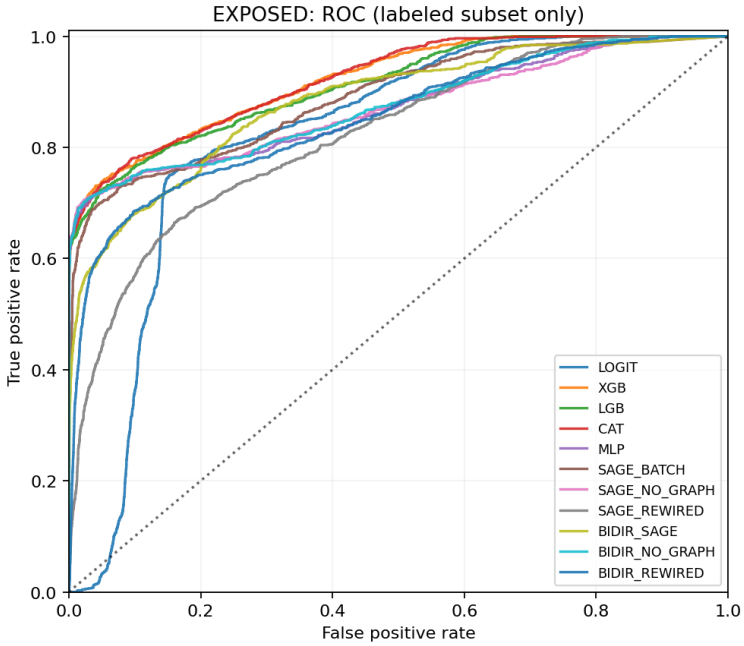
At roughly **6.5%** known-label illicit prevalence in the diagnostic window, the baseline level of precision for a random ranking is low.

This makes PR curves especially informative.

Yet an AP ranking cannot tell investigators how many false alerts they would encounter under a particular staffing constraint, and it does not replace the historical decision-policy analysis.

The curves also represent verified examples only, rather than the label-unknown majority of the transaction network.

Figure 9. Pooled ROC curves and discrimination



Source figure: figures/09_roc.png

The ROC plot compares sensitivity against false-positive rate as the decision threshold moves.

CatBoost attains an ROC-AUC of about **0.922**, XGBoost is close at about **0.921**, and LightGBM follows at about **0.911**.

Even logistic regression has an ROC-AUC above **0.82** despite the very high number of false positives generated by its historically selected F1 threshold.

This apparent discrepancy is a useful lesson rather than a contradiction.

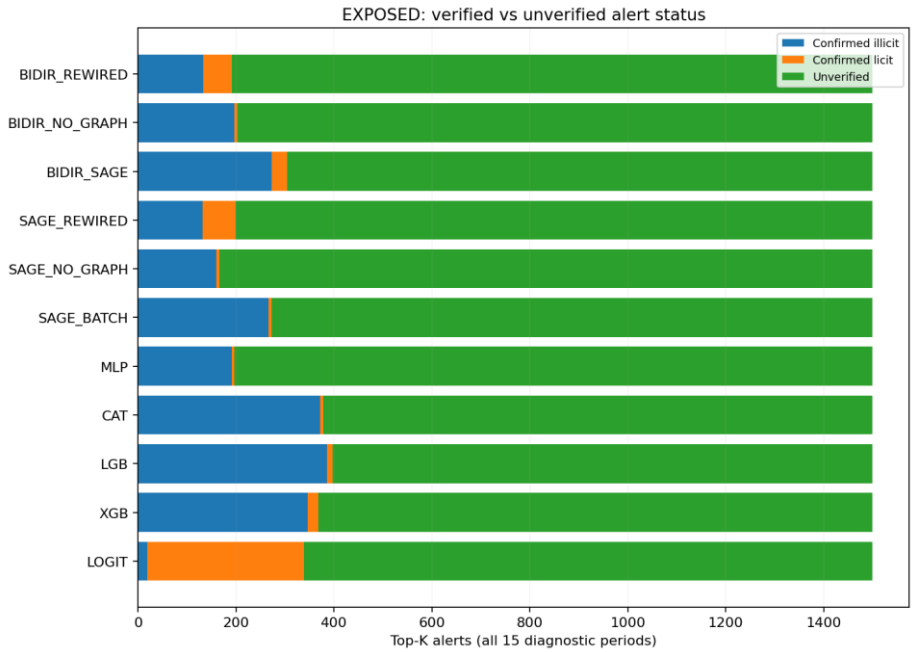
ROC-AUC evaluates ordering across all possible thresholds.

The confusion matrix reports what happens at one concrete threshold.

In an imbalanced dataset, an apparently modest false-positive rate can correspond to hundreds or thousands of licit transactions being flagged.

For an alert-review application, this ROC figure is informative but should be read together with the precision-recall, confusion-matrix, and top-ranked-alert plots.

Figure 10. Verification status of the top-ranked alerts



Source figure: figures/10_topk_alert_status.png

Each model contributes the highest-scoring **100** transactions in each of **15** late diagnostic periods, producing up to **1,500** alerts per model.

The stacked colored bars divide these alerts into verified illicit, verified licit, and label-unknown transactions.

For example, LightGBM's **1,500** selected alerts include **386** verified illicit, **11** verified licit, and **1,103** unknown transactions.

The large unknown segment is the central result of this figure.

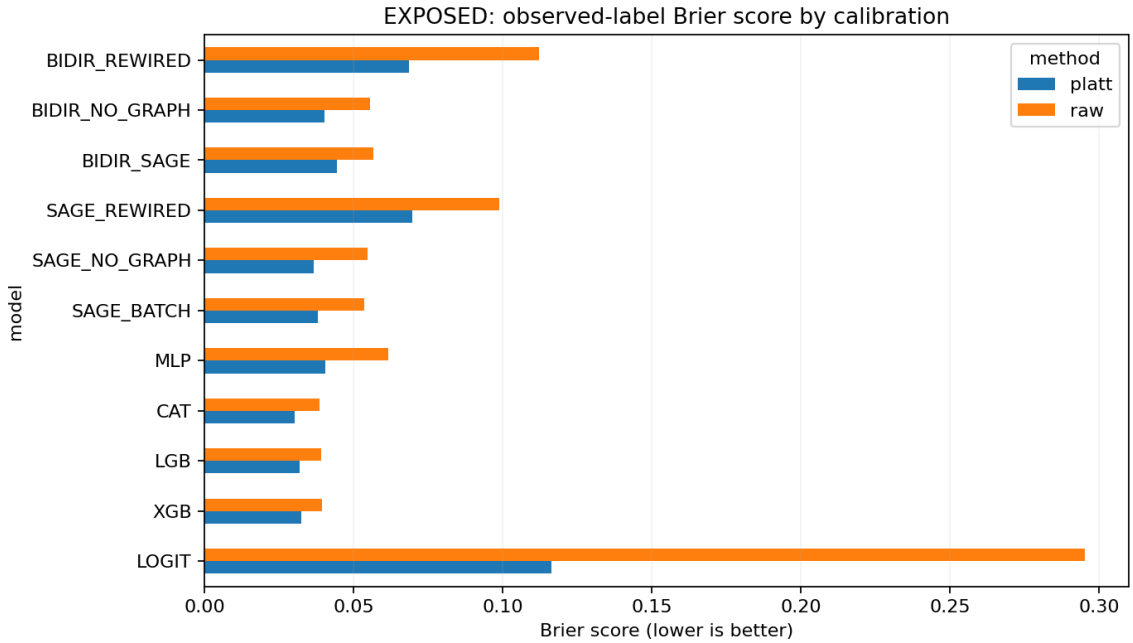
The small number of verified licit alerts should not be used to proclaim near-perfect real-world alert precision.

The majority of selected transactions have no definitive observed outcome.

Treating unknown as wrong would underestimate performance, while treating unknown as illicit would grossly overstate it.

The right interpretation is conditional: among currently verified top-ranked alerts the observed ratio can be high, but the actual precision over all reviewed transactions is not identified by this dataset.

Figure 11. Calibration changes measured by Brier score



Source figure: figures/11_calibration_brier.png

The paired bars compare mean squared probability error before and after the separately fitted calibration transformation.

Lower is better.

For logistic regression, the exposed known-label diagnostic Brier score decreases from roughly **0.295** before calibration to **0.116** after calibration.

XGBoost decreases from approximately 0.0393 to 0.0325, and CatBoost from approximately 0.0385 to 0.0303.

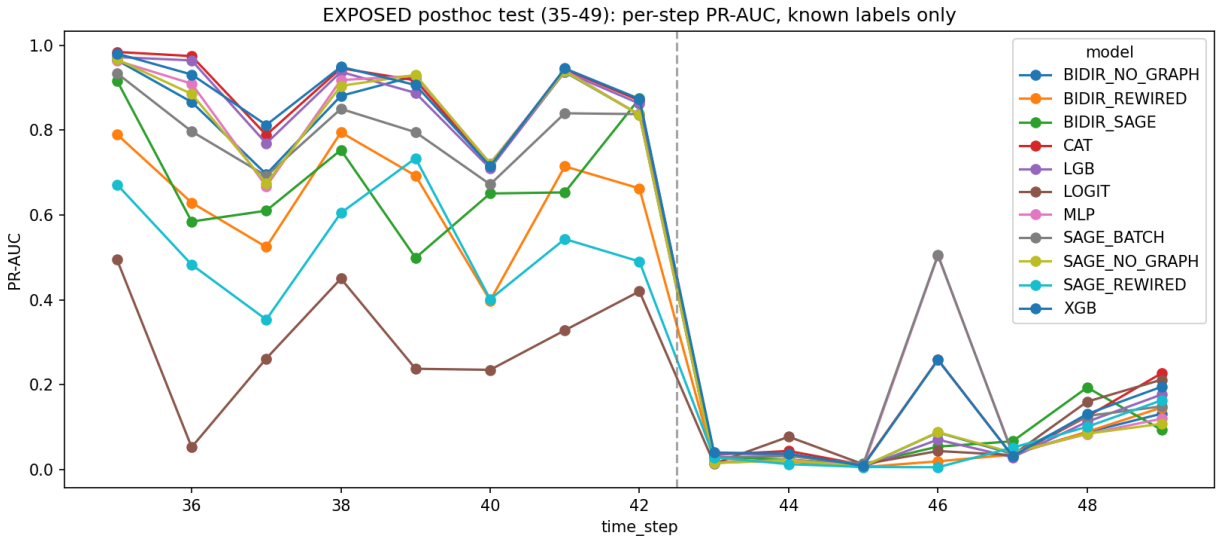
These are visible changes in the numerical quality of estimated probabilities, not necessarily in the ranking of scores.

A strictly increasing calibration function preserves rank-based ROC-AUC and AP when it introduces no new score ties, even as it changes the numerical probabilities.

The improvement is evaluated among known-label transactions and over an exposed diagnostic period.

It cannot guarantee that the same probability estimates will be well calibrated for the unknown-label majority, for a different future time interval, or under a changing class prevalence.

Figure 12. Second view of Average Precision by diagnostic time step



Source figure: diagnostic_ap_by_step.png

This alternate rendering reproduces the broad period-by-period AP picture presented earlier.

Several models have strong discrimination in the early diagnostic periods represented by steps **35-42**, followed by a distinct and largely sustained decline later.

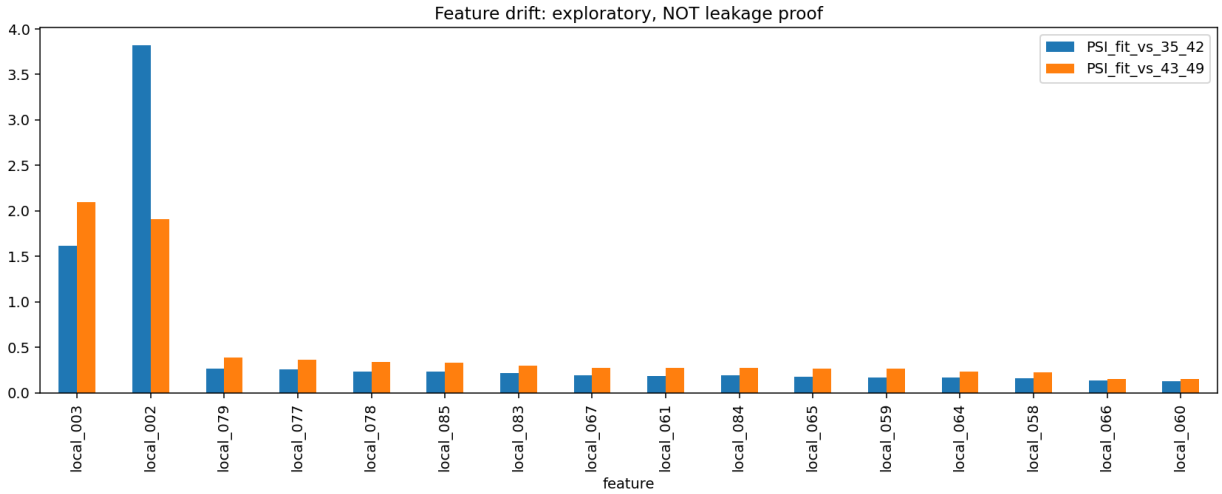
The alternate visual presentation makes individual model trajectories somewhat easier to trace, but does not add an independent experiment or a second source of labels.

When reading very low late-period AP values, recall that a substantial change in observed positive prevalence can affect the level of AP itself.

The right analysis pairs this figure with the label coverage, count-of-positive-cases and input-drift figures.

It is scientifically preferable to describe the transition as an observed degradation rather than assigning a confident causal explanation.

Figure 13. Alternative drift comparison against the training distribution



Source figure: feature_drift.png

Unlike the earlier early-versus-late diagnostic PSI chart, this visualization compares selected diagnostic windows with the historical training feature distribution.

The notable shifts in features such as local_003 and local_002 indicate that a single historical reference can become less representative as the experiment moves forward in time.

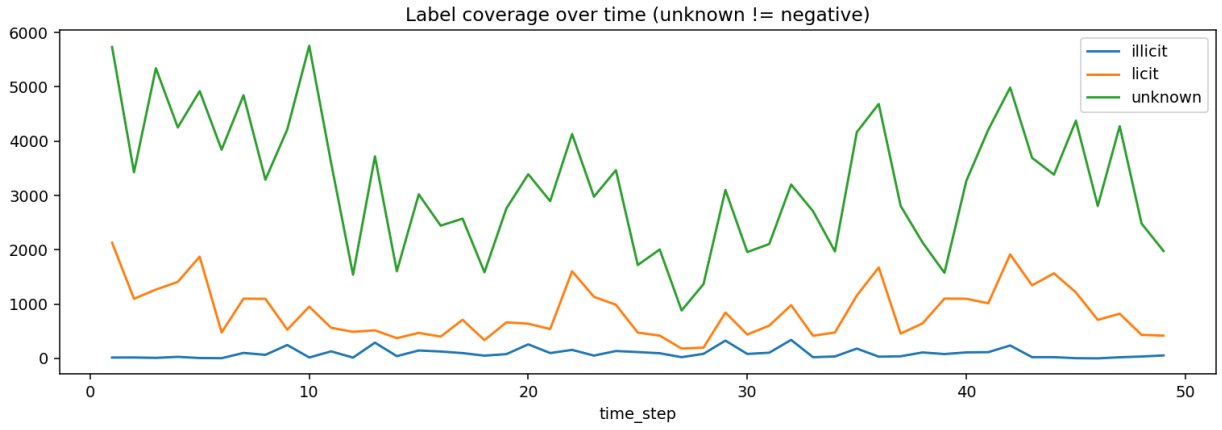
Both training-relative and diagnostic-window-relative comparisons are useful, because they answer slightly different questions.

PSI summarizes histogram differences and is sensitive to bin definitions and the selection of comparison periods.

It is a descriptive alert that a feature has changed, not a proof of label leakage or model failure.

A high drift score is most informative when linked to the model's sensitivity to that feature and to the chronology of the change.

Figure 14. Alternative line depiction of label availability



Source figure: label_coverage.png

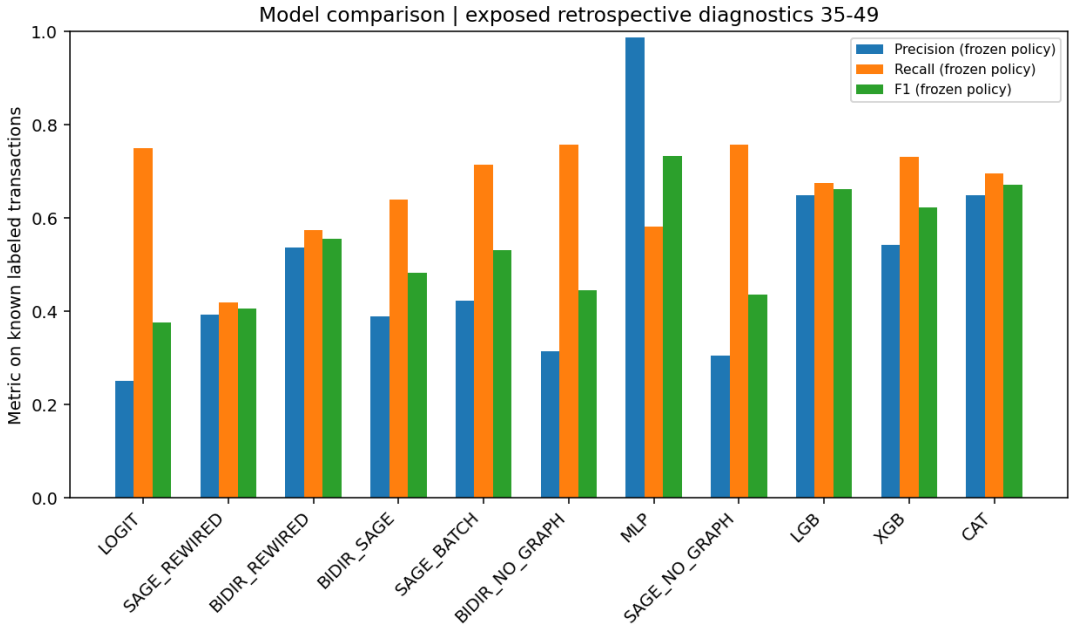
This line figure shows the same label-status counts as the earlier coverage plot, but the separate trajectories make the shifts in the numbers of verified illicit and licit cases easier to follow.

The relatively sparse verified positives in some later periods are particularly important for understanding why period-specific AP and recall may fluctuate so sharply.

Because this is another depiction of the same source data, it is complementary rather than independent evidence.

It reinforces a central boundary of the study: evaluation is conditional on which transactions received a verified label, and coverage varies over time.

Figure 15. Model comparison under the fixed F1 policy



Source figure: FULL_METRICS/plots/COMPARE_RECALL_PRECISION_F1.png

The grouped bars compare precision, recall and F1 at each model's historically selected F1 threshold.

The plot exposes the practical choices hidden by an AP league table.

MLP has exceptionally high precision of approximately **0.987** but a recall of about **0.583**.

Logistic regression has a recall around **0.751** but precision near **0.252**.

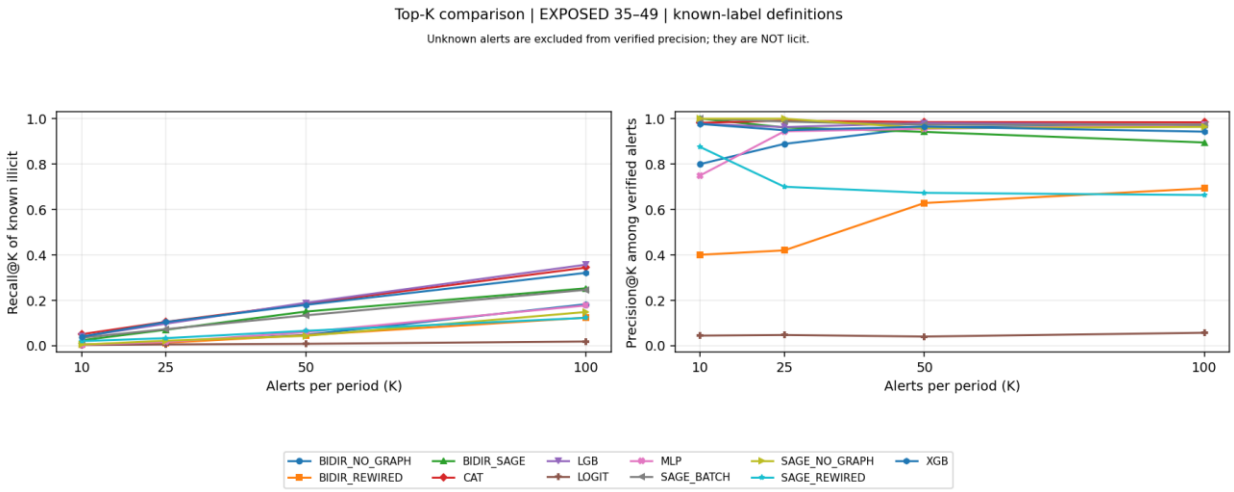
CatBoost and LightGBM offer more balanced operating points within the boosted-tree group.

The visually strongest F1 bar need not belong to the best ranking model.

That is expected, because F1 describes one specific threshold, while AP summarizes behavior over an entire range.

For a research discussion this comparison helps avoid calling a model "best" without identifying the loss function or review objective that gives the claim meaning.

Figure 16. Precision@K and Recall@K across all models



Source figure: FULL_METRICS/plots/COMPARE_PRECISION_RECALL_AT_K.png

This multi-model figure traces retrieval quality under several review budgets: K = **10**, **25**, **50** and **100** per period.

It asks what happens when a user looks at only the highest-scoring transactions rather than making a binary decision about every candidate.

At K = **100**, LightGBM retrieves **386** verified illicit alerts across all periods, corresponding to micro Recall@**100** around **0.356**, CatBoost retrieves **372**, for approximately **0.343**.

One must distinguish two denominators.

Recall@K uses the total number of known illicit transactions in the selected periods, but the reported precision uses only those alerts with known illicit or licit labels.

Because unknown alerts dominate, the precision curve is not directly comparable to precision in a fully adjudicated operational alert stream.

The plot is valuable for comparing ranking strategies under a fixed budget, provided this verification conditioning is stated clearly.

5. Model-by-model interpretation of the remaining 66 figures

Each of the eleven configurations has six dedicated figures in the sections that follow: a confusion matrix at its historically selected F1 operating point, a confusion matrix at the false-positive-budget operating point, a paired precision-recall/ROC figure, a time-resolved recall and precision figure, a post-hoc threshold sensitivity figure, and a probability reliability diagram.

Instead of repeating the mathematical definition without interpretation, the accompanying text explains the specific trade-off observed for that model and connects the two threshold policies to its broader ranking and calibration behavior.

5.1 Logistic Regression (LOGIT)

Logistic Regression records pooled diagnostic AP of **0.171** and ROC-AUC of **0.821**.

At the historical F1 threshold (**0.440**), it flags **3,232** verified transactions, of which **813** are correctly identified as illicit.

Its recall is **75.1%**, precision **25.2%**, and F1 **0.377**.

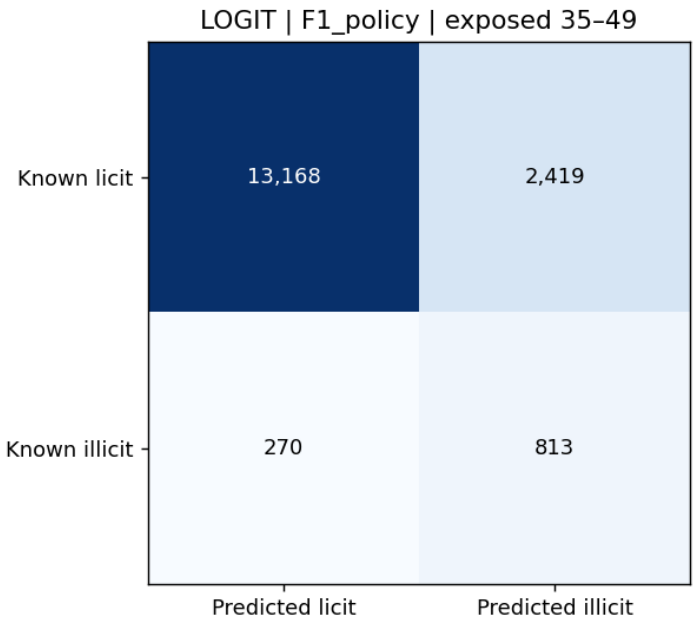
The separate historical false-positive-budget threshold is **0.990**.

As a linear baseline, this model helps show how much nonlinear and graph-based representations add.

It retrieves a relatively large proportion of the verified illicit class, but it also generates the largest false-alarm burden under the F1 policy.

The fact that its ROC-AUC remains meaningfully above chance is a reminder that ranking discrimination and a deployable binary operating point are not the same property.

Figure 17. LOGIT: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_LOGIT_F1_policy.png

The matrix records **13,168** true negatives, **2,419** false positives, **270** false negatives and **813** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

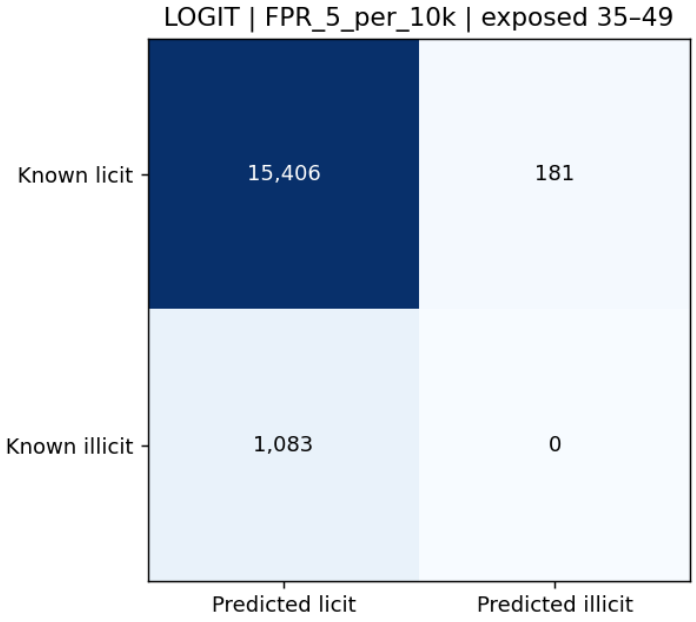
The model detects **813** of the **1,083** known illicit transactions, producing recall of **75.1%**.

Among its **3,232** positive alerts with verified labels, **25.2%** are illicit.

These counts explain its F1 of **0.377** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 18. LOGIT: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_LOGIT_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.990**) yields **0** true positives, **1,083** false negatives, **181** false positives and **15,406** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **2,419** to **181**, while detected illicit transactions change from **813** to **0**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

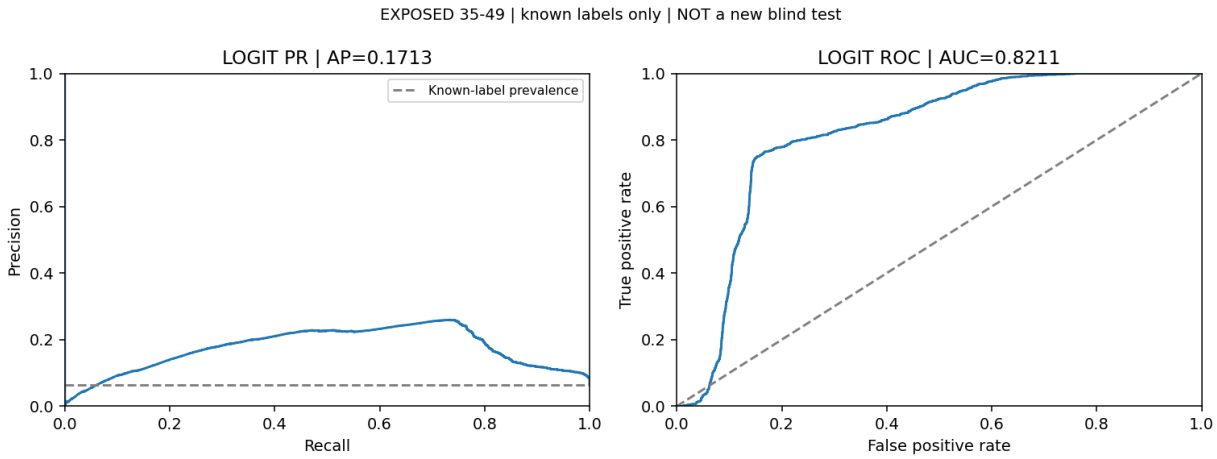
The observed budget-policy false-positive rate equals **116.12** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **0.0%**.

The diagnostic value exceeds the historical target, showing that a threshold selected for a historical false-positive objective need not meet that objective in later periods.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 19. LOGIT: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_LOGIT.png

The paired panels show how the LOGIT scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.171** and ROC-AUC = **0.821**.

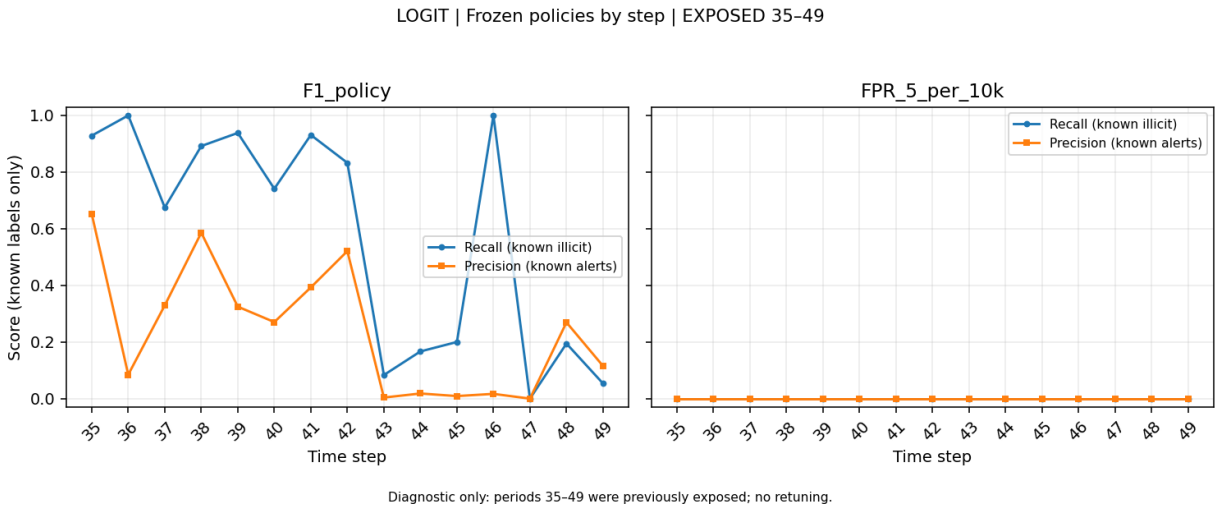
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 20. LOGIT: recall and precision over diagnostic time



Source figure: FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_LOGIT.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For LOGIT, the unweighted average of period-specific AP falls from **0.310** over periods **35-42** to **0.079** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **1** of the seven late periods.

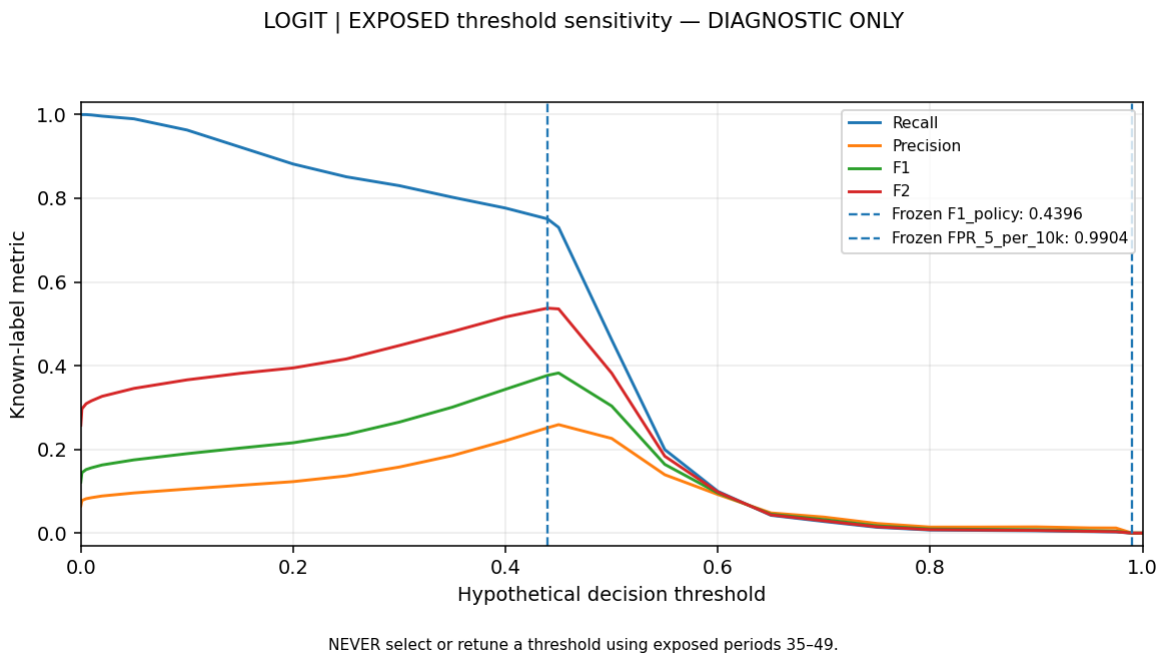
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 21. LOGIT: post-hoc threshold sensitivity (diagnostic only)



Source figure:
FULL_METRICS/plots/THRESHOLD_SENSITIVITY_LOGIT_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.440**) and false-positive-budget threshold (**0.990**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.450** with observed F1 of **0.383**, but this number is a post-hoc descriptive result.

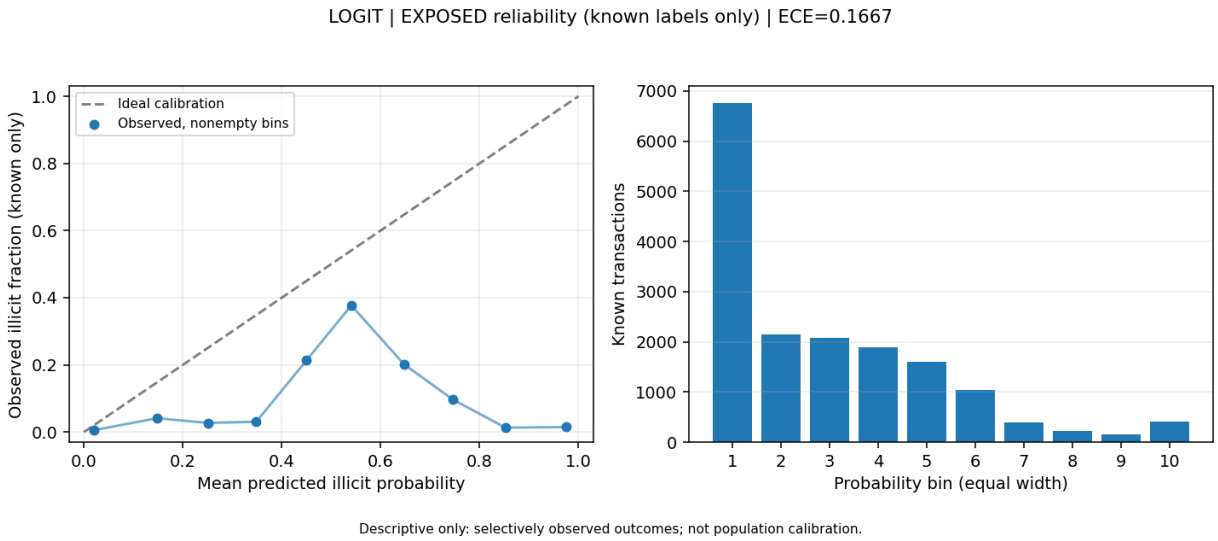
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 22. LOGIT: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_LOGIT.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.167 for LOGIT.

Its diagnostic Brier score is **0.2953** with uncalibrated probabilities and **0.1162** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.2 XGBoost (XGB)

XGBoost records pooled diagnostic AP of **0.752** and ROC-AUC of **0.921**.

At the historical F1 threshold (**0.297**), it flags **1,460** verified transactions, of which **792** are correctly identified as illicit.

Its recall is **73.1%**, precision **54.2%**, and F1 **0.623**.

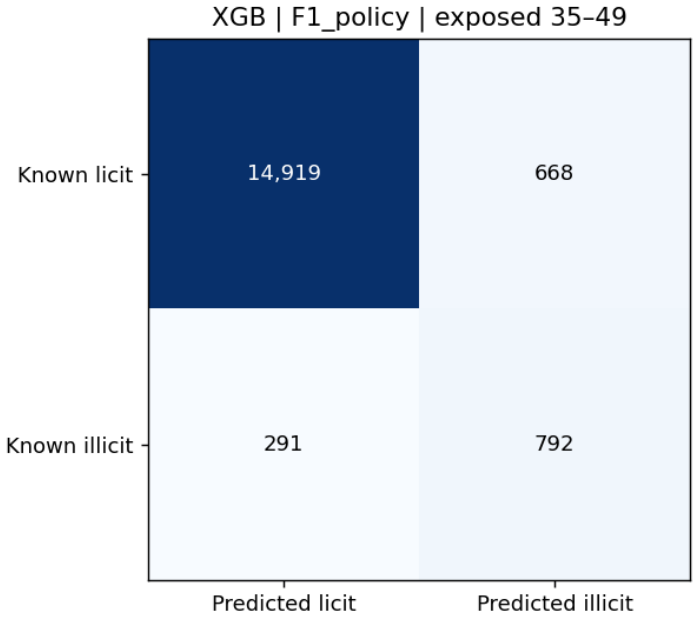
The separate historical false-positive-budget threshold is **0.763**.

This gradient-boosted decision-tree model belongs to the strongest cluster of feature-based ranking systems in the diagnostic window.

Its relative position changes with the metric being emphasized: score ranking, thresholded recall, false-positive workload and calibration each highlight a different aspect of performance.

Its results are especially useful when compared with the other two boosted-tree implementations rather than interpreted in isolation.

Figure 23. XGB: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_XGB_F1_policy.png

The matrix records **14,919** true negatives, **668** false positives, **291** false negatives and **792** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

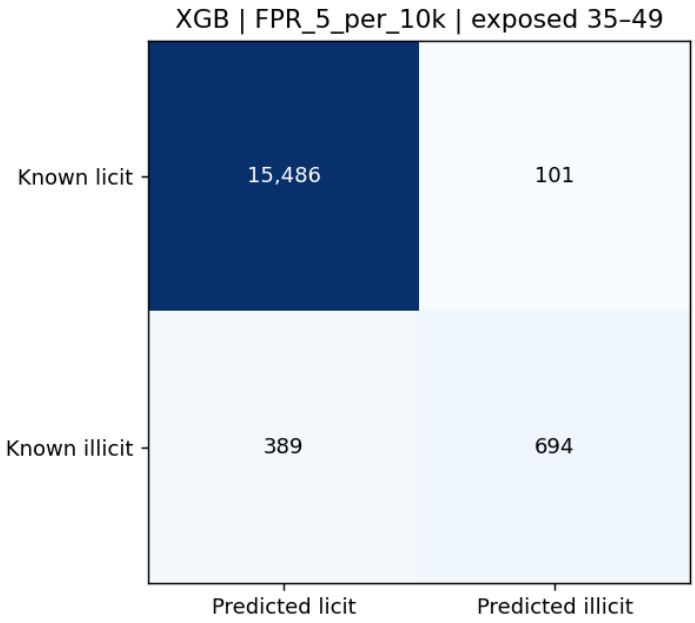
The model detects **792** of the **1,083** known illicit transactions, producing recall of **73.1%**.

Among its **1,460** positive alerts with verified labels, **54.2%** are illicit.

These counts explain its F1 of **0.623** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 24. XGB: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_XGB_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.763**) yields **694** true positives, **389** false negatives, **101** false positives and **15,486** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **668** to **101**, while detected illicit transactions change from **792** to **694**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

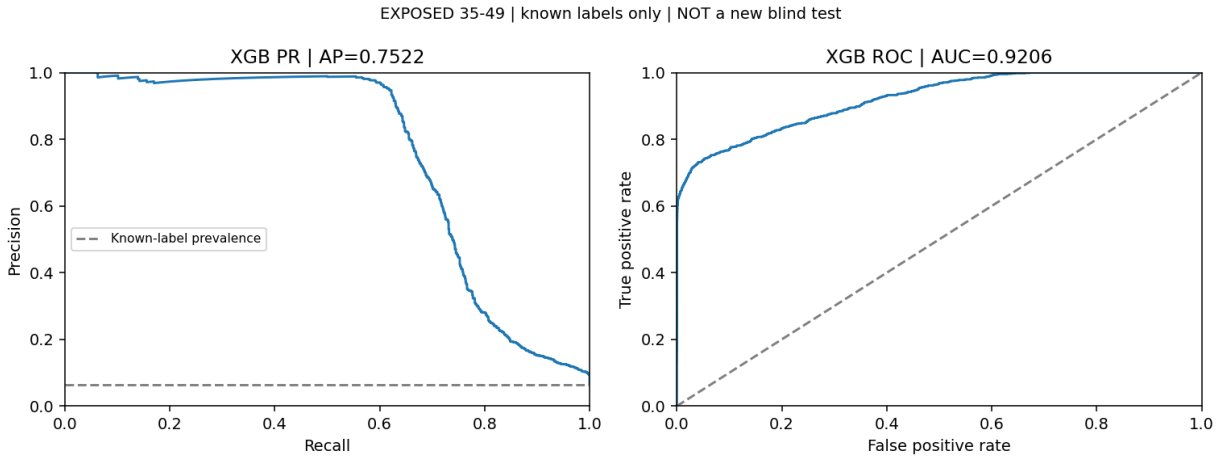
The observed budget-policy false-positive rate equals **64.80** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **64.1%**.

The diagnostic value exceeds the historical target, showing that a threshold selected for a historical false-positive objective need not meet that objective in later periods.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 25. XGB: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_XGB.png

The paired panels show how the XGB scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.752** and ROC-AUC = **0.921**.

The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 26. XGB: recall and precision over diagnostic time



Source figure: FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_XGB.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For XGB, the unweighted average of period-specific AP falls from **0.889** over periods **35-42** to **0.101** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **3** of the seven late periods.

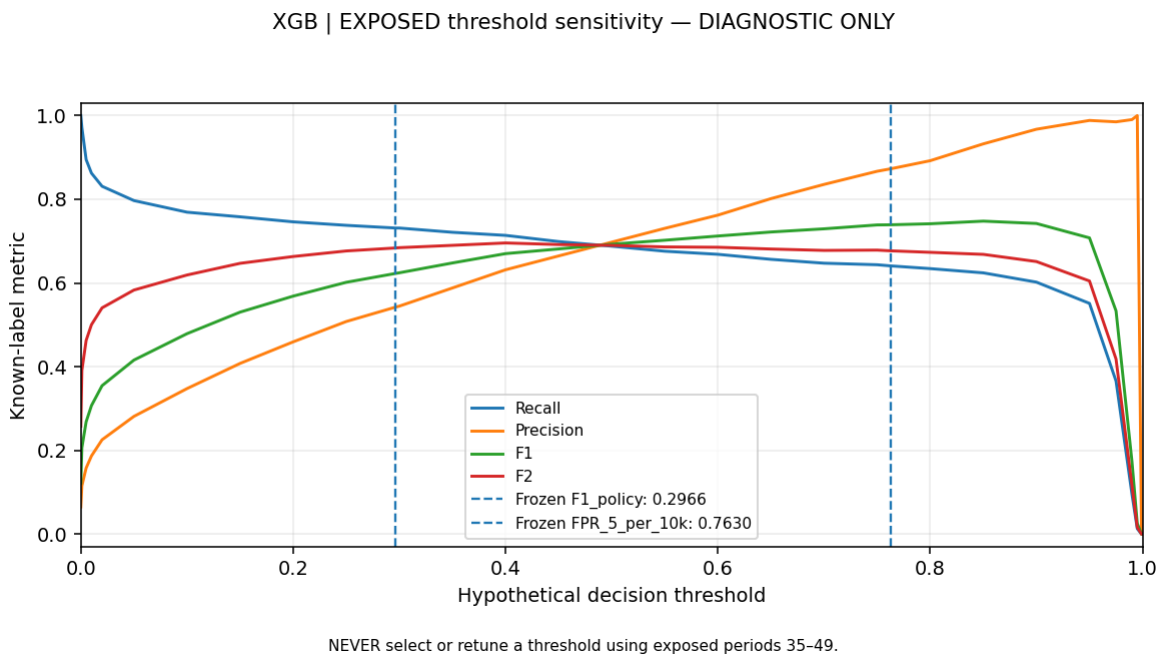
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 27. XGB: post-hoc threshold sensitivity (diagnostic only)



Source figure:
FULL_METRICS/plots/THRESHOLD_SENSITIVITY_XGB_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.297**) and false-positive-budget threshold (**0.763**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.850** with observed F1 of **0.748**, but this number is a post-hoc descriptive result.

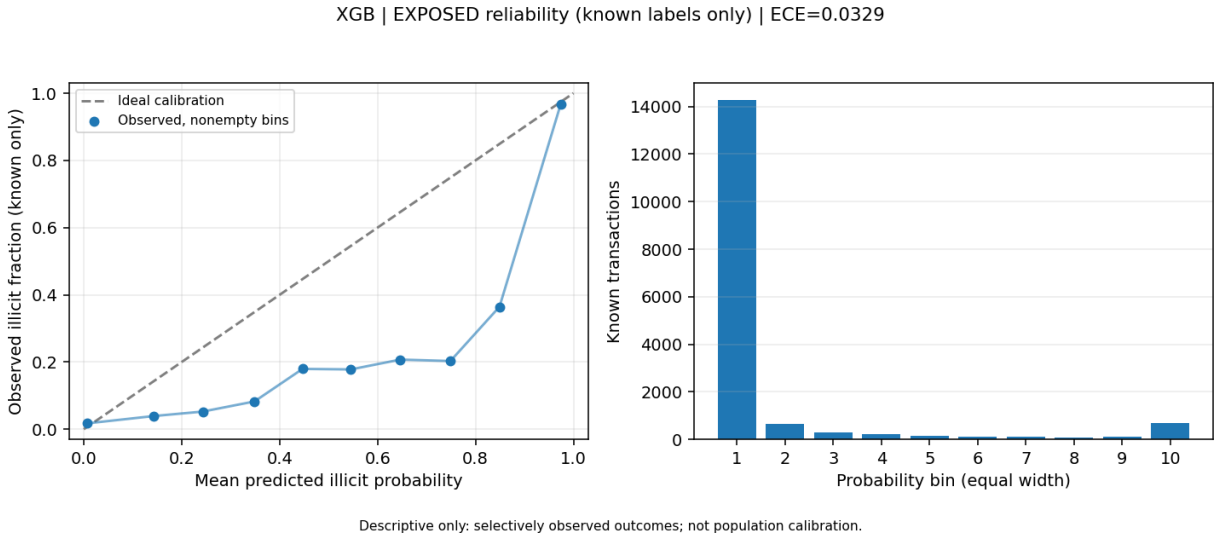
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 28. XGB: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_XGB.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.033 for XGB.

Its diagnostic Brier score is **0.0393** with uncalibrated probabilities and **0.0325** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.3 LightGBM (LGB)

LightGBM records pooled diagnostic AP of **0.745** and ROC-AUC of **0.911**.

At the historical F1 threshold (**0.399**), it flags **1,127** verified transactions, of which **732** are correctly identified as illicit.

Its recall is **67.6%**, precision **65.0%**, and F1 **0.662**.

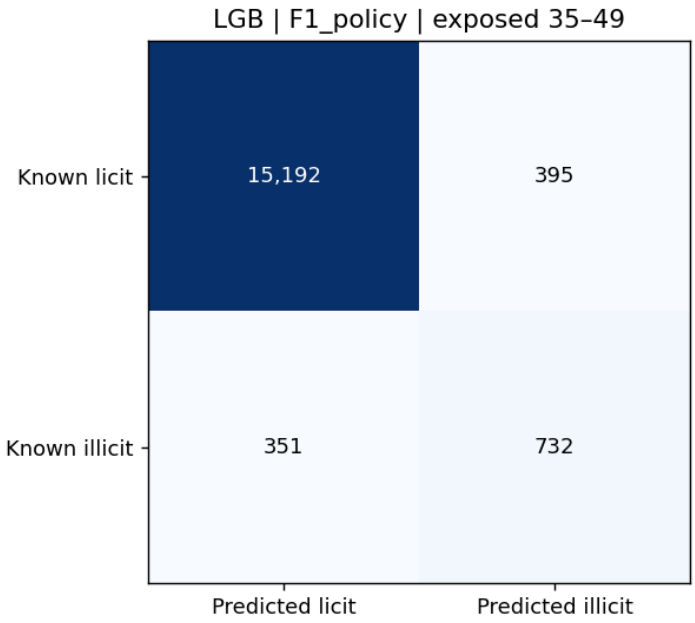
The separate historical false-positive-budget threshold is **0.919**.

This gradient-boosted decision-tree model belongs to the strongest cluster of feature-based ranking systems in the diagnostic window.

Its relative position changes with the metric being emphasized: score ranking, thresholded recall, false-positive workload and calibration each highlight a different aspect of performance.

Its results are especially useful when compared with the other two boosted-tree implementations rather than interpreted in isolation.

Figure 29. LGB: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_LGB_F1_policy.png

The matrix records **15,192** true negatives, **395** false positives, **351** false negatives and **732** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

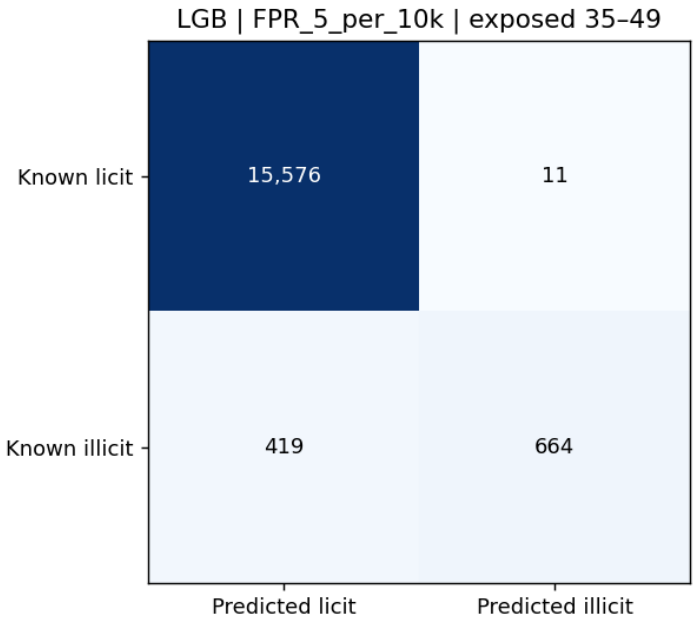
The model detects **732** of the **1,083** known illicit transactions, producing recall of **67.6%**.

Among its **1,127** positive alerts with verified labels, **65.0%** are illicit.

These counts explain its F1 of **0.662** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 30. LGB: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_LGB_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.919**) yields **664** true positives, **419** false negatives, **11** false positives and **15,576** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **395** to **11**, while detected illicit transactions change from **732** to **664**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

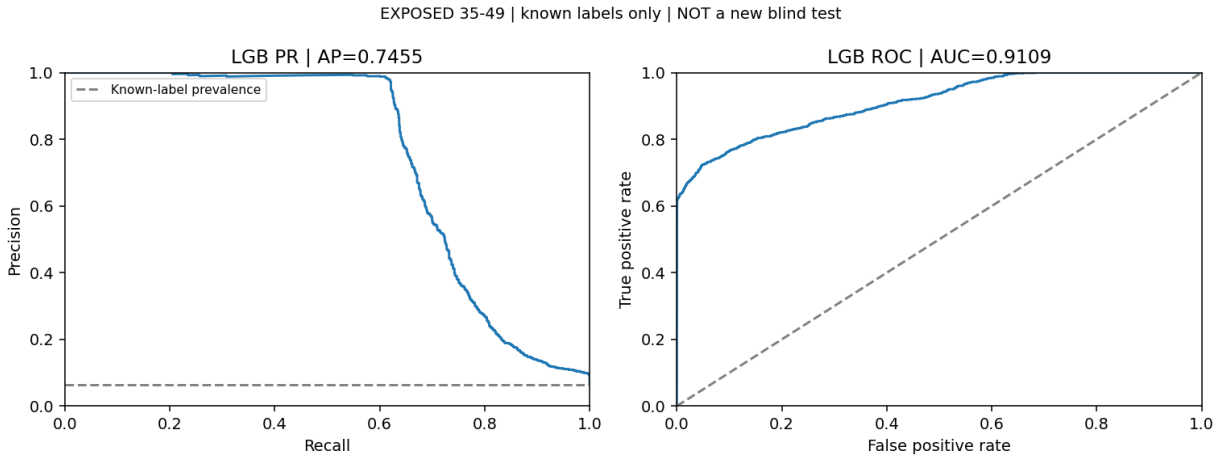
The observed budget-policy false-positive rate equals **7.06** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **61.3%**.

The diagnostic value exceeds the historical target, showing that a threshold selected for a historical false-positive objective need not meet that objective in later periods.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 31. LGB: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_LGB.png

The paired panels show how the LGB scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.745** and ROC-AUC = **0.911**.

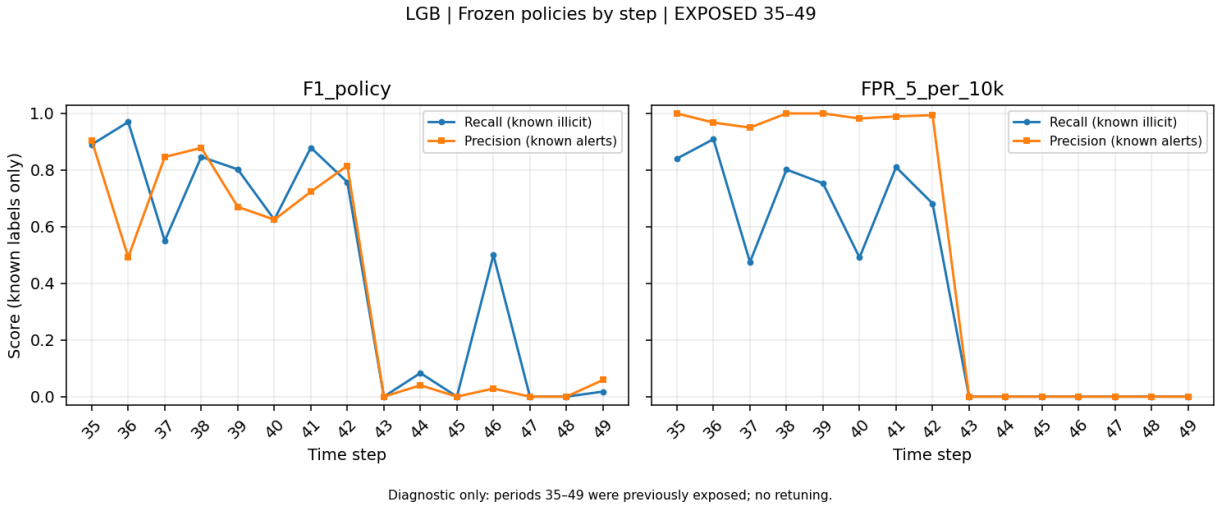
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 32. LGB: recall and precision over diagnostic time



Source figure: FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_LGB.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For LGB, the unweighted average of period-specific AP falls from **0.880** over periods **35-42** to **0.067** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **4** of the seven late periods.

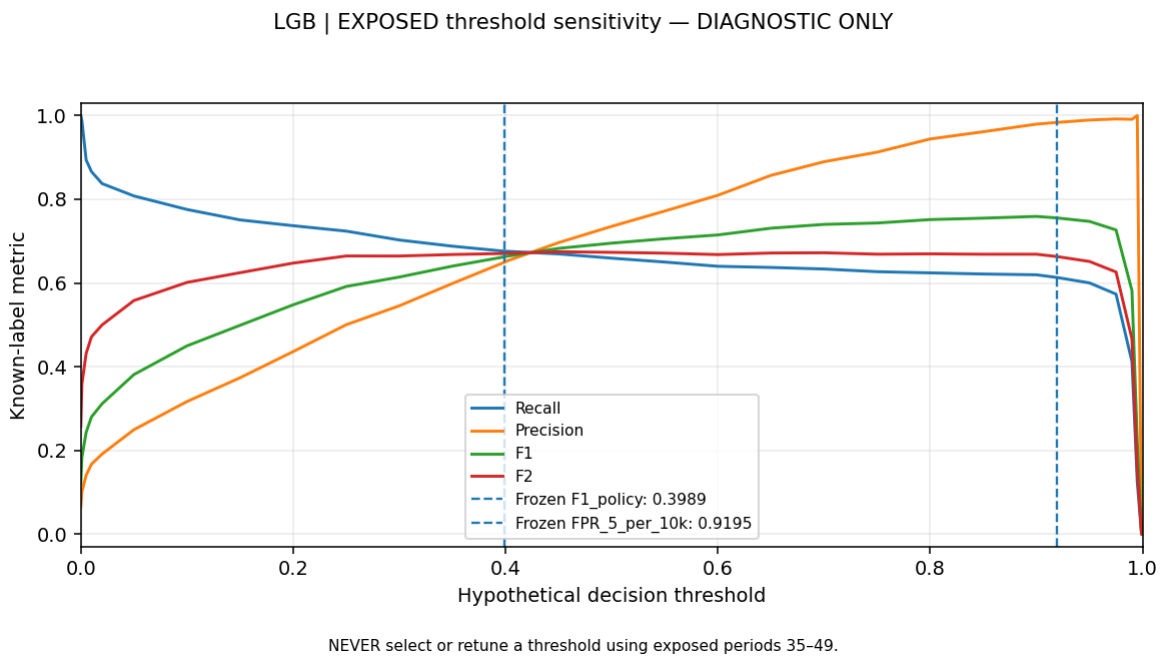
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 33. LGB: post-hoc threshold sensitivity (diagnostic only)



Source figure:
FULL_METRICS/plots/THRESHOLD_SENSITIVITY_LGB_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.399**) and false-positive-budget threshold (**0.919**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.900** with observed F1 of **0.759**, but this number is a post-hoc descriptive result.

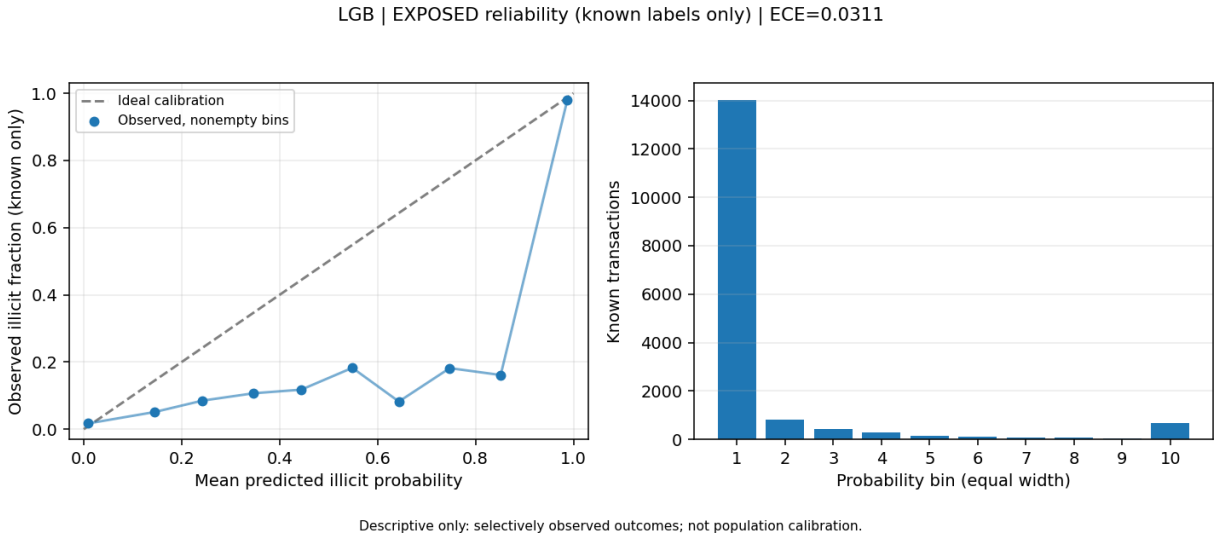
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 34. LGB: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_LGB.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.031 for LGB.

Its diagnostic Brier score is **0.0392** with uncalibrated probabilities and **0.0318** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.4 CatBoost (CAT)

CatBoost records pooled diagnostic AP of **0.757** and ROC-AUC of **0.922**.

At the historical F1 threshold (**0.380**), it flags **1,159** verified transactions, of which **753** are correctly identified as illicit.

Its recall is **69.5%**, precision **65.0%**, and F1 **0.672**.

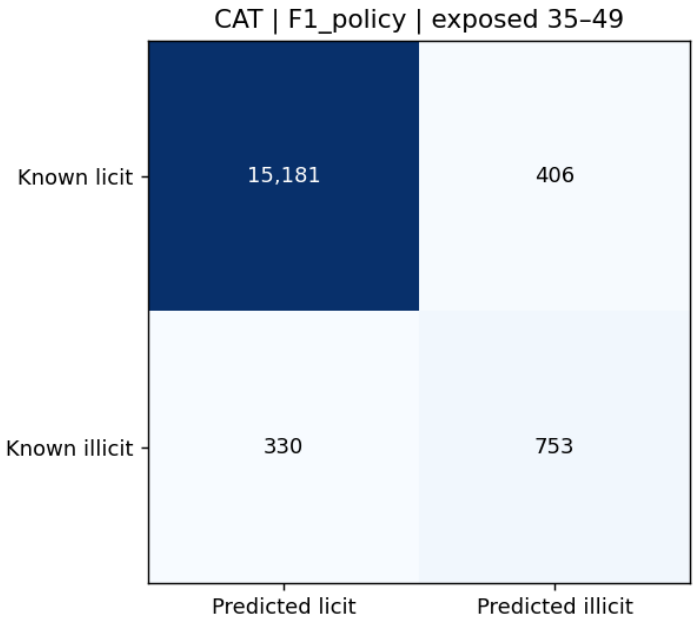
The separate historical false-positive-budget threshold is **0.728**.

This gradient-boosted decision-tree model belongs to the strongest cluster of feature-based ranking systems in the diagnostic window.

Its relative position changes with the metric being emphasized: score ranking, thresholded recall, false-positive workload and calibration each highlight a different aspect of performance.

Its results are especially useful when compared with the other two boosted-tree implementations rather than interpreted in isolation.

Figure 35. CAT: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_CAT_F1_policy.png

The matrix records **15,181** true negatives, **406** false positives, **330** false negatives and **753** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

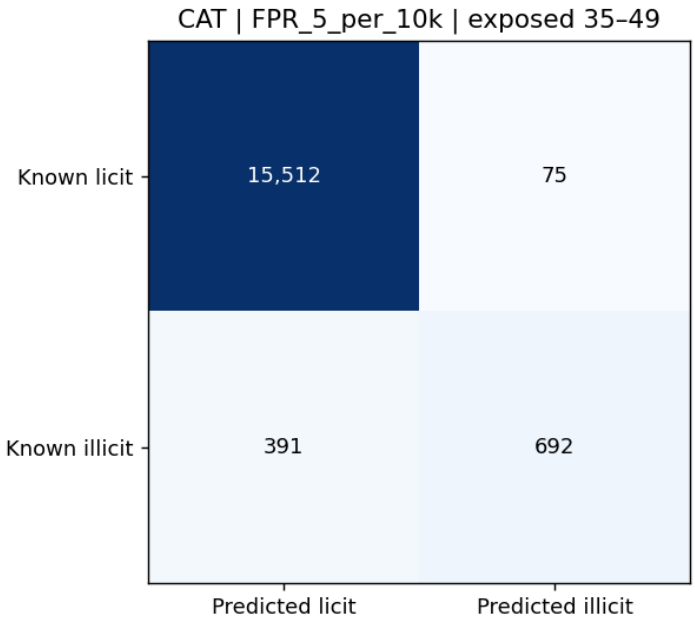
The model detects **753** of the **1,083** known illicit transactions, producing recall of **69.5%**.

Among its **1,159** positive alerts with verified labels, **65.0%** are illicit.

These counts explain its F1 of **0.672** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 36. CAT: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_CAT_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.728**) yields **692** true positives, **391** false negatives, **75** false positives and **15,512** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **406** to **75**, while detected illicit transactions change from **753** to **692**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

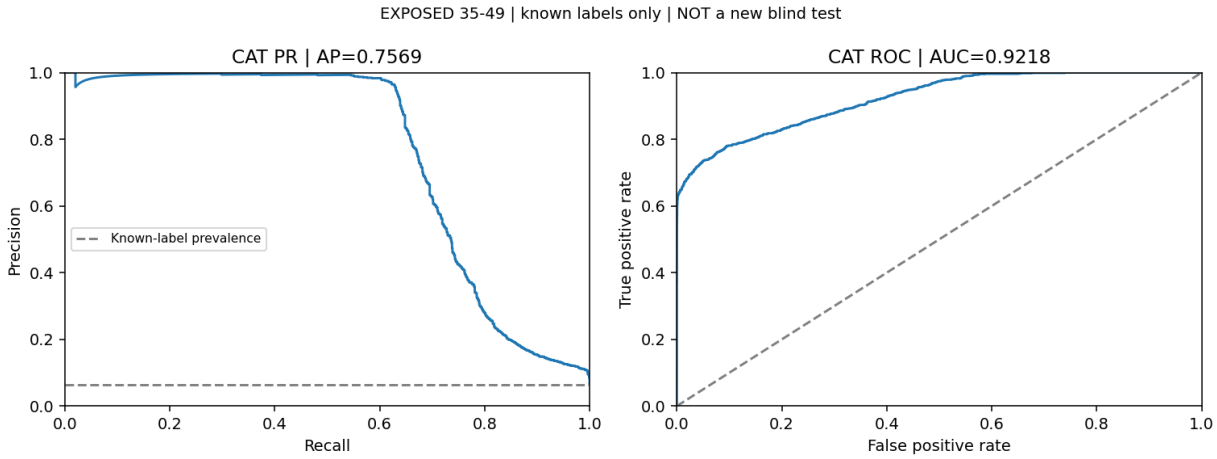
The observed budget-policy false-positive rate equals **48.12** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **63.9%**.

The diagnostic value exceeds the historical target, showing that a threshold selected for a historical false-positive objective need not meet that objective in later periods.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 37. CAT: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_CAT.png

The paired panels show how the CAT scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.757** and ROC-AUC = **0.922**.

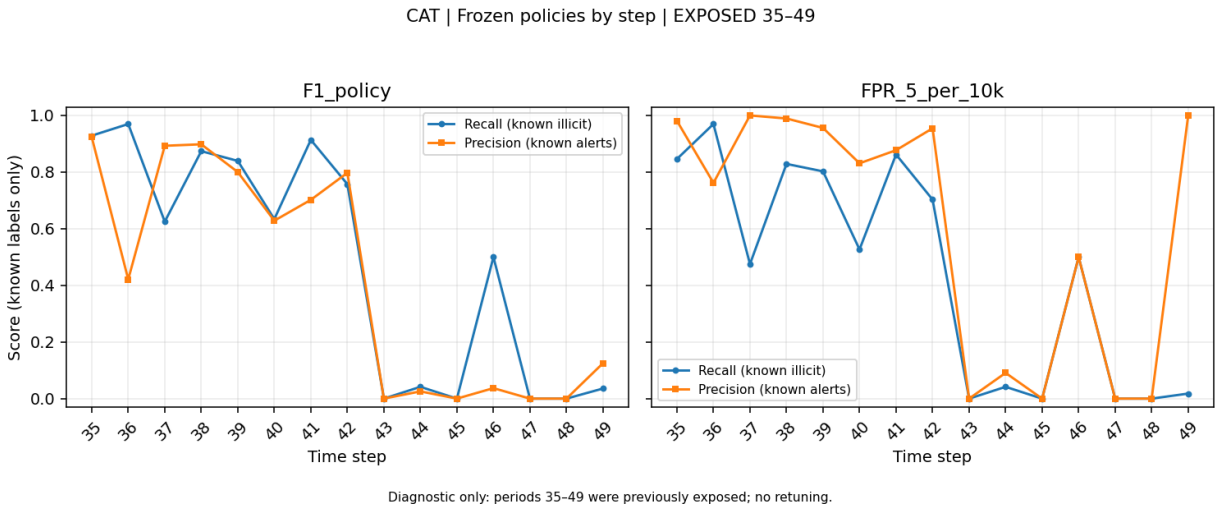
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 38. CAT: recall and precision over diagnostic time



Source figure: FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_CAT.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For CAT, the unweighted average of period-specific AP falls from **0.892** over periods **35-42** to **0.105** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **4** of the seven late periods.

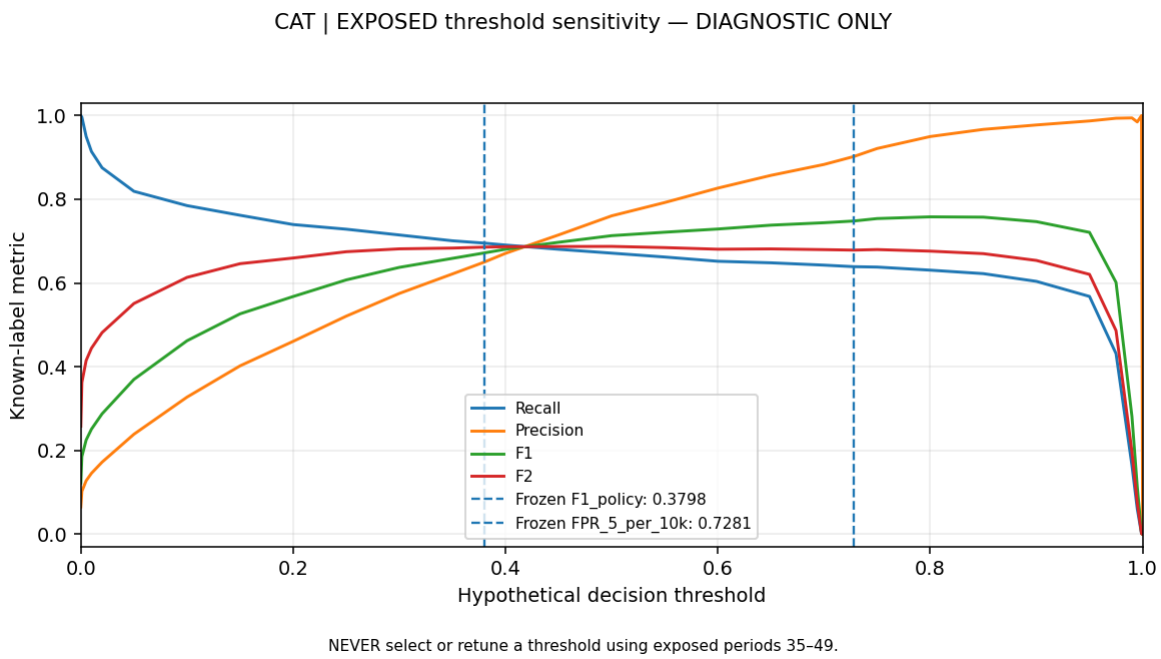
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 39. CAT: post-hoc threshold sensitivity (diagnostic only)



Source figure:
FULL_METRICS/plots/THRESHOLD_SENSITIVITY_CAT_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.380**) and false-positive-budget threshold (**0.728**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.800** with observed F1 of **0.758**, but this number is a post-hoc descriptive result.

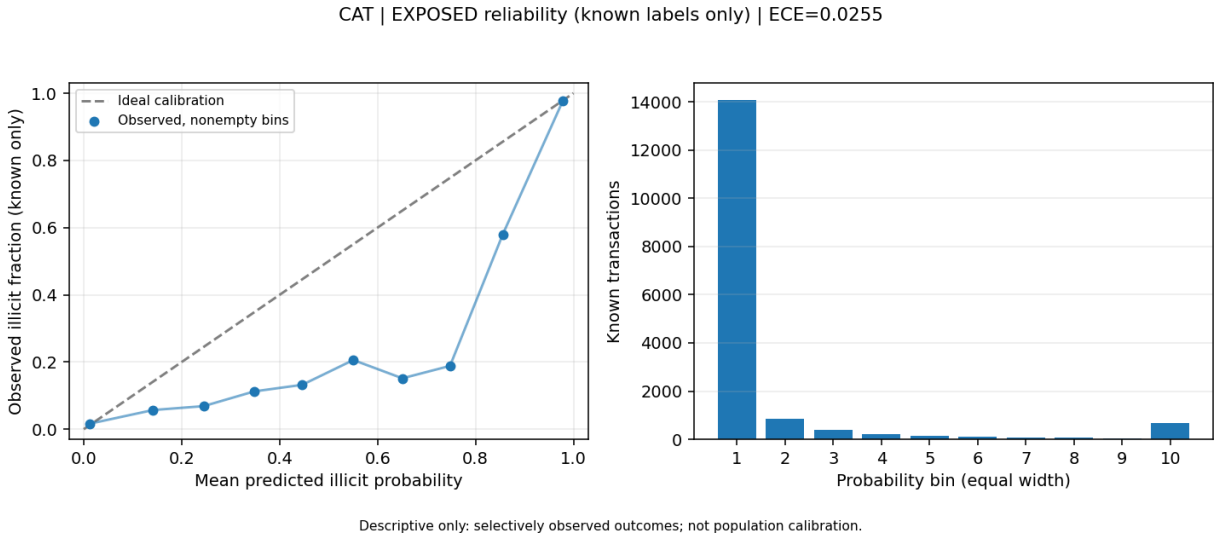
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 40. CAT: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_CAT.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.025 for CAT.

Its diagnostic Brier score is **0.0385** with uncalibrated probabilities and **0.0303** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.5 Multi-Layer Perceptron (MLP)

Multi-Layer Perceptron records pooled diagnostic AP of **0.732** and ROC-AUC of **0.872**.

At the historical F1 threshold (**0.774**), it flags **639** verified transactions, of which **631** are correctly identified as illicit.

Its recall is **58.3%**, precision **98.7%**, and F1 **0.733**.

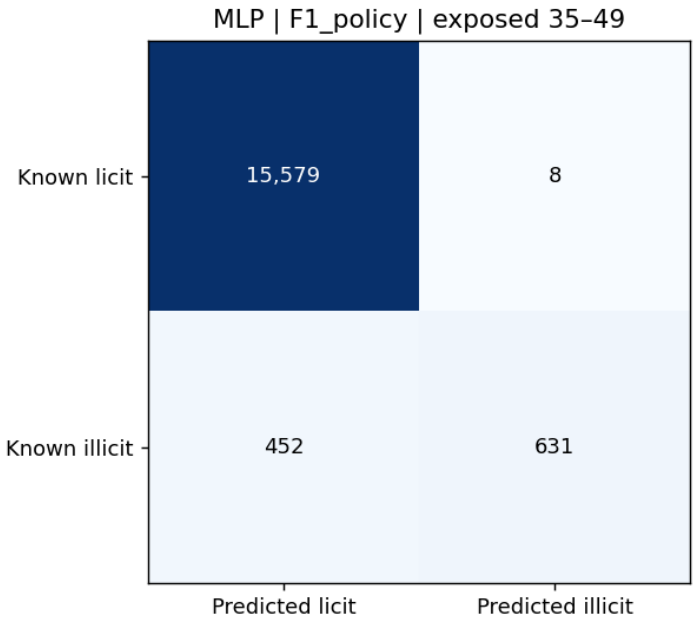
The separate historical false-positive-budget threshold is **0.803**.

This dense neural feature baseline is unusually conservative at the historical F1 operating point.

It provides an informative counterexample to the assumption that the highest precision must imply the most effective illicit-transaction search.

The model can appear highly reliable among the transactions it does flag while failing to retrieve a substantial number of verified illicit examples.

Figure 41. MLP: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_MLP_F1_policy.png

The matrix records **15,579** true negatives, **8** false positives, **452** false negatives and **631** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

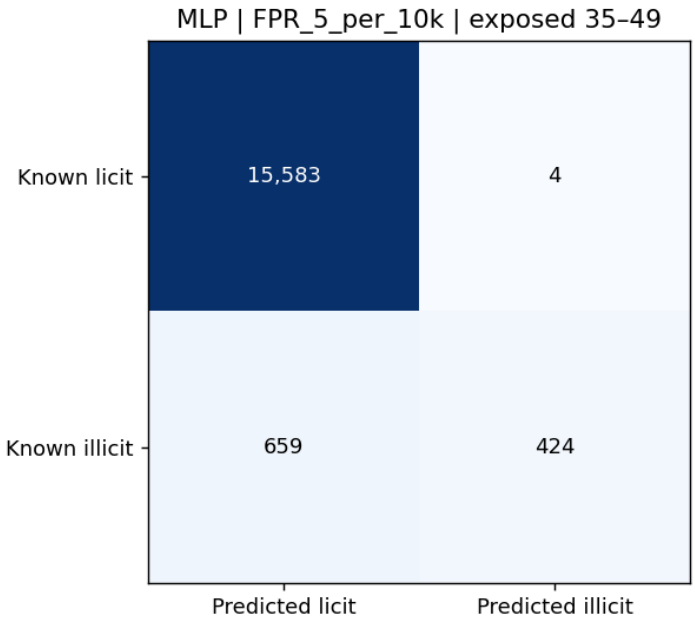
The model detects **631** of the **1,083** known illicit transactions, producing recall of **58.3%**.

Among its **639** positive alerts with verified labels, **98.7%** are illicit.

These counts explain its F1 of **0.733** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 42. MLP: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_MLP_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.803**) yields **424** true positives, **659** false negatives, **4** false positives and **15,583** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **8** to **4**, while detected illicit transactions change from **631** to **424**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

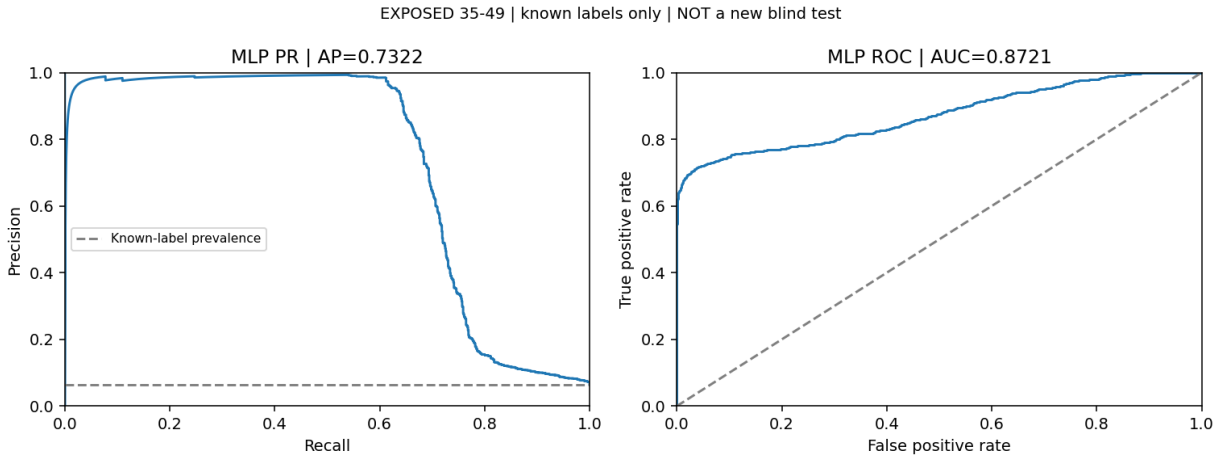
The observed budget-policy false-positive rate equals **2.57** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **39.2%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 43. MLP: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_MLP.png

The paired panels show how the MLP scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.732** and ROC-AUC = **0.872**.

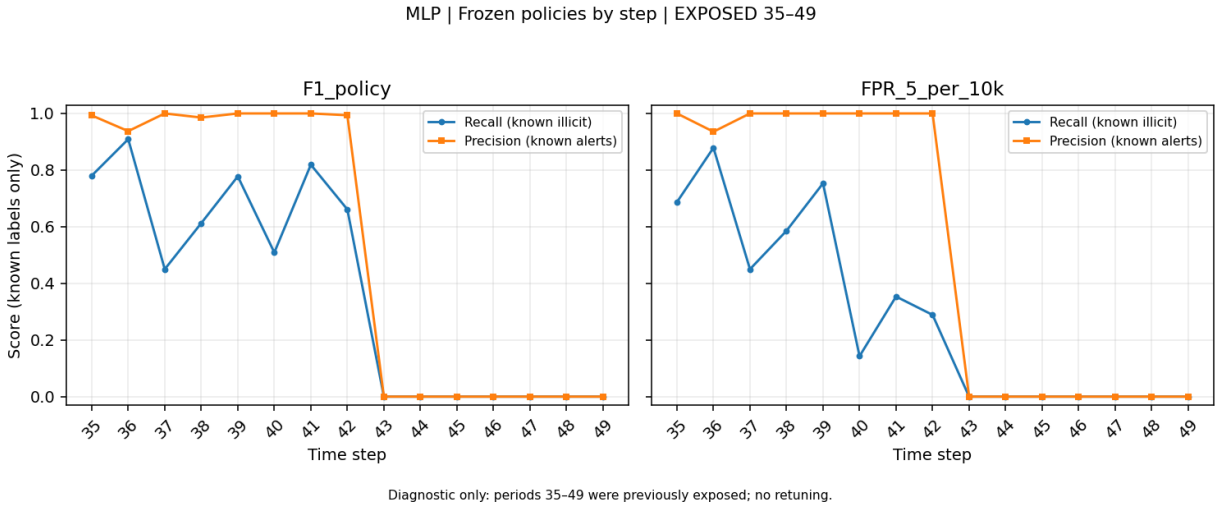
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 44. MLP: recall and precision over diagnostic time



Source figure: FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_MLP.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For MLP, the unweighted average of period-specific AP falls from **0.860** over periods **35-42** to **0.114** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **7** of the seven late periods.

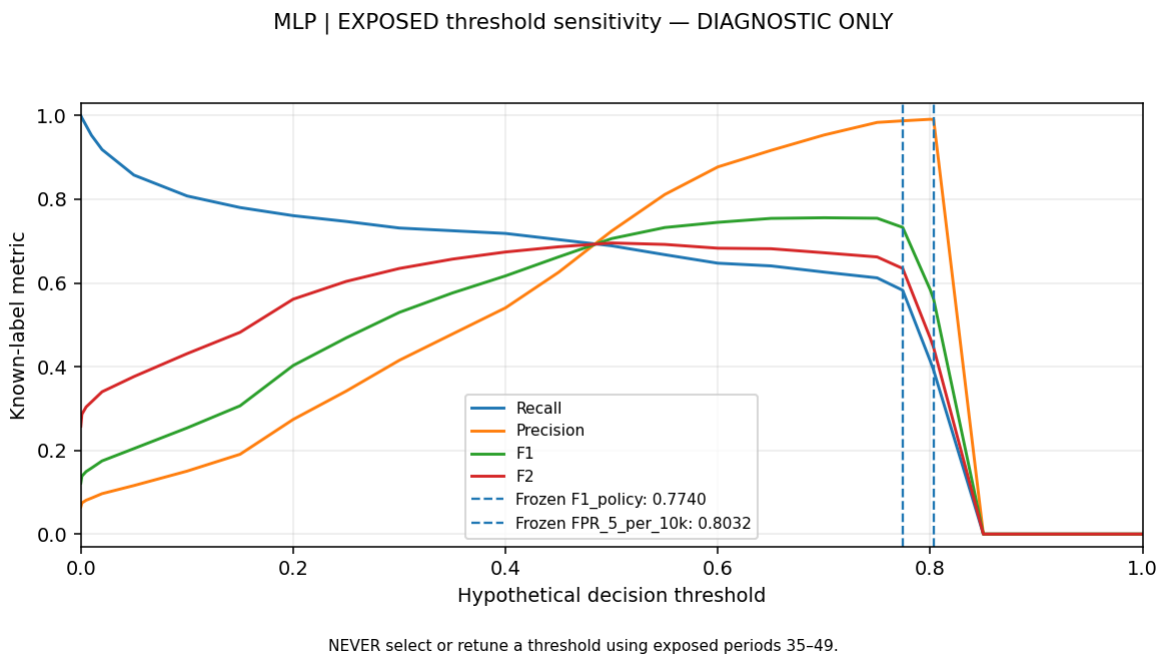
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 45. MLP: post-hoc threshold sensitivity (diagnostic only)



Source figure:
FULL_METRICS/plots/THRESHOLD_SENSITIVITY_MLP_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.774**) and false-positive-budget threshold (**0.803**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.700** with observed F1 of **0.756**, but this number is a post-hoc descriptive result.

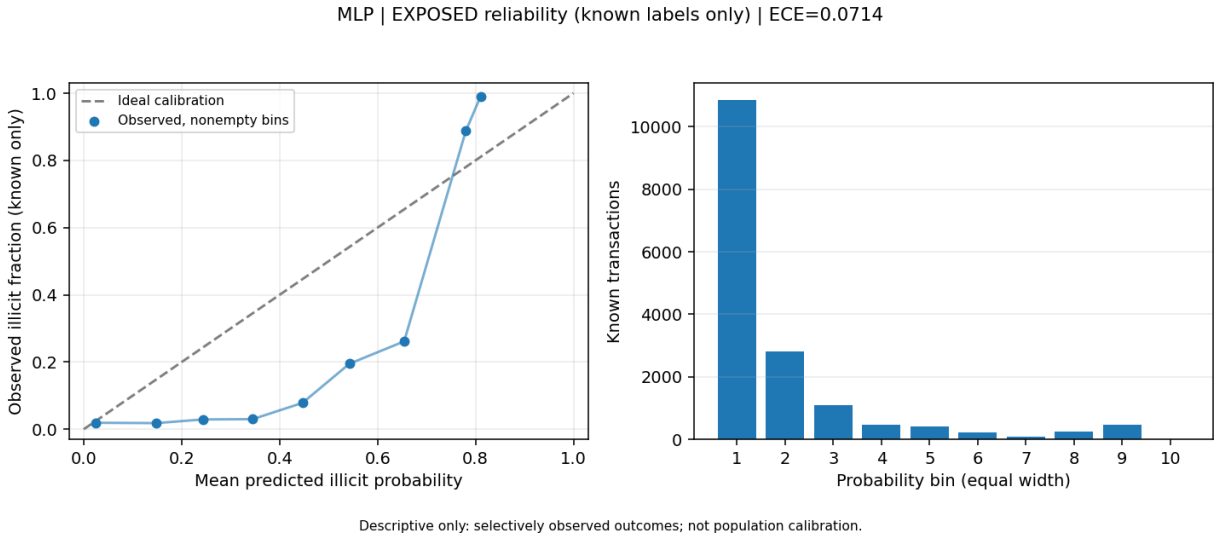
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 46. MLP: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_MLP.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.071 for MLP.

Its diagnostic Brier score is **0.0617** with uncalibrated probabilities and **0.0406** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.6 Incoming-edge GraphSAGE (SAGE_BATCH)

Incoming-edge GraphSAGE records pooled diagnostic AP of **0.693** and ROC-AUC of **0.889**.

At the historical F1 threshold (**0.261**), it flags **1,830** verified transactions, of which **774** are correctly identified as illicit.

Its recall is **71.5%**, precision **42.3%**, and F1 **0.531**.

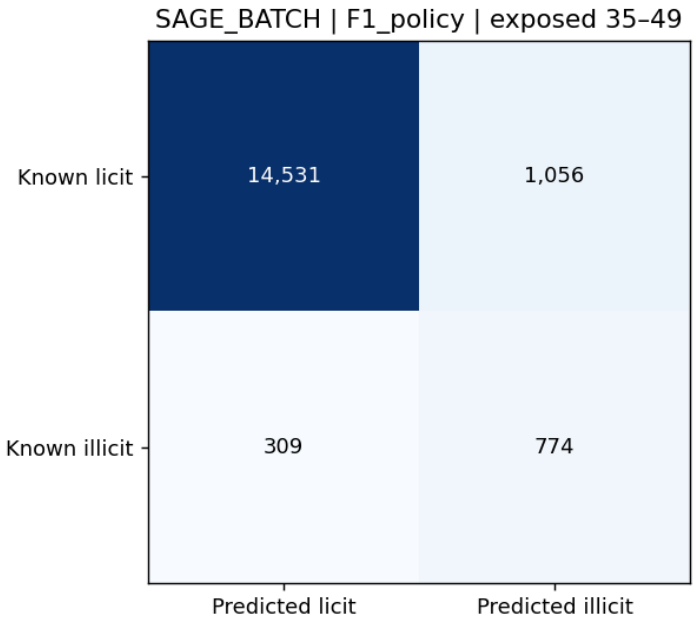
The separate historical false-positive-budget threshold is **0.914**.

This model aggregates information from the transaction graph and should be compared with its no-message and rewired counterparts.

Strong results may reflect transaction features, neighborhood composition, connectivity direction or interactions among them.

The matched comparisons make the research more informative than a simple graph-versus-tree leaderboard, although no single diagnostic window can establish a universal graph benefit.

Figure 47. SAGE_BATCH: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_SAGE_BATCH_F1_policy.png

The matrix records **14,531** true negatives, **1,056** false positives, **309** false negatives and **774** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

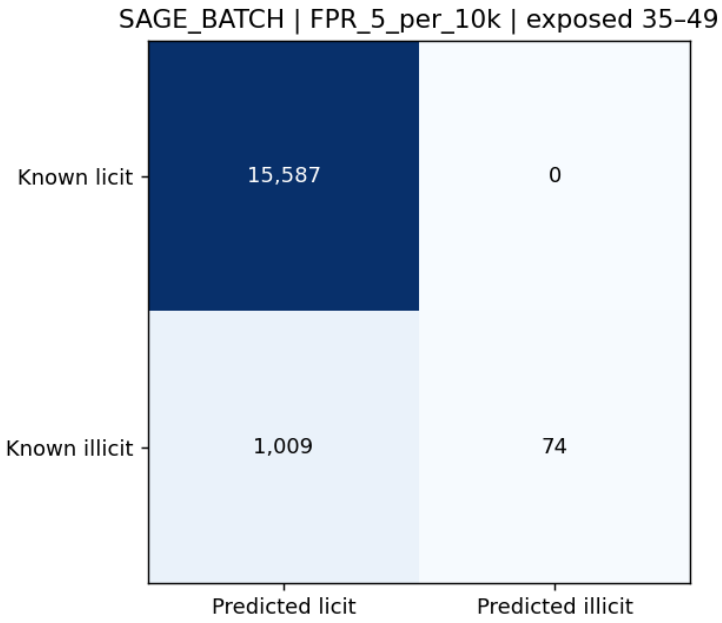
The model detects **774** of the **1,083** known illicit transactions, producing recall of **71.5%**.

Among its **1,830** positive alerts with verified labels, **42.3%** are illicit.

These counts explain its F1 of **0.531** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 48. SAGE_BATCH: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_SAGE_BATCH_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.914**) yields **74** true positives, **1,009** false negatives, **0** false positives and **15,587** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **1,056** to **0**, while detected illicit transactions change from **774** to **74**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

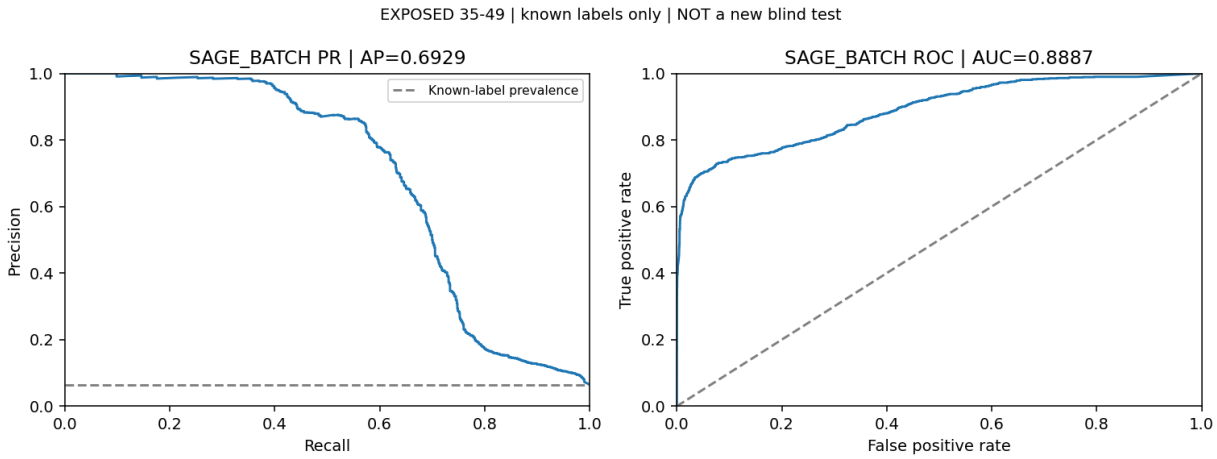
The observed budget-policy false-positive rate equals **0.00** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **6.8%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 49. SAGE_BATCH: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_SAGE_BATCH.png

The paired panels show how the SAGE_BATCH scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.693** and ROC-AUC = **0.889**.

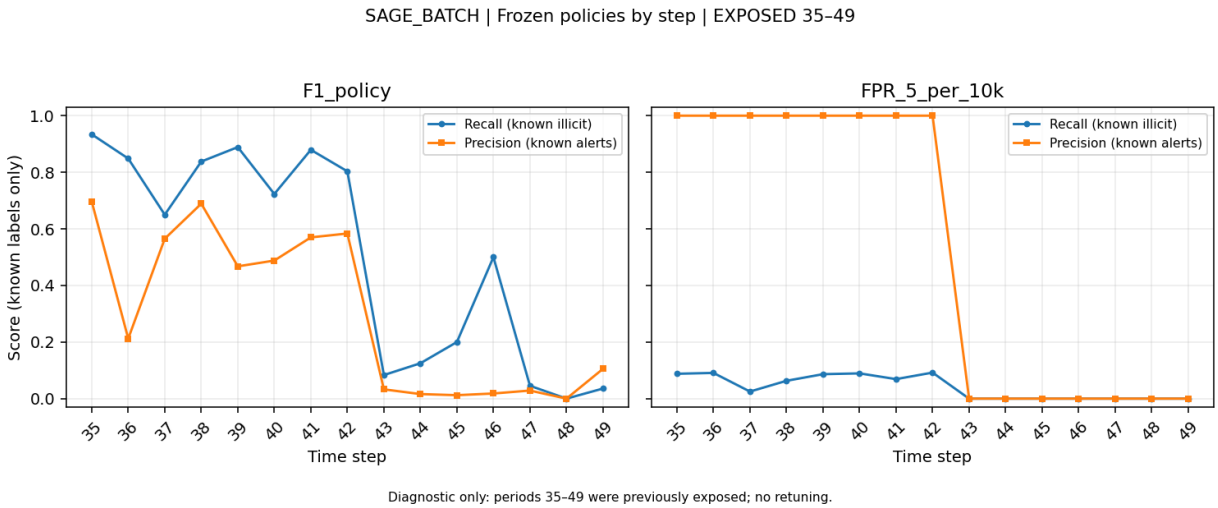
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 50. SAGE_BATCH: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_SAGE_BATCH.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For SAGE_BATCH, the unweighted average of period-specific AP falls from **0.802** over periods **35-42** to **0.128** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **1** of the seven late periods.

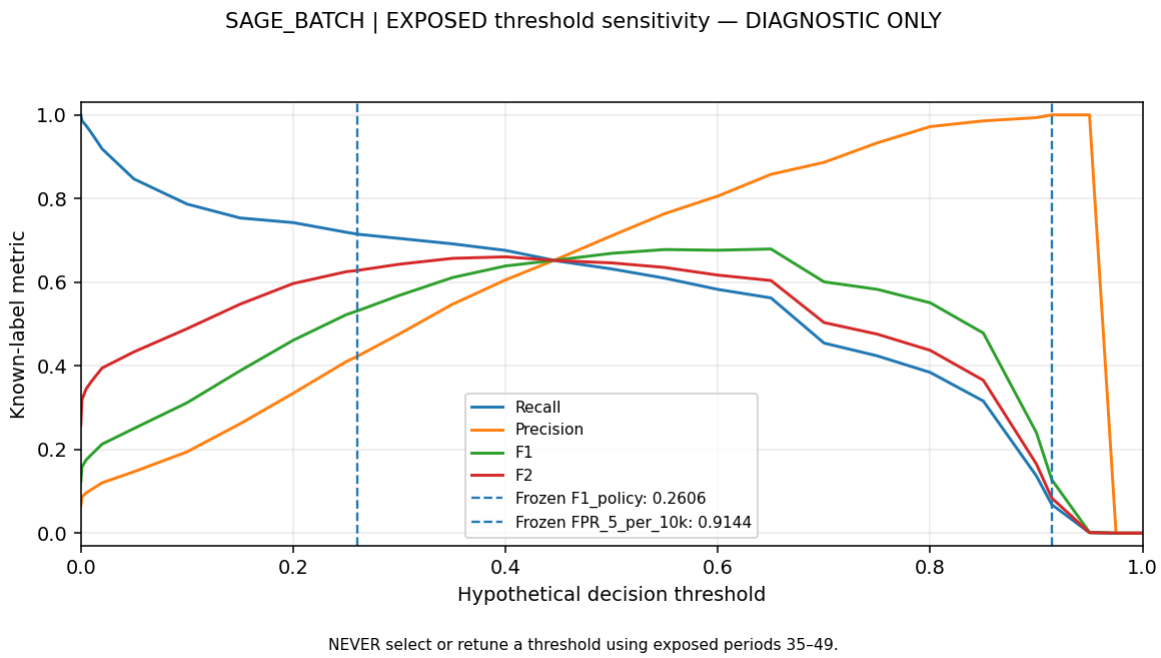
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 51. SAGE_BATCH: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_SAGE_BATCH_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.261**) and false-positive-budget threshold (**0.914**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.650** with observed F1 of **0.679**, but this number is a post-hoc descriptive result.

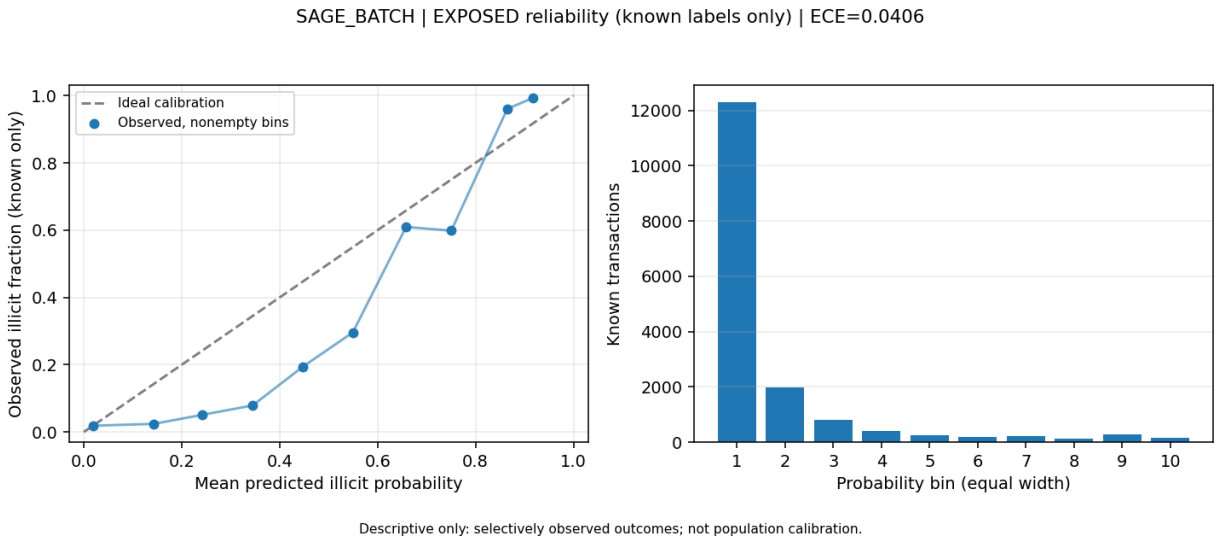
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 52. SAGE_BATCH: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_SAGE_BATCH.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.041 for SAGE_BATCH.

Its diagnostic Brier score is **0.0535** with uncalibrated probabilities and **0.0379** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.7 GraphSAGE without neighbor messages (SAGE_NO_GRAPH)

GraphSAGE without neighbor messages records pooled diagnostic AP of **0.737** and ROC-AUC of **0.871**.

At the historical F1 threshold (**0.192**), it flags **2,684** verified transactions, of which **821** are correctly identified as illicit.

Its recall is **75.8%**, precision **30.6%**, and F1 **0.436**.

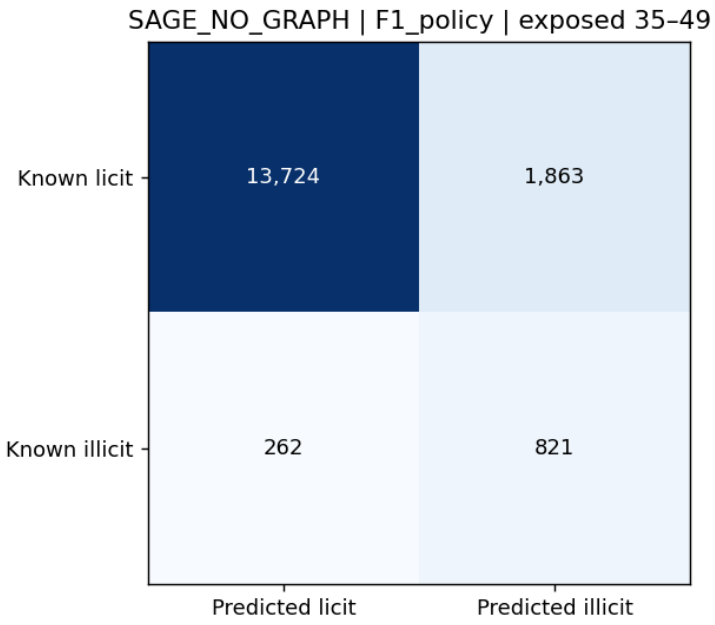
The separate historical false-positive-budget threshold is **0.844**.

This configuration is an architecture-matched control in which neighbor messages are disabled.

Its function in the research design is not merely to compete for the highest score, but to distinguish improvements attributable to graph connectivity from improvements attributable to neural network size, optimization and transaction features.

A strong control result makes a graph-effect claim harder - and more scientifically meaningful - to establish.

Figure 53. SAGE_NO_GRAPH: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_SAGE_NO_GRAPH_F1_policy.png

The matrix records **13,724** true negatives, **1,863** false positives, **262** false negatives and **821** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

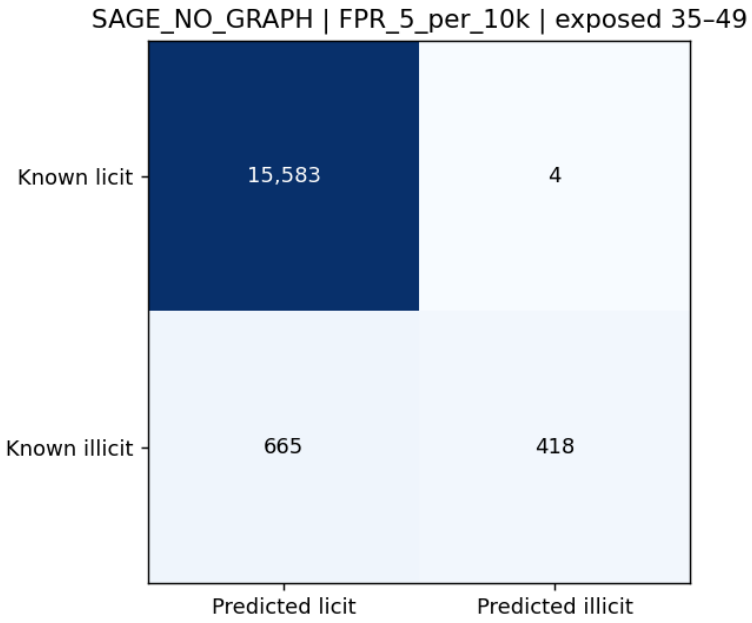
The model detects **821** of the **1,083** known illicit transactions, producing recall of **75.8%**.

Among its **2,684** positive alerts with verified labels, **30.6%** are illicit.

These counts explain its F1 of **0.436** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 54. SAGE_NO_GRAPH: confusion matrix at the false-positive-budget policy



Source figure:
FULL_METRICS/plots/CONFUSION_SAGE_NO_GRAPH_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.844**) yields **418** true positives, **665** false negatives, **4** false positives and **15,583** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **1,863** to **4**, while detected illicit transactions change from **821** to **418**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

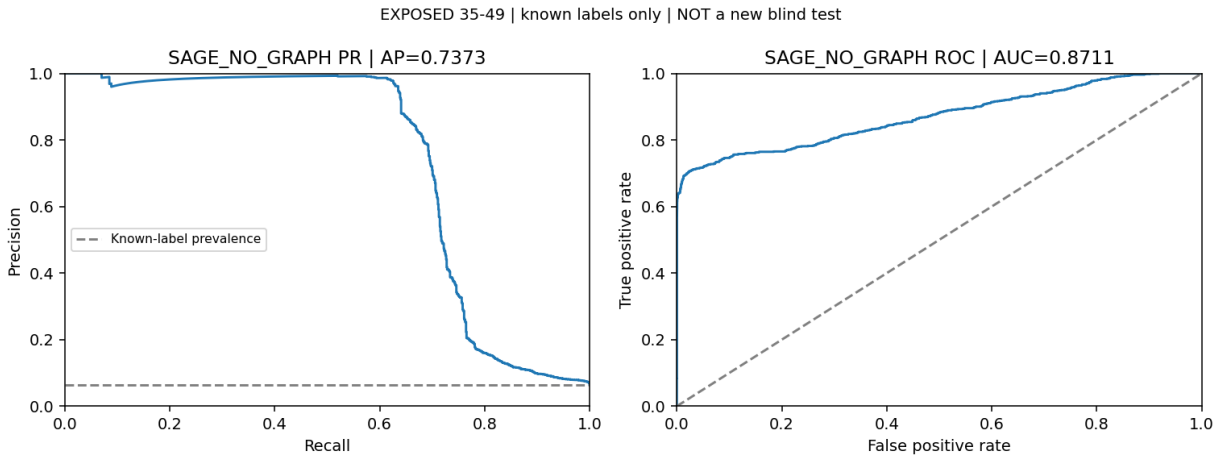
The observed budget-policy false-positive rate equals **2.57** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **38.6%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 55. SAGE_NO_GRAPH: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_SAGE_NO_GRAPH.png

The paired panels show how the SAGE_NO_GRAPH scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.737** and ROC-AUC = **0.871**.

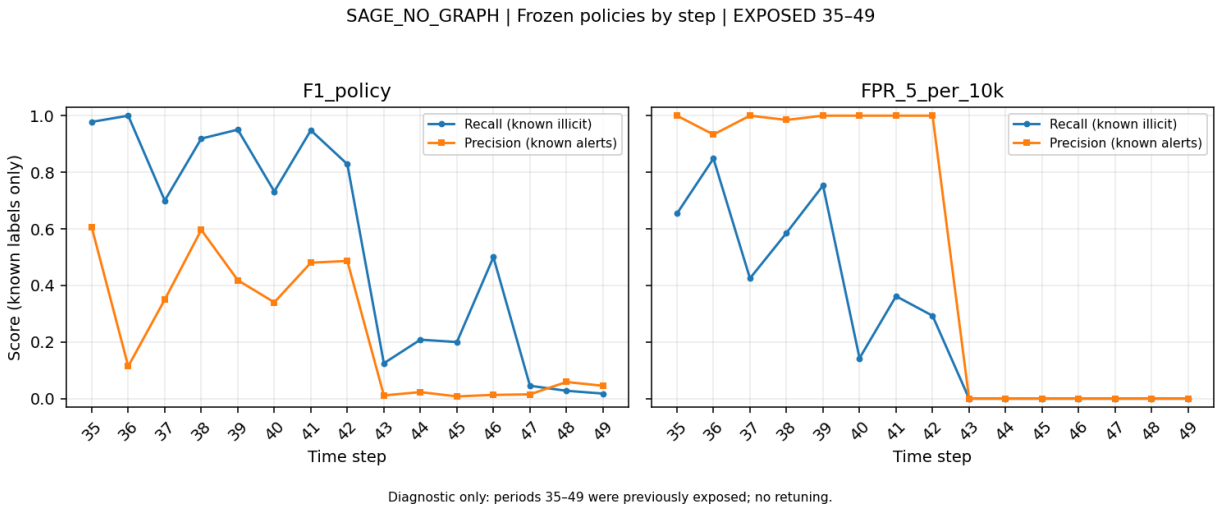
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 56. SAGE_NO_GRAPH: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_SAGE_NO_GRAPH.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For SAGE_NO_GRAPH, the unweighted average of period-specific AP falls from **0.857** over periods **35-42** to **0.052** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **0** of the seven late periods.

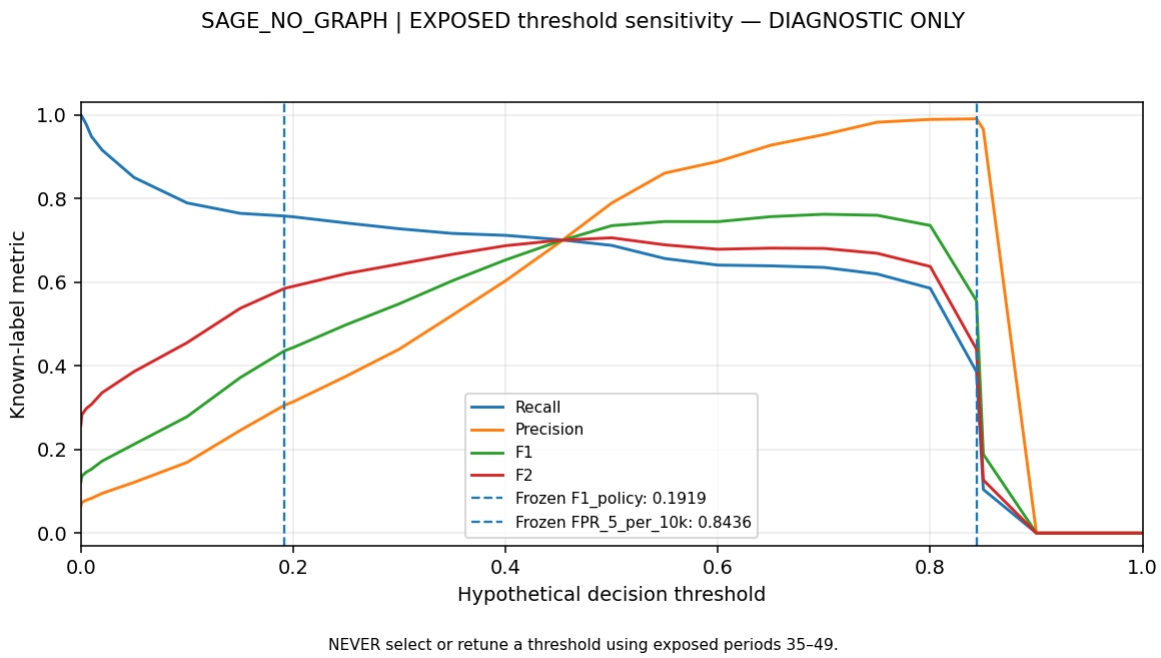
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 57. SAGE_NO_GRAPH: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_SAGE_NO_GRAPH_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.192**) and false-positive-budget threshold (**0.844**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.700** with observed F1 of **0.762**, but this number is a post-hoc descriptive result.

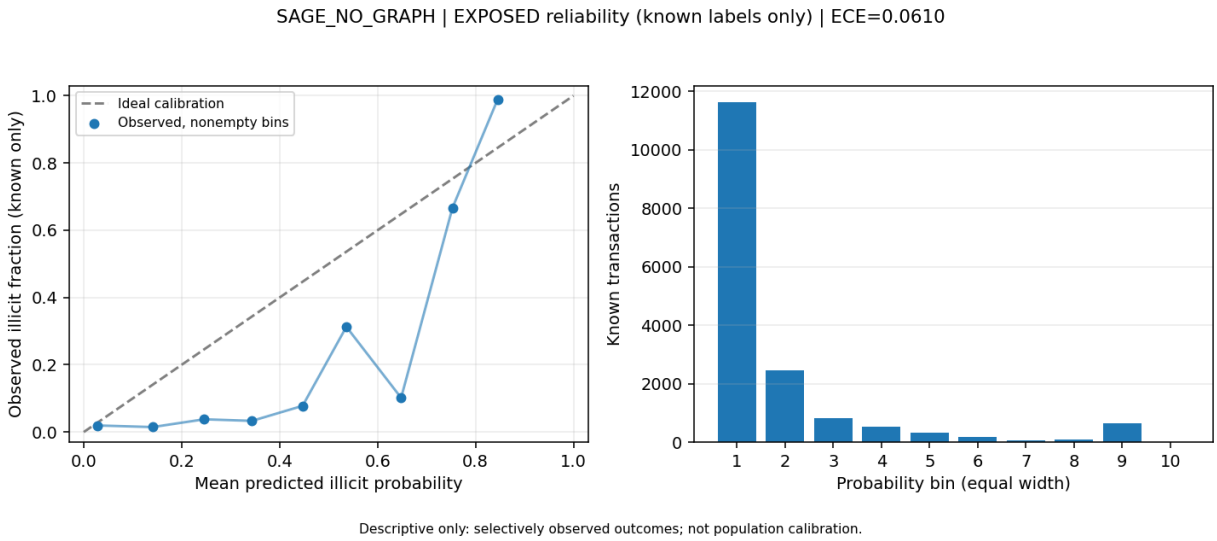
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 58. SAGE_NO_GRAPH: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_SAGE_NO_GRAPH.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.061 for SAGE_NO_GRAPH.

Its diagnostic Brier score is **0.0547** with uncalibrated probabilities and **0.0366** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.8 GraphSAGE with degree-preserving rewiring (SAGE_REWIRED)

GraphSAGE with degree-preserving rewiring records pooled diagnostic AP of **0.361** and ROC-AUC of **0.822**.

At the historical F1 threshold (**0.343**), it flags **1,156** verified transactions, of which **454** are correctly identified as illicit.

Its recall is **41.9%**, precision **39.3%**, and F1 **0.406**.

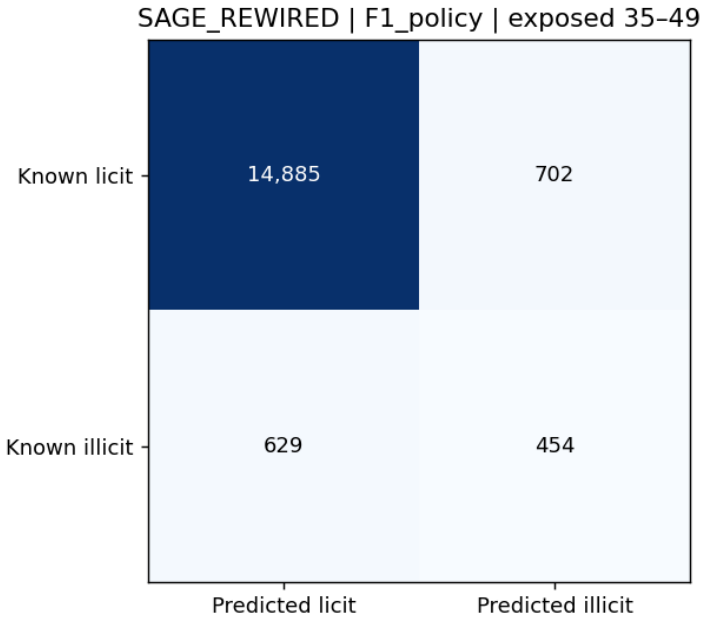
The separate historical false-positive-budget threshold is **0.389**.

This graph-control configuration replaces the observed connectivity with a degree-preserving rewiring strategy within time periods.

Differences from the original graph indicate sensitivity to the particular edge arrangement, while the preserved degree structure limits some trivial structural explanations.

A rewired model is still a meaningful learning system, not an ordinary random-guessing baseline.

Figure 59. SAGE_REWIRED: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_SAGE_REWIRED_F1_policy.png

The matrix records **14,885** true negatives, **702** false positives, **629** false negatives and **454** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

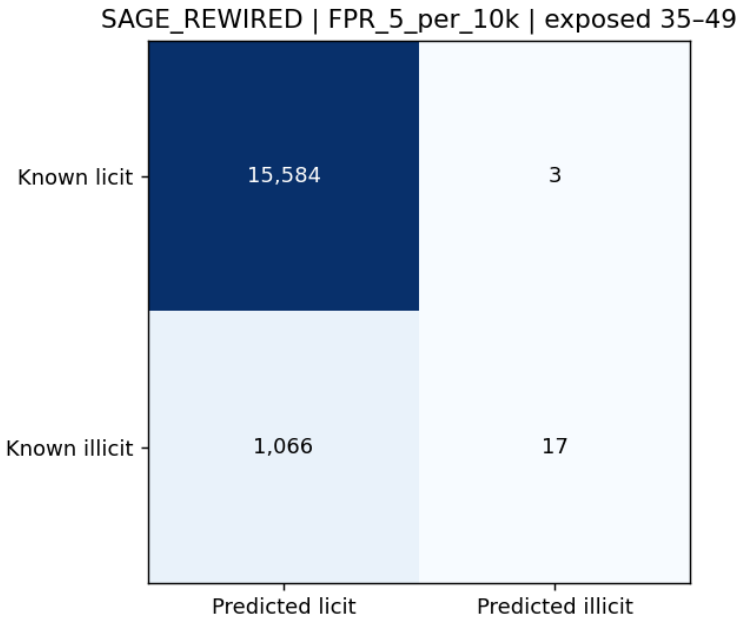
The model detects **454** of the **1,083** known illicit transactions, producing recall of **41.9%**.

Among its **1,156** positive alerts with verified labels, **39.3%** are illicit.

These counts explain its F1 of **0.406** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 60. SAGE_REWIRED: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_SAGE_REWIRED_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.389**) yields **17** true positives, **1,066** false negatives, **3** false positives and **15,584** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **702** to **3**, while detected illicit transactions change from **454** to **17**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

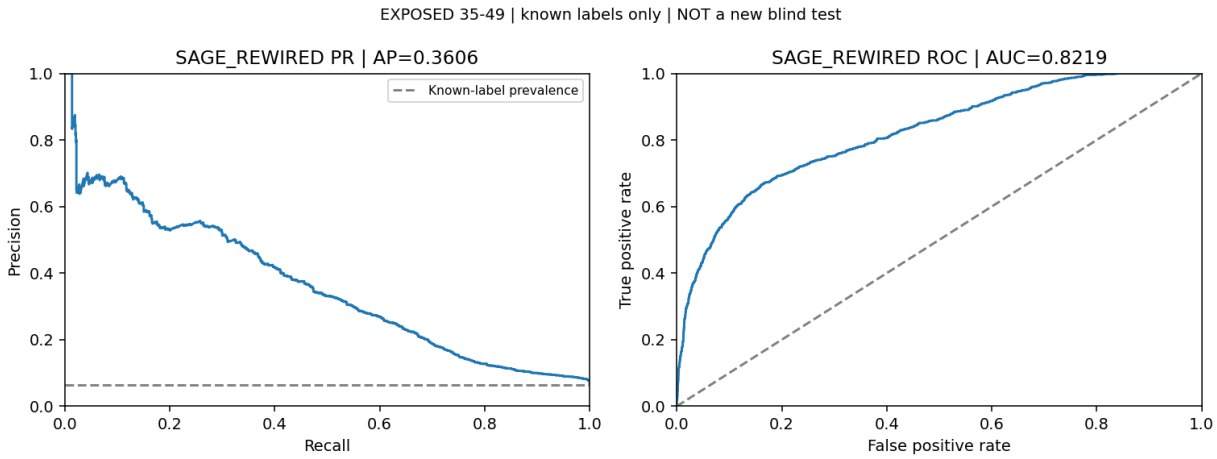
The observed budget-policy false-positive rate equals **1.92** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **1.6%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 61. SAGE_REWIRED: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_SAGE_REWIRED.png

The paired panels show how the SAGE_REWIRED scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.361** and ROC-AUC = **0.822**.

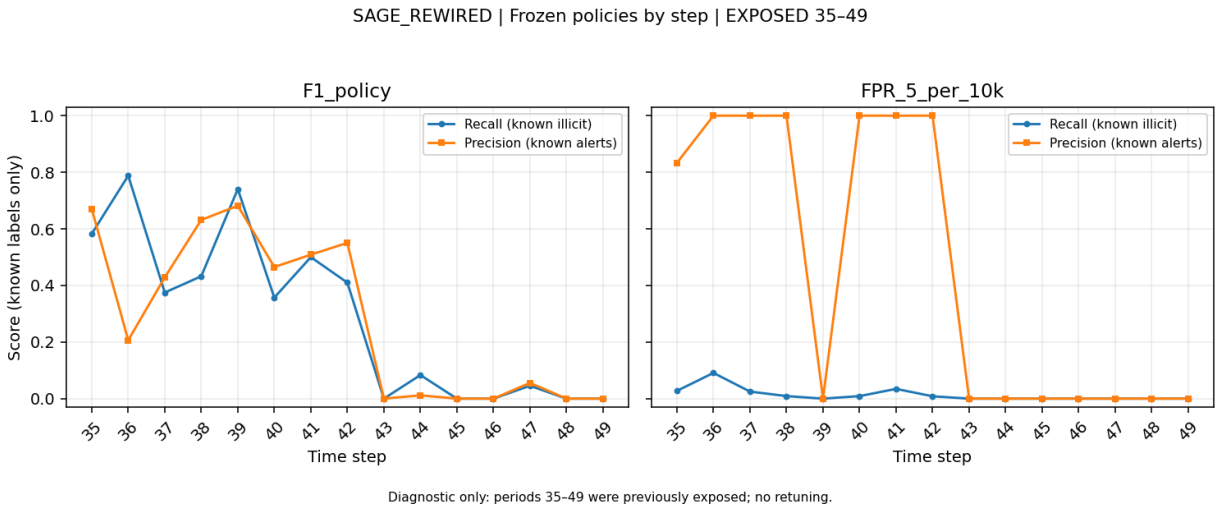
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 62. SAGE_REWIRED: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_SAGE_REWIRED.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For SAGE_REWIRED, the unweighted average of period-specific AP falls from **0.535** over periods **35-42** to **0.053** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **5** of the seven late periods.

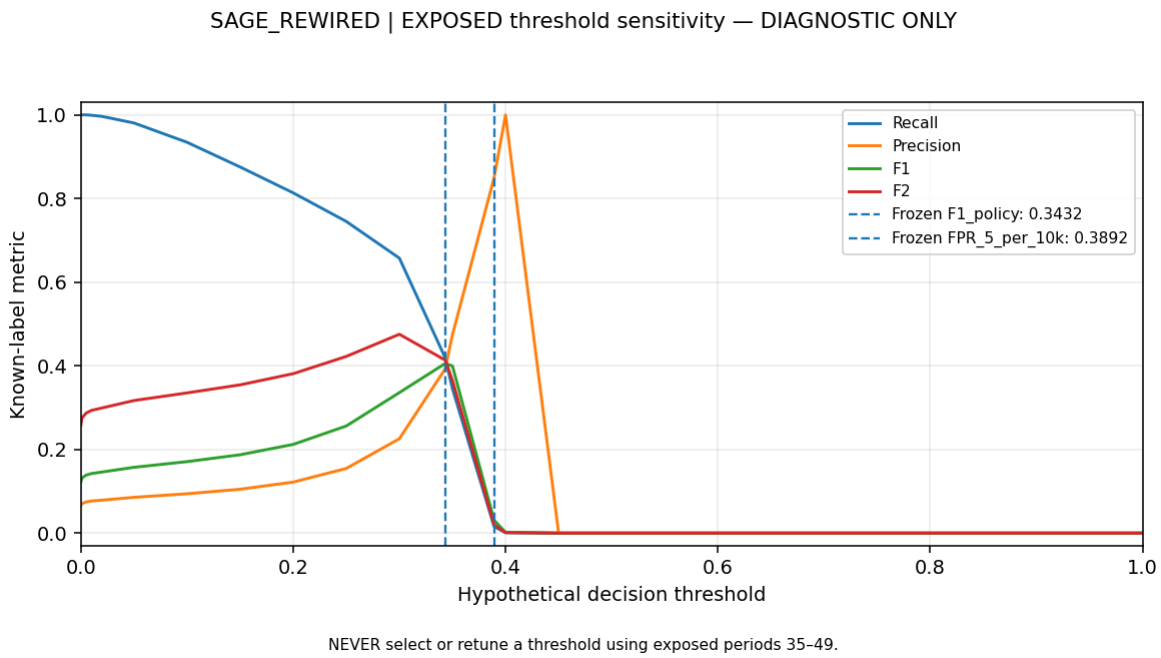
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 63. SAGE_REWIRED: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_SAGE_REWIRED_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.343**) and false-positive-budget threshold (**0.389**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.343** with observed F1 of **0.406**, but this number is a post-hoc descriptive result.

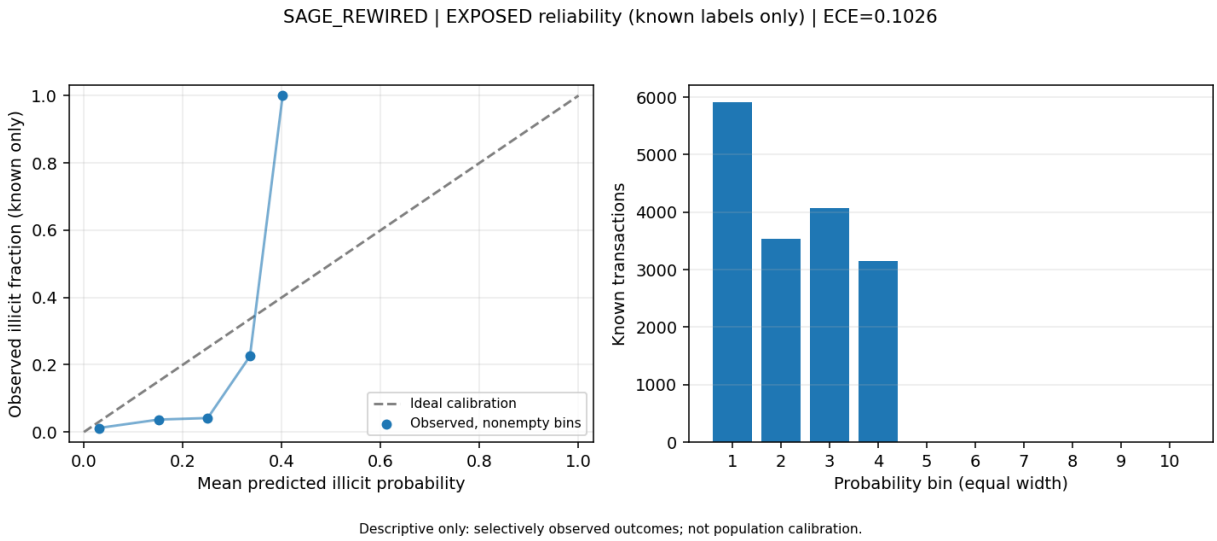
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 64. SAGE_REWIRED: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_SAGE_REWIRED.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.103 for SAGE_REWIRED.

Its diagnostic Brier score is **0.0989** with uncalibrated probabilities and **0.0696** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.9 Bidirectional GraphSAGE (BIDIR_SAGE)

Bidirectional GraphSAGE records pooled diagnostic AP of **0.620** and ROC-AUC of **0.881**.

At the historical F1 threshold (**0.298**), it flags **1,783** verified transactions, of which **693** are correctly identified as illicit.

Its recall is **64.0%**, precision **38.9%**, and F1 **0.484**.

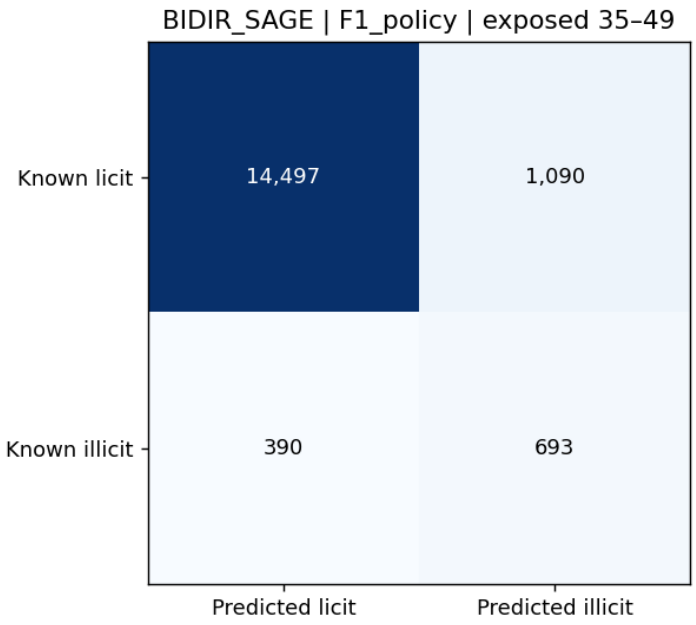
The separate historical false-positive-budget threshold is **0.927**.

This model aggregates information from the transaction graph and should be compared with its no-message and rewired counterparts.

Strong results may reflect transaction features, neighborhood composition, connectivity direction or interactions among them.

The matched comparisons make the research more informative than a simple graph-versus-tree leaderboard, although no single diagnostic window can establish a universal graph benefit.

Figure 65. BIDIR_SAGE: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_BIDIR_SAGE_F1_policy.png

The matrix records **14,497** true negatives, **1,090** false positives, **390** false negatives and **693** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

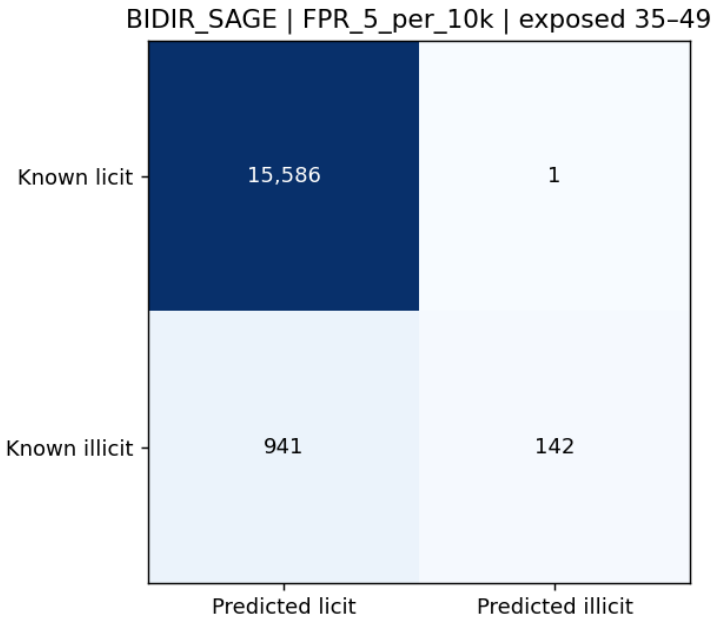
The model detects **693** of the **1,083** known illicit transactions, producing recall of **64.0%**.

Among its **1,783** positive alerts with verified labels, **38.9%** are illicit.

These counts explain its F1 of **0.484** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 66. BIDIR_SAGE: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_BIDIR_SAGE_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.927**) yields **142** true positives, **941** false negatives, **1** false positives and **15,586** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **1,090** to **1**, while detected illicit transactions change from **693** to **142**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

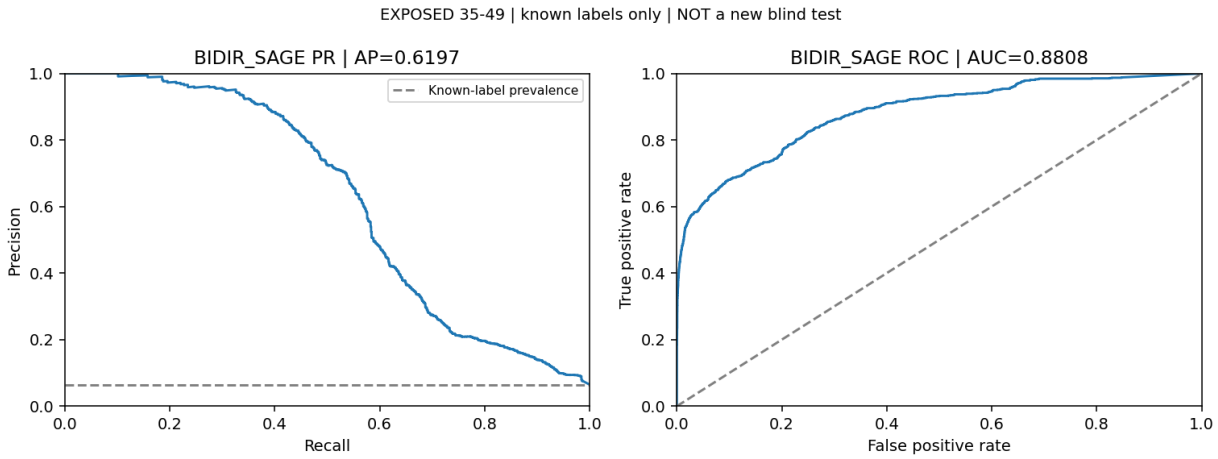
The observed budget-policy false-positive rate equals **0.64** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **13.1%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 67. BIDIR_SAGE: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_BIDIR_SAGE.png

The paired panels show how the BIDIR_SAGE scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.620** and ROC-AUC = **0.881**.

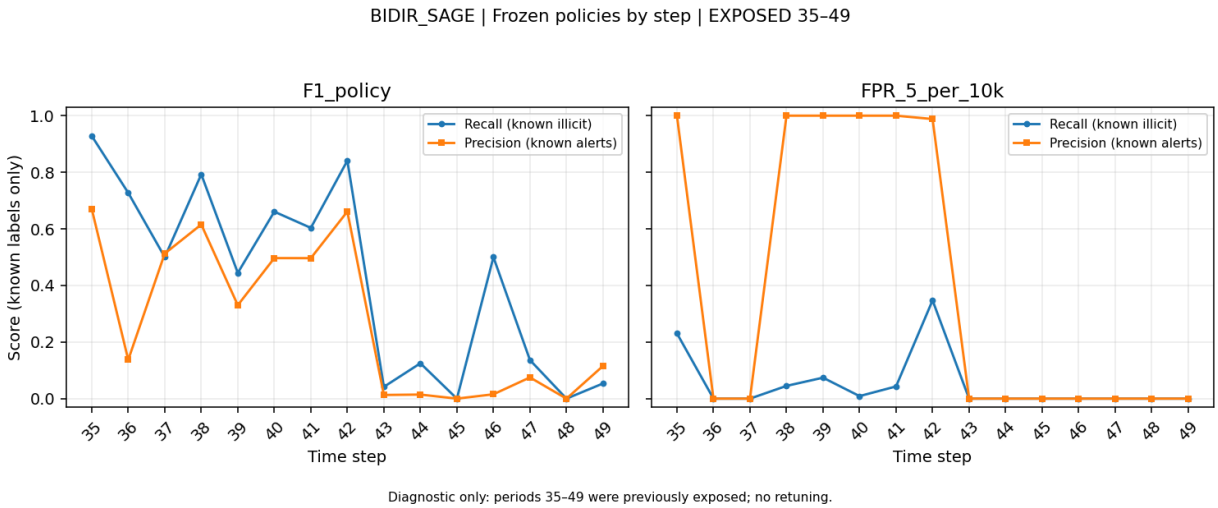
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 68. BIDIR_SAGE: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_BIDIR_SAGE.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For BIDIR_SAGE, the unweighted average of period-specific AP falls from **0.693** over periods **35-42** to **0.068** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **2** of the seven late periods.

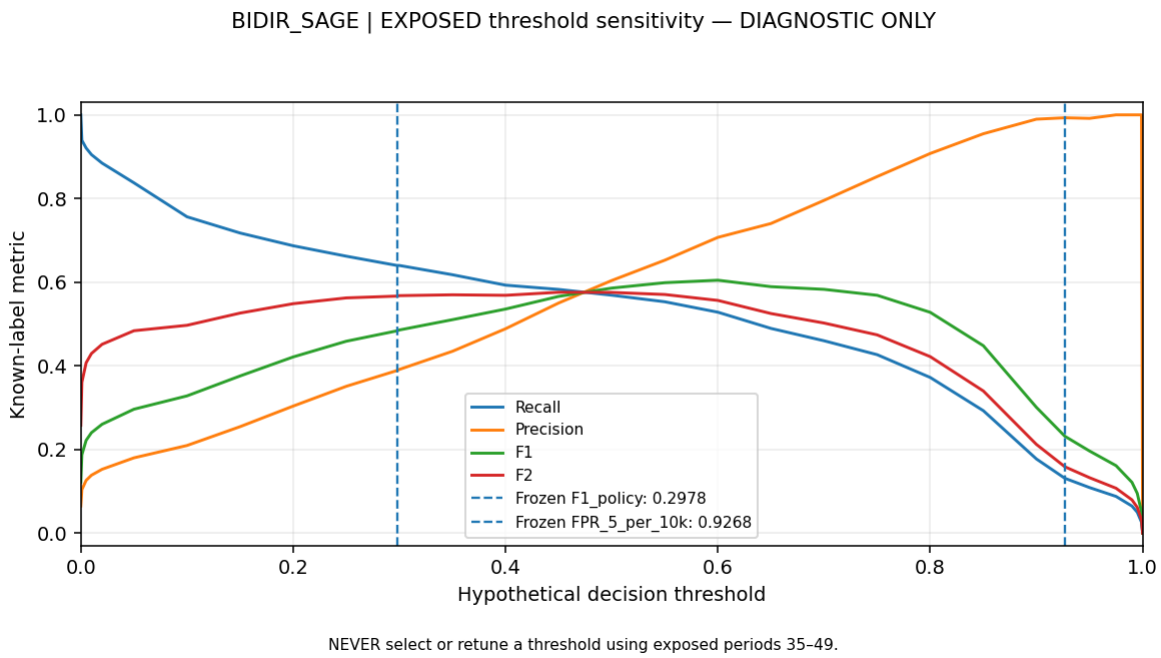
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 69. BIDIR_SAGE: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_BIDIR_SAGE_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.298**) and false-positive-budget threshold (**0.927**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.600** with observed F1 of **0.605**, but this number is a post-hoc descriptive result.

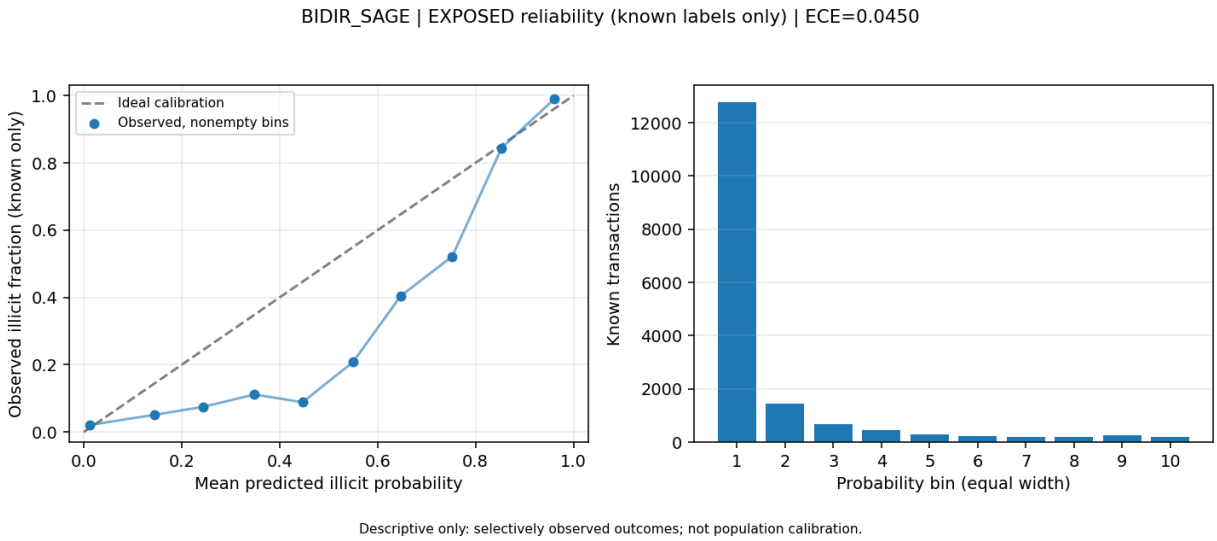
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 70. BIDIR_SAGE: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_BIDIR_SAGE.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.045 for BIDIR_SAGE.

Its diagnostic Brier score is **0.0567** with uncalibrated probabilities and **0.0445** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.10 Bidirectional network without graph messages (BIDIR_NO_GRAPH)

Bidirectional network without graph messages records pooled diagnostic AP of **0.731** and ROC-AUC of **0.875**.

At the historical F1 threshold (**0.231**), it flags **2,604** verified transactions, of which **821** are correctly identified as illicit.

Its recall is **75.8%**, precision **31.5%**, and F1 **0.445**.

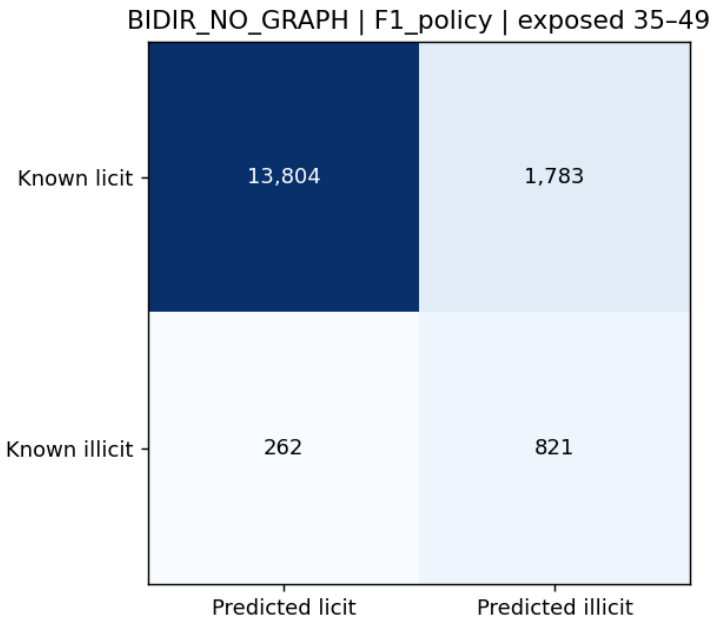
The separate historical false-positive-budget threshold is **0.815**.

This configuration is an architecture-matched control in which neighbor messages are disabled.

Its function in the research design is not merely to compete for the highest score, but to distinguish improvements attributable to graph connectivity from improvements attributable to neural network size, optimization and transaction features.

A strong control result makes a graph-effect claim harder - and more scientifically meaningful - to establish.

Figure 71. BIDIR_NO_GRAPH: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_BIDIR_NO_GRAPH_F1_policy.png

The matrix records **13,804** true negatives, **1,783** false positives, **262** false negatives and **821** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

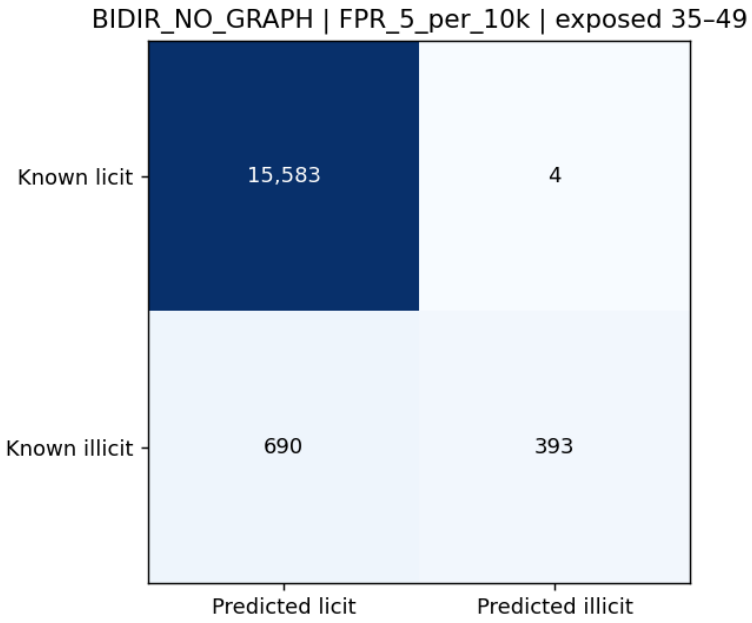
The model detects **821** of the **1,083** known illicit transactions, producing recall of **75.8%**.

Among its **2,604** positive alerts with verified labels, **31.5%** are illicit.

These counts explain its F1 of **0.445** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 72. BIDIR_NO_GRAPH: confusion matrix at the false-positive-budget policy



Source figure:
FULL_METRICS/plots/CONFUSION_BIDIR_NO_GRAPH_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.815**) yields **393** true positives, **690** false negatives, **4** false positives and **15,583** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **1,783** to **4**, while detected illicit transactions change from **821** to **393**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

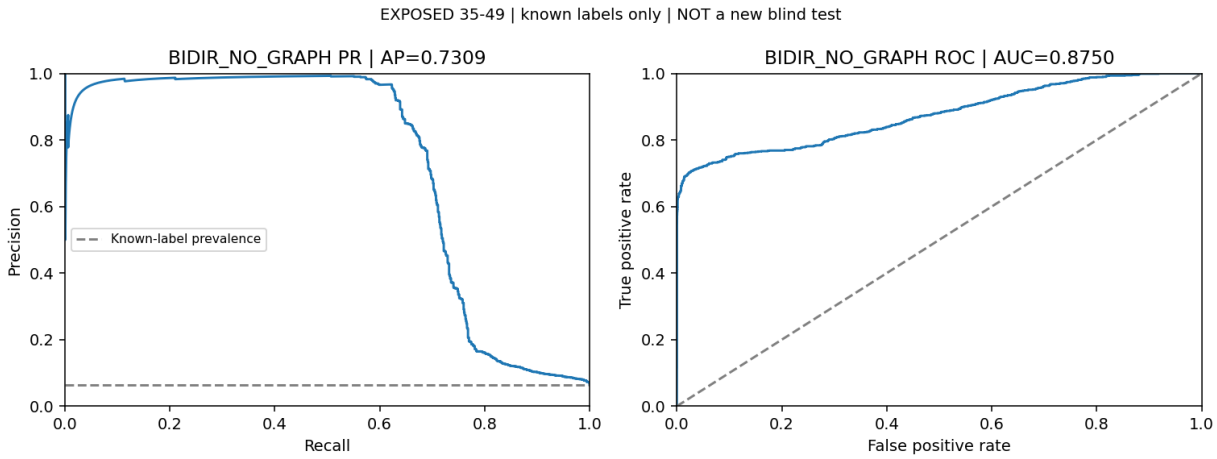
The observed budget-policy false-positive rate equals **2.57** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **36.3%**.

The diagnostic value is below the historical target, but this descriptive success does not guarantee performance in an unseen future cohort.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 73. BIDIR_NO_GRAPH: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_BIDIR_NO_GRAPH.png

The paired panels show how the BIDIR_NO_GRAPH scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.731** and ROC-AUC = **0.875**.

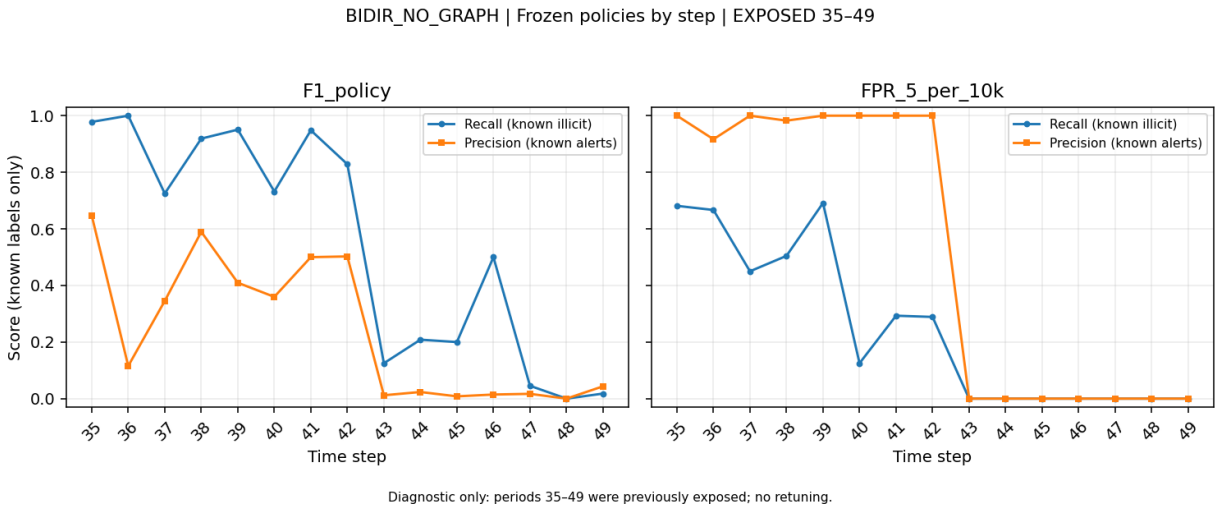
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 74. BIDIR_NO_GRAPH: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_BIDIR_NO_GRAPH.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For BIDIR_NO_GRAPH, the unweighted average of period-specific AP falls from **0.854** over periods **35-42** to **0.056** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **1** of the seven late periods.

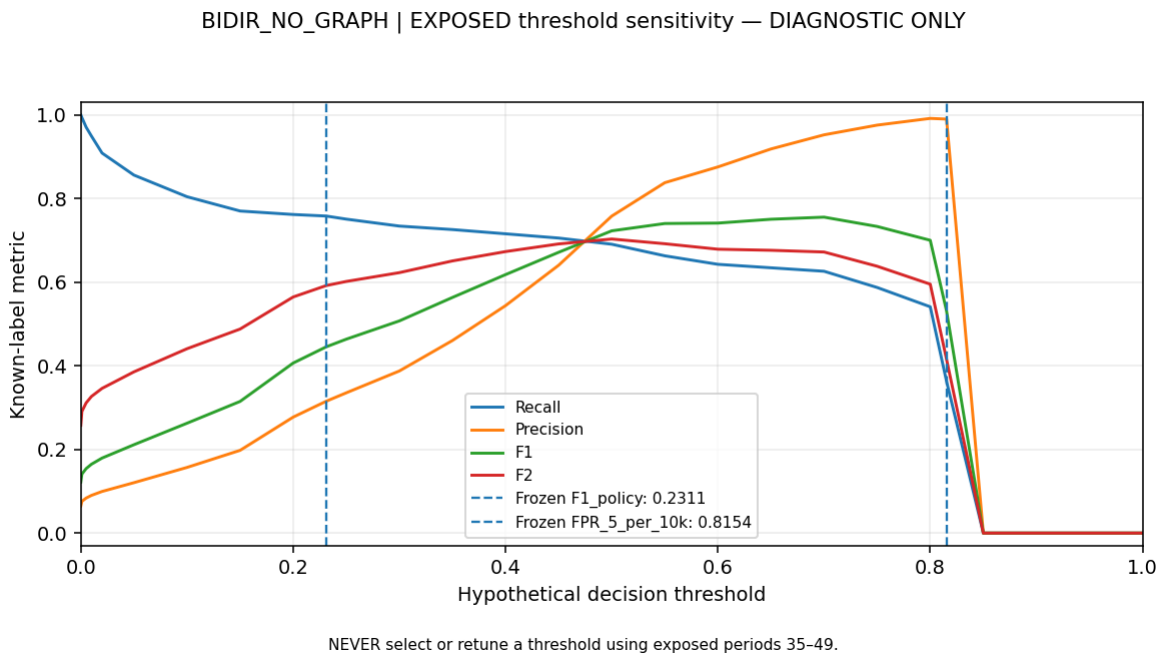
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 75. BIDIR_NO_GRAPH: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_BIDIR_NO_GRAPH_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.231**) and false-positive-budget threshold (**0.815**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.700** with observed F1 of **0.755**, but this number is a post-hoc descriptive result.

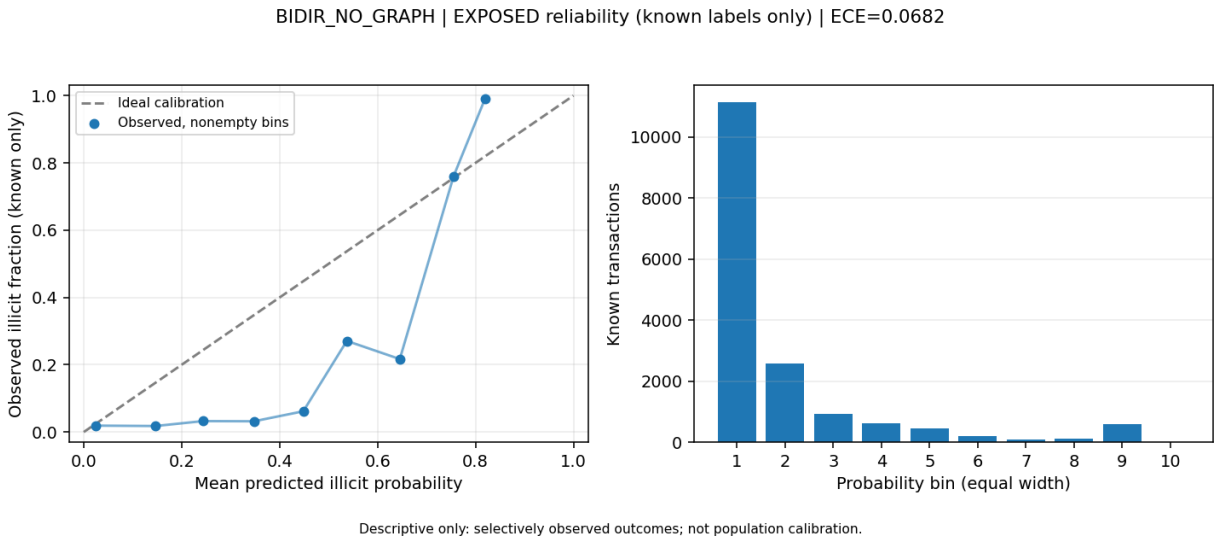
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 76. BIDIR_NO_GRAPH: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_BIDIR_NO_GRAPH.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.068 for BIDIR_NO_GRAPH.

Its diagnostic Brier score is **0.0554** with uncalibrated probabilities and **0.0403** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

5.11 Bidirectional GraphSAGE with rewired edges (BIDIR_REWIRED)

Bidirectional GraphSAGE with rewired edges records pooled diagnostic AP of **0.479** and ROC-AUC of **0.852**.

At the historical F1 threshold (**0.571**), it flags **1,160** verified transactions, of which **623** are correctly identified as illicit.

Its recall is **57.5%**, precision **53.7%**, and F1 **0.556**.

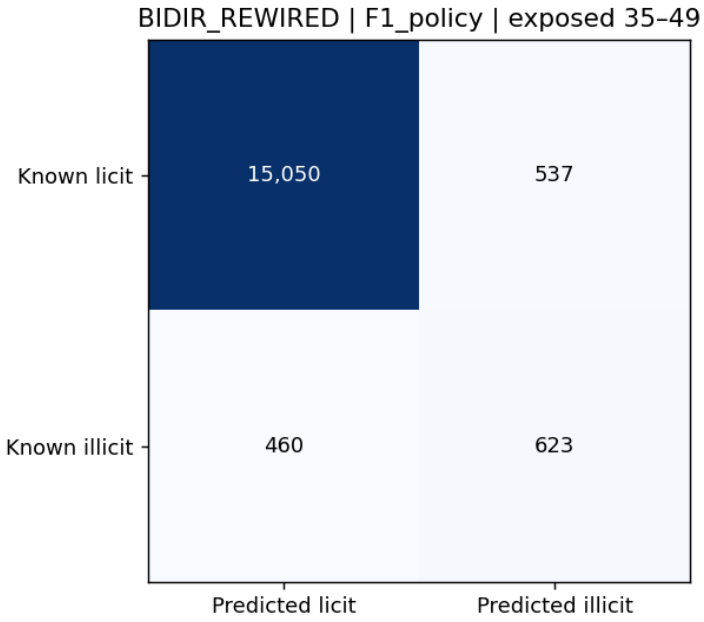
The separate historical false-positive-budget threshold is **0.841**.

This graph-control configuration replaces the observed connectivity with a degree-preserving rewiring strategy within time periods.

Differences from the original graph indicate sensitivity to the particular edge arrangement, while the preserved degree structure limits some trivial structural explanations.

A rewired model is still a meaningful learning system, not an ordinary random-guessing baseline.

Figure 77. BIDIR_REWIRED: confusion matrix at the F1 policy



Source figure: FULL_METRICS/plots/CONFUSION_BIDIR_REWIRED_F1_policy.png

The matrix records **15,050** true negatives, **537** false positives, **460** false negatives and **623** true positives.

Rows identify the known true class and columns the predicted class.

The upper-right cell is the false-alarm burden, the lower-left cell represents illicit transactions the system failed to flag.

These are distinct errors that can carry very different consequences in an investigative workflow.

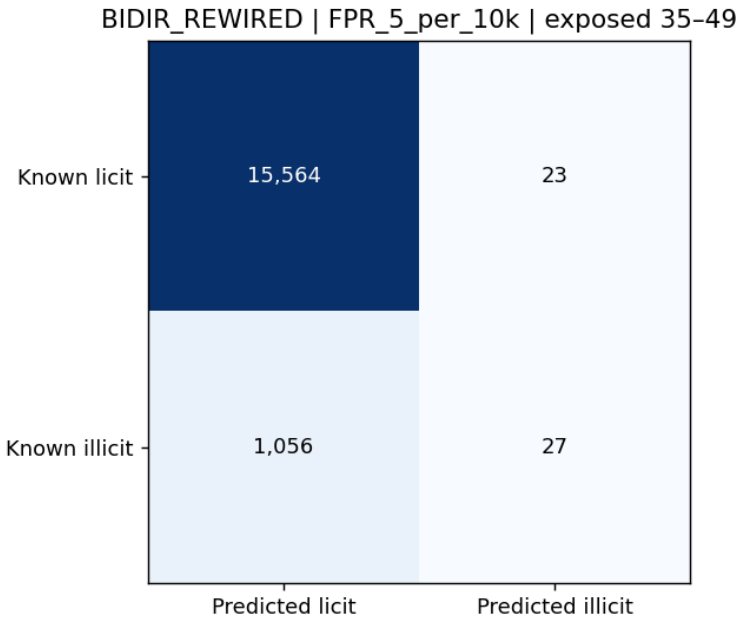
The model detects **623** of the **1,083** known illicit transactions, producing recall of **57.5%**.

Among its **1,160** positive alerts with verified labels, **53.7%** are illicit.

These counts explain its F1 of **0.556** more transparently than the score alone.

Because the diagnostic sample contains 15,587 known licit transactions, raw true-negative counts are naturally much larger than raw true-positive counts, normalized metrics are necessary for a fair reading.

Figure 78. BIDIR_REWIRED: confusion matrix at the false-positive-budget policy



Source figure: FULL_METRICS/plots/CONFUSION_BIDIR_REWIRED_FPR_5_per_10k.png

Changing only the decision threshold to the frozen budget policy (**0.841**) yields **27** true positives, **1,056** false negatives, **23** false positives and **15,564** true negatives.

Relative to the historical F1 policy, the number of false positives changes from **537** to **23**, while detected illicit transactions change from **623** to **27**.

This is a concrete demonstration of the detection-versus-investigation-workload trade-off.

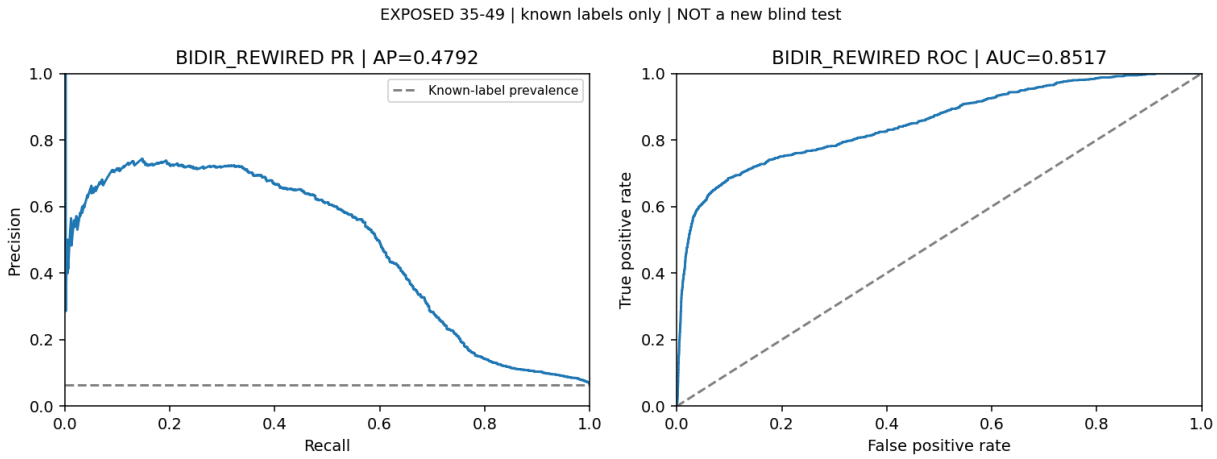
The observed budget-policy false-positive rate equals **14.76** per **10,000** known licit transactions in the exposed window, compared with the historical objective of five.

The resulting recall is **2.5%**.

The diagnostic value exceeds the historical target, showing that a threshold selected for a historical false-positive objective need not meet that objective in later periods.

The correct scientific practice is to keep this threshold frozen and report what actually happened.

Figure 79. BIDIR_REWIRED: precision-recall and ROC curves



Source figure: FULL_METRICS/plots/PR_ROC_BIDIR_REWIRED.png

The paired panels show how the BIDIR_REWIRED scores behave over a continuum of possible classification cutoffs.

In the Precision-Recall panel, one examines the precision available at each recall level, in the ROC panel, sensitivity is plotted against the false-positive rate.

The area-style summaries are AP = **0.479** and ROC-AUC = **0.852**.

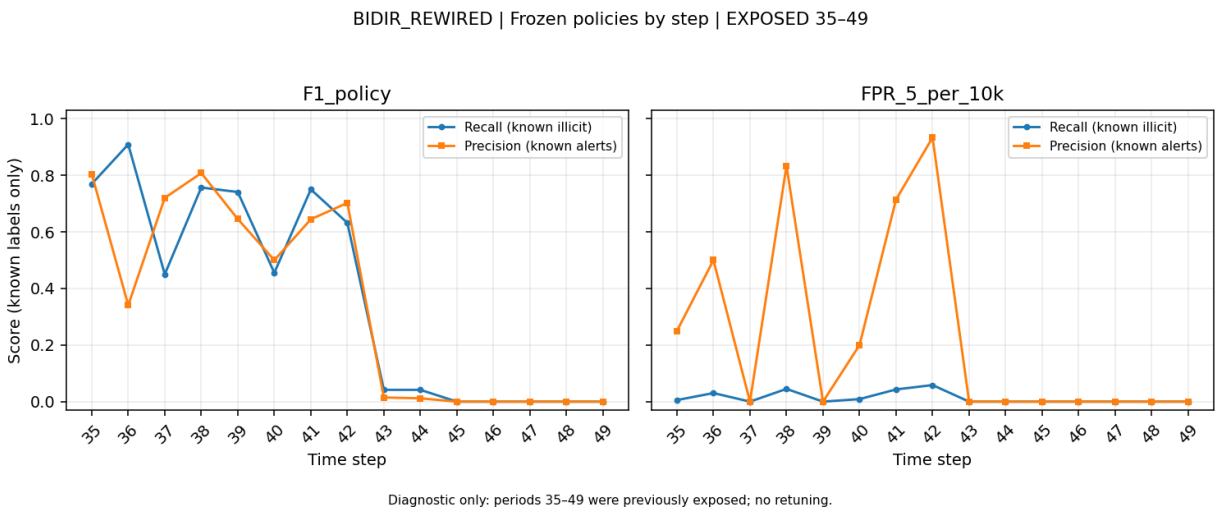
The PR reference level is related to the known-label positive prevalence of about **6.5%** in this diagnostic cohort.

These curves place the model's fixed operating points in context without replacing them.

The ranking results can be stronger or weaker than the F1-policy confusion matrix suggests, because AP and ROC-AUC aggregate behavior across thresholds.

Their interpretation is limited to the verified labels used to construct the curves, and they cannot establish the model's precision among the many unverified transactions that may nevertheless receive high scores.

Figure 80. BIDIR_REWIRED: recall and precision over diagnostic time



Source figure:
FULL_METRICS/plots/TIME_SERIES_RECALL_PRECISION_BIDIR_REWIRED.png

The four traces show recall and precision under the two historical decision policies as the experiment moves from step **35** to step **49**.

They make clear whether performance is stable or is concentrated in a small subset of periods.

For BIDIR_REWIRED, the unweighted average of period-specific AP falls from **0.651** over periods **35-42** to **0.048** over periods **43-49**.

Under the F1 policy, recall is exactly zero in **5** of the seven late periods.

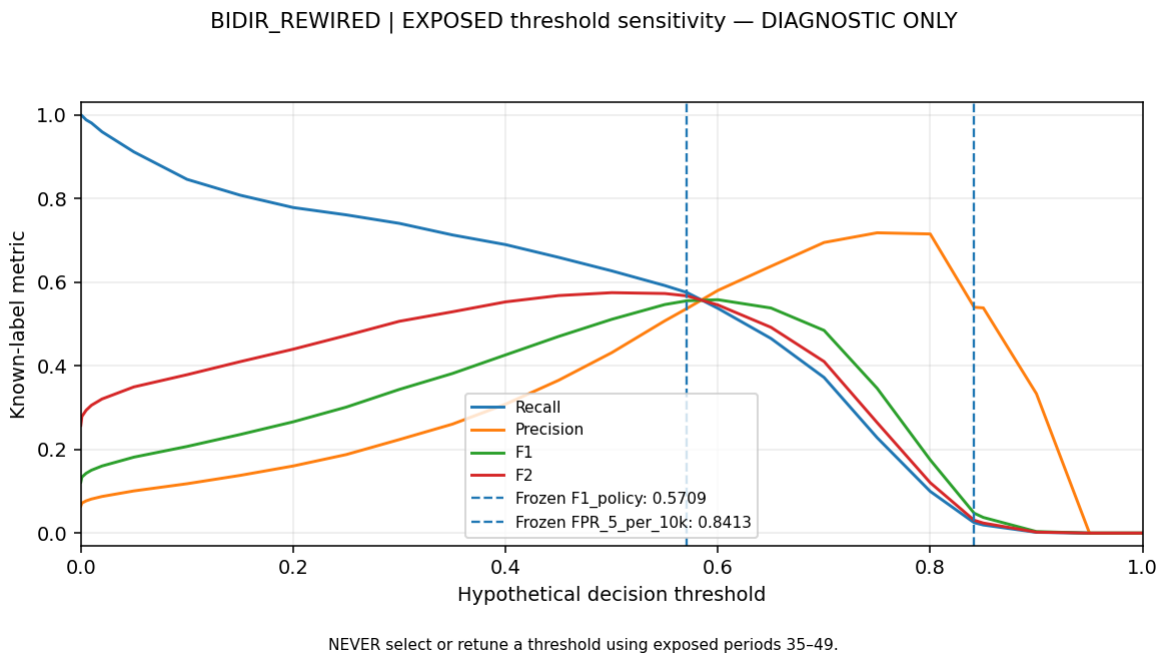
An apparent spike or collapse in a single period needs to be assessed against the number of verified illicit cases present in that step.

Period-level precision may also be undefined or set to a reporting convention when a model makes no positive predictions.

This figure should be read together with the label coverage charts and with the model's pooled confusion matrix.

A pooled score can remain respectable even when the model is unreliable in the last several periods.

Figure 81. BIDIR_REWIRED: post-hoc threshold sensitivity (diagnostic only)



Source figure: FULL_METRICS/plots/THRESHOLD_SENSITIVITY_BIDIR_REWIRED_DIAGNOSTIC_ONLY.png

The horizontal axis varies the classification threshold, while the curves show the associated precision, recall and F-score behavior measured using the already-exposed diagnostic labels.

The original F1 policy threshold (**0.571**) and false-positive-budget threshold (**0.841**) provide historically meaningful reference points.

The diagnostic sweep contains a threshold around **0.600** with observed F1 of **0.558**, but this number is a post-hoc descriptive result.

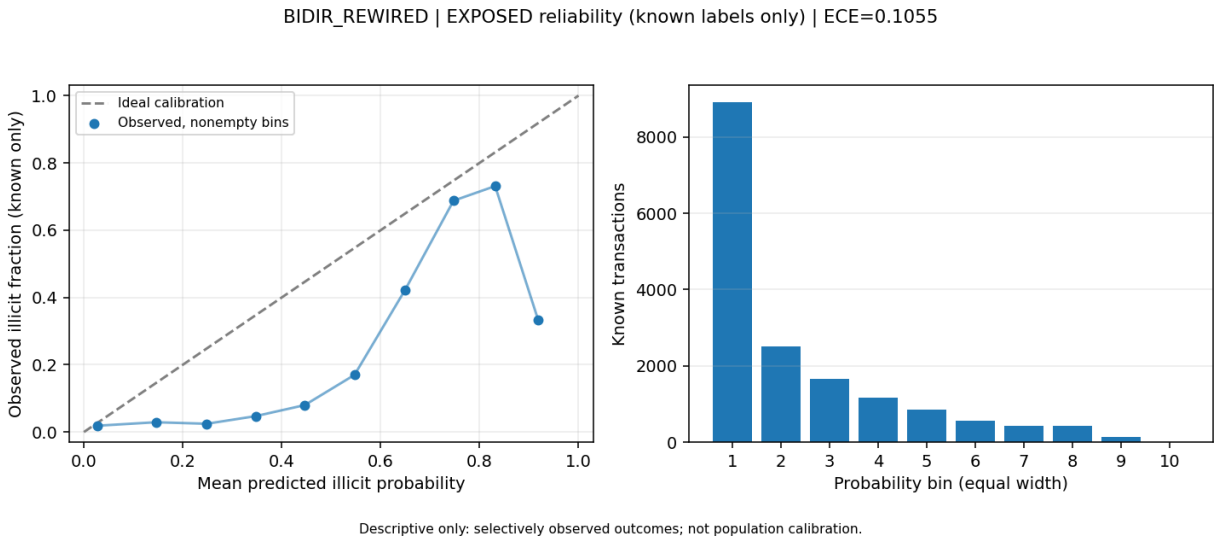
The distinction between looking and selecting is crucial here.

The entire curve can be useful for explaining why a model is sensitive or insensitive to threshold movement.

It does not authorize moving the operating threshold to the strongest point on the exposed-data curve and reporting the new result as an independent test.

Any new threshold policy would need to be fixed in a valid development window and evaluated subsequently on genuinely untouched observations.

Figure 82. BIDIR_REWIRED: reliability diagram and probability calibration



Source figure: FULL_METRICS/plots/RELIABILITY_BIDIR_REWIRED.png

Each reliability bin compares the average predicted illicit probability with the observed illicit fraction among verified transactions in that bin.

Points on the diagonal would be perfectly calibrated under that conditional population and binning method.

The saved report gives an expected calibration error of approximately 0.105 for BIDIR_REWIRED.

Its diagnostic Brier score is **0.1122** with uncalibrated probabilities and **0.0686** after the historical Platt-style calibration transformation.

The post-calibration Brier score is lower, suggesting improved squared probability accuracy for the known-label diagnostic cohort.

A reliability plot can also be misleading when a bin contains very few verified examples.

More importantly, matching the observed illicit frequency among labeled transactions does not prove that probability estimates are calibrated on the unknown-label majority or in a new prospective period.

6. Interpretation across model families

The combined evidence favors a multi-metric view over a single leaderboard.

Boosted trees have strong pooled ranking scores and remain important benchmarks for transaction-feature classification.

The MLP demonstrates that highly selective positive predictions are possible in the exposed diagnostic cohort, but its high precision comes with a material number of missed illicit cases.

Linear logistic regression provides a useful reference for broad detection at the cost of a large false-alarm burden.

Graph networks add scientific value through their matched no-message and rewired comparisons even when the graph models do not top the pooled AP table.

The graph experiments are particularly instructive because “graph structure helps” is not a single, context-free proposition.

The incoming-only model, the bidirectional model, and their controls operate with different neighborhood information.

It is reasonable to describe evidence that bidirectional message passing is beneficial under certain development comparisons.

It is not reasonable to infer that every form of graph aggregation necessarily outperforms well-tuned feature-based boosting, or that the observed benefits would generalize across all future chains and labeling policies.

7. The late-period performance break

Across the period-wise AP charts, the most conspicuous pattern is the reduction in ranking quality after step **42**.

This is analytically separate from the question of which model has the highest pooled AP.

A large pooled AP can be influenced by strong early periods and by the number of verified observations in each interval.

It can coexist with near-zero retrieval quality in subsequent, sparsely labeled periods.

For this reason, the temporal figures should have a prominent place in any research presentation.

Three distinct sources of difficulty appear in the reports.

First, the distributions of some numerical features differ markedly between early and late periods.

Second, the prevalence and number of known illicit cases change, reducing the stability of ranking and confusion-matrix estimates.

Third, the decision thresholds are carried forward from an earlier historical window, so calibration and error-rate behavior can change under shift.

These observations are related, but no single figure identifies which factor is the dominant cause.

A credible extension would require a new prospectively separated time window and carefully defined feature and labeling availability.

8. What the negative controls do and do not establish

The global and within-period label-permutation controls test different null hypotheses.

Their distinctly different AP results show that the time cohort itself can carry information that survives some forms of randomization.

Similarly, comparisons between original and rewired graphs test sensitivity to the particular edge pattern while preserving selected graph statistics.

The no-graph architectures test whether message passing matters beyond model capacity and transaction features.

Together these controls strengthen the scientific interpretation, but no finite control suite proves the total absence of leakage, hidden confounders or retrospective feature-availability problems.

The independent artifact audit serves a complementary role.

Recomputing classification metrics from saved transaction predictions, checking the presence of **22** policy-by-model summaries and **330** time-specific rows, and verifying SHA-**256** hashes help establish reproducibility of the reported computation.

They cannot demonstrate that previously exposed labels were never consulted in informal research choices, nor can they certify that the model will be deployed without distribution shift.

Computational integrity and external validity are separate achievements.

9. What claims are supported by these results?

It is supported to say that, within the recorded labeled retrospective diagnostic cohort, CatBoost has the highest pooled AP of the eleven configurations, the historical F1-operating-point MLP has the highest F1, and different models produce sharply different false-alarm burdens.

It is supported to say that the mean per-period AP of several models falls substantially after period **42**, that feature distributions change between the relevant windows, and that much of the highest-ranked alert population remains unverified.

It is also supported to describe the observed historical matched-architecture differences for particular graph variants.

It is not supported to describe periods **35-49** as a newly held-out, untouched test, to equate unknown labels with legal activity, to claim that verified-alert precision is the same as precision across all top-ranked alerts, or to assert that the name of the false-positive-budget policy guarantees a specific false-positive rate in later periods.

It is likewise unsupported to infer from the existing figures that graph structure causes all observed gains or that any model's probabilities are calibrated for every unlabeled transaction in a future operational deployment.

10. Choosing the Best Model for Different Research and Operational Objectives

The data do not support a single unconditional winner. A model can lead in score ranking, fixed-threshold classification, false-positive control or top-ranked alert retrieval, and still be inferior on a different objective.

Two kinds of evidence must be kept separate. **Historical development performance** is relevant to choosing a research model without consulting the exposed diagnostic window. **Performance observed in periods 35-49** describes the behavior of already-frozen models and policies, it cannot retrospectively redefine the primary model or create a fresh blind-test claim.

The table below summarizes the appropriate conditional choices. All diagnostic figures refer to the **16,670 known-label transactions** in periods **35-49**, including **1,083 known illicit** and **15,587 known licit** transactions. Top-K results use the full ranked candidate pool but compute verified outcomes only where labels are known.

Evaluation objective	Model supported by the observed evidence	What the evidence actually establishes
Primary, historically selected research model	**LightGBM (LGB)**	Top mean development AP, **0.8997** across 3 historical folds. This is the protocol-respecting primary choice.
Best pooled score ranking in the exposed diagnostic window	**CatBoost (CAT)**	Highest pooled AP, **0.7569**, and ROC-AUC, **0.9218**. These are exposed descriptive results.
Best F1 under the preselected F1 policy	**MLP**	Highest pooled F1, **0.7329**, with **631 TP**, **8 FP**, **452 FN** and recall **0.5826**.
Greatest recall with the F1 policy	**BIDIR_NO_GRAPH**	Joint highest recall, **0.7581**, matching SAGE_NO_GRAPH, with fewer FP (**1,783 vs 1,863**).
Best F2 under the F1 policy	**CatBoost (CAT)**	Highest pooled F2, **0.6857**, narrowly above XGBoost at **0.6837**.
Best F1 under the strict-policy-labeled threshold	**LightGBM (LGB)**	F1 **0.7554**, but its observed FP rate is **7.06 per 10,000**, above the nominal historical target of 5.
Lowest false-positive count at the strict policy	**SAGE_BATCH**	**0 FP**, but only **74 TP** and recall **0.0683**. Zero observed FP does not imply a future guarantee.
Highest recall among models meeting the observed <=5/10,000 FP constraint	**MLP**	Recall **0.3915** with **4 FP**, or **2.57 FP per 10,000** known licit observations. This comparison is post hoc.
Top-10 and Top-25 review budget	**CatBoost (CAT)**	Identified **54** and **113** verified illicit alerts across 15 periods, respectively.
Top-50 and Top-100 review budget	**LightGBM (LGB)**	Identified **204** and **386** verified illicit alerts, respectively. At Top-100, verified recall was **0.3564**.
Lowest pooled Brier score	**CatBoost (CAT)**	Brier **0.0303** on known labels. This does not establish calibration on unknown-label transactions.
Graph structure research	**BIDIR_SAGE with matched controls**	Historical comparisons test value from directed neighbor messages, not universal superiority to boosted trees.

10.1 Selecting a Model Before Looking at Exposed Diagnostic Outcomes

Recommended primary research model: LightGBM. Its historical development mean Average Precision is **0.8997**, compared with **0.8961** for XGBoost and **0.8933** for CatBoost.

The margins between these boosted-tree models are small and development folds overlap, so this ranking should not be described as statistically decisive evidence that LightGBM is inherently better.

LightGBM remains the defensible primary choice because the selection rule was applied to the earlier development stage, rather than being revised to match the exposed outcomes. The trained model, preprocessing, calibrator and decision policies should remain frozen for the retrospective analysis.

CatBoost and XGBoost should be retained as central pre-specified comparators. Their value is to show how sensitive conclusions are to reasonable alternatives, not to replace the primary model after inspecting periods 35-49.

10.2 Best Model for Overall Ranking

CatBoost has the highest pooled diagnostic Average Precision, AP = 0.7569, narrowly ahead of **XGBoost at 0.7522** and **LightGBM at 0.7455**.

It also has the highest pooled diagnostic ROC-AUC of **0.9218**. These metrics assess the ordering of known illicit and known licit cases over many possible thresholds, rather than performance at a single operating decision.

CatBoost is therefore the observed ranking leader in the exposed labeled cohort. It should not be called the independently verified best future model, because the evaluation period is not a fresh blind test, the label observation process is selective, and the differences between boosted-tree models are modest.

10.3 Best Model for F1 and High Positive Predictive Value

MLP leads on pooled F1 under the preselected F1 policy, F1 = 0.7329. Its corresponding precision is **0.9875**, with **631 true positives**, **8 false positives** and **452 false negatives**.

This operating point is exceptionally selective among known-label cases. It produces few confirmed false alerts, but it retrieves only **58.26%** of the known illicit transactions. The remaining **41.74%** are missed in the labeled diagnostic sample.

MLP is attractive when the review process is strongly constrained by false-alert burden and missed detections are comparatively less costly. It is not the preferred choice if maximizing the total number of detected known illicit transactions is the overriding objective.

An F1 result also depends on the frozen historical decision threshold. A model with a higher AP can have a lower F1 at that particular threshold without any arithmetic inconsistency.

10.4 Highest Recall and Detection-Oriented Evaluation

Under the fixed F1 policy, **SAGE_NO_GRAPH** and **BIDIR_NO_GRAPH** tie for the highest recall, **0.7581**, detecting **821** of **1,083** known illicit transactions.

Between these two matched no-message architectures, **BIDIR_NO_GRAPH** has fewer false positives, **1,783 rather than 1,863**, and slightly higher precision, **0.3153 rather than 0.3059**. It is therefore the more efficient of the two at their recorded fixed operating points.

The false-positive burden is still substantial. For comparison, XGBoost detects **792** known illicit transactions with **668** false positives, and CatBoost detects **753** with **406** false positives.

If review capacity is limited, the model with maximum recall may be impractical. A high-recall decision must always be reported together with its number of alerts and its false-positive cost, not recall alone.

10.5 Recall-Weighted F2 and the Precision-Recall Trade-off

CatBoost leads on pooled F2 under the historical F1 policy, F2 = 0.6857, with XGBoost close behind at **0.6837**.

F2 places greater weight on recall than F1, but it does not maximize recall in isolation. The result reflects a compromise between missed illicit cases and the number of known licit transactions incorrectly flagged.

CatBoost may be the most attractive model in this exposed comparison when recall matters more than precision, but the expected false-alert burden remains relevant. The small F2 gap to XGBoost does not justify a strong claim of stable superiority.

10.6 Very Low False-Positive Budgets

The policy labeled **FPR_5_per_10k** was selected historically to target no more than five false positives per 10,000 known licit transactions in the policy-selection window. Its name is not a guarantee of the same rate in later data.

In the exposed diagnostic window, LightGBM achieves **F1 = 0.7554**, with **664 TP** and **11 FP**, but its false-positive rate is **7.06 per 10,000**, above the numerical target.

MLP generates **4 FP**, or **2.57 per 10,000**, and detects **424** known illicit cases, giving recall **0.3915**. Among the models that actually satisfy the numerical target on this already-exposed window, MLP has the highest recall.

SAGE_BATCH produces **0 observed FP**, but detects only **74** known illicit cases and reaches recall **0.0683**. This is a highly conservative but low-coverage operating point, not proof of perfect future specificity.

This scenario illustrates why the constraints must be applied before maximizing an objective. **LightGBM has the highest F1 under this named policy, but it does not satisfy the observed five-per-10,000 criterion.** Rechoosing its threshold from the exposed window would invalidate a claim of independent historical evaluation.

A future operational selection under a hard false-positive budget would require a newly defined historical policy window and a separate untouched evaluation period, with acceptable uncertainty bounds for false positives.

10.7 Limited Investigation Capacity and Top-K Alerts

A limited review capacity changes the problem from classifying all transactions to ranking them for an investigator. The model should then be compared at the actual number of alerts an analyst can review per period.

For **K = 10** and **K = 25** in each of the 15 diagnostic periods, CatBoost identifies the largest number of verified illicit cases, **54** and **113**, respectively.

For **K = 50** and **K = 100** per period, LightGBM moves ahead with **204** and **386** verified illicit alerts, corresponding to micro Recall@K of **0.1884** and **0.3564**.

At Top-100, CatBoost identifies **372** verified illicit alerts, with verified-alert precision **0.9841**, compared with **386** for LightGBM at verified-alert precision **0.9723**.

The top-ranked alert population contains many unknown-label cases. For LightGBM at Top-100, **1,103 of the 1,500** selected alerts have no verified label. The precision figures therefore describe only verified alerts, not the true precision of all 1,500 reviewed items.

Practical interpretation: CatBoost is the observed leader for very small review queues, while LightGBM leads for larger queues of 50 or 100 alerts per period. This comparison remains descriptive because the diagnostic labels were already exposed.

10.8 Calibration Quality and Probability Interpretation

CatBoost achieves the lowest pooled Brier score, 0.0303, followed by **LightGBM at 0.0318** and **XGBoost at 0.0325**.

A lower Brier score indicates a smaller average squared difference between predicted probability and observed binary outcome among known-label transactions. It reflects both calibration and discrimination, so it should be interpreted alongside reliability diagrams and Log Loss.

A model with the best ranking does not automatically produce the most reliable probabilities, and calibration measured on selected known-label cases does not establish calibration in the mostly unknown-label transaction population.

For decisions based directly on predicted probability rather than relative ranking, calibration quality becomes part of the model-selection criterion. The available data favor CatBoost descriptively, but do not establish future calibration robustness under temporal shift.

10.9 Graph-Based Models and the Scientific Question of Network Value

Graph neural networks are valuable in this study because they test whether directed transaction relationships carry predictive information beyond transaction-level features.

The matched-architecture comparisons provide more relevant evidence about the contribution of message passing than a simple ranking of unrelated architectures. Bidirectional aggregation shows favorable historical differences over its corresponding no-message control in some development comparisons, while the incoming-only variant does not yield a consistent advantage.

For a study whose objective is **explaining when graph structure helps**, BIDIR_SAGE and its matched no-graph and rewired controls should be retained as the central experimental family. They should not displace LightGBM as the primary prediction benchmark solely on the basis of these comparisons.

The rewired-graph controls strengthen the analysis by testing sensitivity to the particular edge arrangement, but neither a positive nor a negative graph difference proves a universal causal effect across later data.

10.10 When No Reliable Single Best Model Can Be Claimed

Temporal performance changes are large. For example, LightGBM's unweighted mean period AP falls from approximately **0.880** in periods **35-42** to approximately **0.067** in periods **43-49**.

This shift, together with sparse and selectively observed labels in some late periods, limits the reliability of a pooled winner-takes-all conclusion. Strong earlier cohorts may dominate pooled results, while later cohorts present a substantially different ranking problem.

No model in this report is established as the universally best model for future data. The reported leadership changes with the objective, the fixed threshold, the review budget and the time window.

11. Final Model Selection and Reproducible Decision Rule

Final selected primary research model: LightGBM (LGB). This choice is grounded in the historical development ranking, not in the exposed diagnostic standings.

The selection rule is to maximize mean Average Precision across the designated chronological development folds, then freeze the chosen model family and the later, historically specified training, calibration and decision-policy protocol. Under that rule, LightGBM obtains **mean development AP = 0.8997**, the largest among the **11** evaluated configurations.

The study should report CatBoost and XGBoost as strong boosted-tree comparators, MLP as the leading observed F1 operating-point alternative, and BIDIR_SAGE with its matched controls as the graph-structure research family.

The primary model should not be switched to CatBoost just because CatBoost obtains the highest exposed pooled AP, nor to MLP simply because it has the largest exposed fixed-threshold F1. Such a switch would make model selection conditional on data already inspected.

Any deployment-oriented choice should begin with a **written target objective and constraint**, such as maximizing verified recall under an alert budget, controlling known-label false positives, or minimizing probabilistic loss. That objective should be evaluated using separate historical validation data before inspection of a new untouched test set.

If a single model must be named in the report, name LightGBM as the primary, historically selected research model, and describe CatBoost, MLP and the graph models as conditional alternatives rather than replacing the declared primary winner.

Selection level	Chosen model or interpretation	Basis and qualification
Primary model for the reproducible study	**LightGBM**	Historical development mean AP, **0.8997**, determined before the exposed diagnostic comparison
Post-hoc best ranking	**CatBoost**	Highest pooled exposed AP, **0.7569**, descriptive and not an independent final selection
Post-hoc best F1 at the F1 policy	**MLP**	**F1 = 0.7329**, precision **0.9875**, recall **0.5826**
Graph research comparator	**BIDIR_SAGE with controls**	Necessary to study the value of directed neighbor information
Future production choice	**Not established**	Requires new, untouched temporal data and a prospective review-cost objective

This is a model-selection statement about the research protocol, not a claim that LightGBM dominates every diagnostic metric. The distinction is essential to preserving the scientific meaning of the results.

12. Final Research Summary and Conclusion

The Elliptic++ study illustrates both the promise and the methodological difficulty of illicit transaction classification within a partially labeled transaction network.

Transaction-level boosted-tree models produce strong ranking discrimination within many labeled historical cohorts.

Neural models can occupy very different operating points, with the MLP showing a particularly striking precision-recall compromise.

Graph neural networks contribute an additional dimension by testing whether directed transaction relationships add predictive information beyond node-level features.

Yet the most consequential finding may be the temporal fragility of these results: model rankings, classification thresholds and calibration metrics all need to be interpreted in light of

the sharp performance change in the later periods and the incomplete verification of the transaction population.

For academic communication, the strongest sequence is to establish the chronological protocol and label limitations first, present the development model choice second, then show pooled ranking and confusion-matrix trade-offs, followed by the temporal degradation, graph controls and calibration diagnostics.

The study is valuable precisely when those constraints remain visible.

Its contribution is a transparent, reproducible, controlled retrospective analysis - not a claim of independently verified, prospective illicit-transaction detection.

12.1 Final Decision and Main Takeaways

Primary final research choice: LightGBM. It wins the pre-exposure historical development selection criterion and remains the appropriate fixed reference model in the report.

Best diagnostic ranking: CatBoost. It achieves the highest pooled AP and ROC-AUC among known-label transactions, but these are exposed retrospective observations.

Best diagnostic F1: MLP. It combines very high precision with very few false positives, while missing **452** known illicit transactions at the selected F1 policy.

Highest diagnostic recall at the F1 policy: BIDIR_NO_GRAPH and SAGE_NO_GRAPH. Both reach recall **0.7581**. BIDIR_NO_GRAPH incurs fewer false positives at that observed operating point.

Best top-ranked alert retrieval depends on K. CatBoost leads at **K = 10** and **25**, LightGBM leads at **K = 50** and **100**.

Strict false-positive policies require an explicit feasibility check. LightGBM leads in F1 under the low-FPR-named threshold but exceeds its numerical target in the diagnostic data. MLP has the highest recall among models meeting that observed target.

No single model is validated as universally best for future transactions. Large temporal changes, incomplete labels and the exposed status of the diagnostic period make such a claim unjustified.

The study's most defensible contribution is a reproducible comparison of ranking, thresholded classification, selective verification, graph effects and temporal robustness under a transparent chronological protocol.

Data provenance and source material

All 82 images reproduced in this document were read from EXPOSED_DIAGNOSTIC_20261029T100035Z.zip without changing their plotted data.

Numerical values were drawn from the same archive, principally FULL_METRICS/CLASSIFICATION_REPORT_ALL_MODELS.csv, FULL_METRICS/CLASSIFICATION_BY_TIME_STEP.csv, FULL_METRICS/RECALL_PRECISION_AT_K_SUMMARY.csv, FULL_METRICS/CALIBRATION_BINS.csv, FULL_METRICS/THRESHOLD_SWEEP_DIAGNOSTIC_ONLY.csv, CALIBRATION_DIAGNOSTICS.csv, DEV_SUMMARY.csv and the related research control outputs.

The archive completion marker identifies the run as an official Elliptic++ transaction-data execution and flags its diagnostic-only status.

No plots were retrained, no model scores were altered, and no thresholds were revised to improve late-period findings while preparing this report.

The interpretation has been expanded into explanatory prose, the underlying research results remain those supplied in the archive.